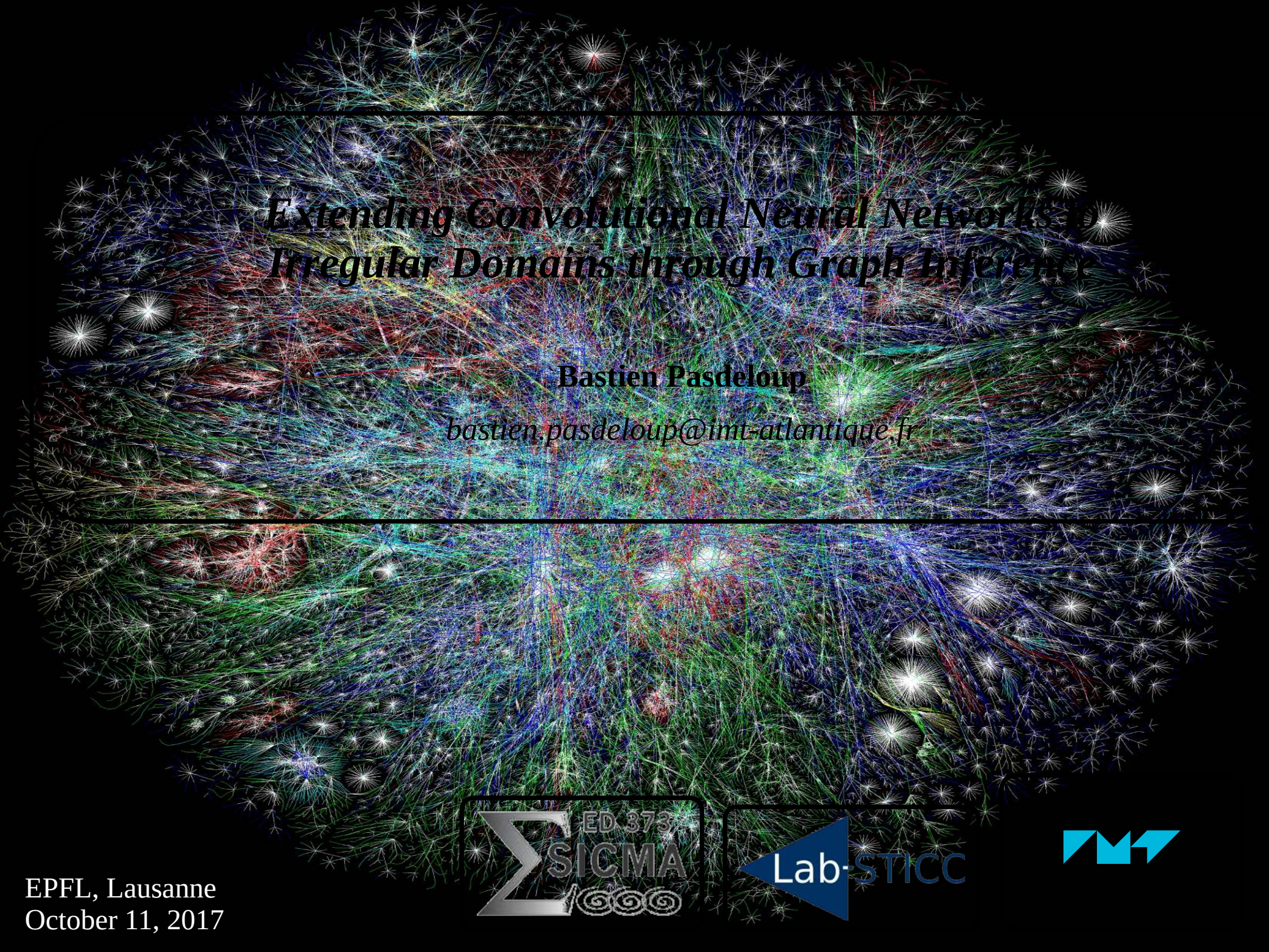




Extending Convolutional Neural Networks to Irregular Domains through Graph Inference

Bastien Padeloup

bastien.padeloup@imt-atlantique.fr







Methods on graph signal processing

- What is it?

- The main ideas

- Some problems and applications

- Graph inference from signals

- Why it may be necessary

- The case of stationary signals

- Translations on irregular graphs

- How we define them

- How we identify them

- Example on a random graph

- Extension of convolutional neural network

- Simulation of the method

Methods on graph signal processing

What is it?

The main ideas

Some problems and applications

► Graph inference from signals

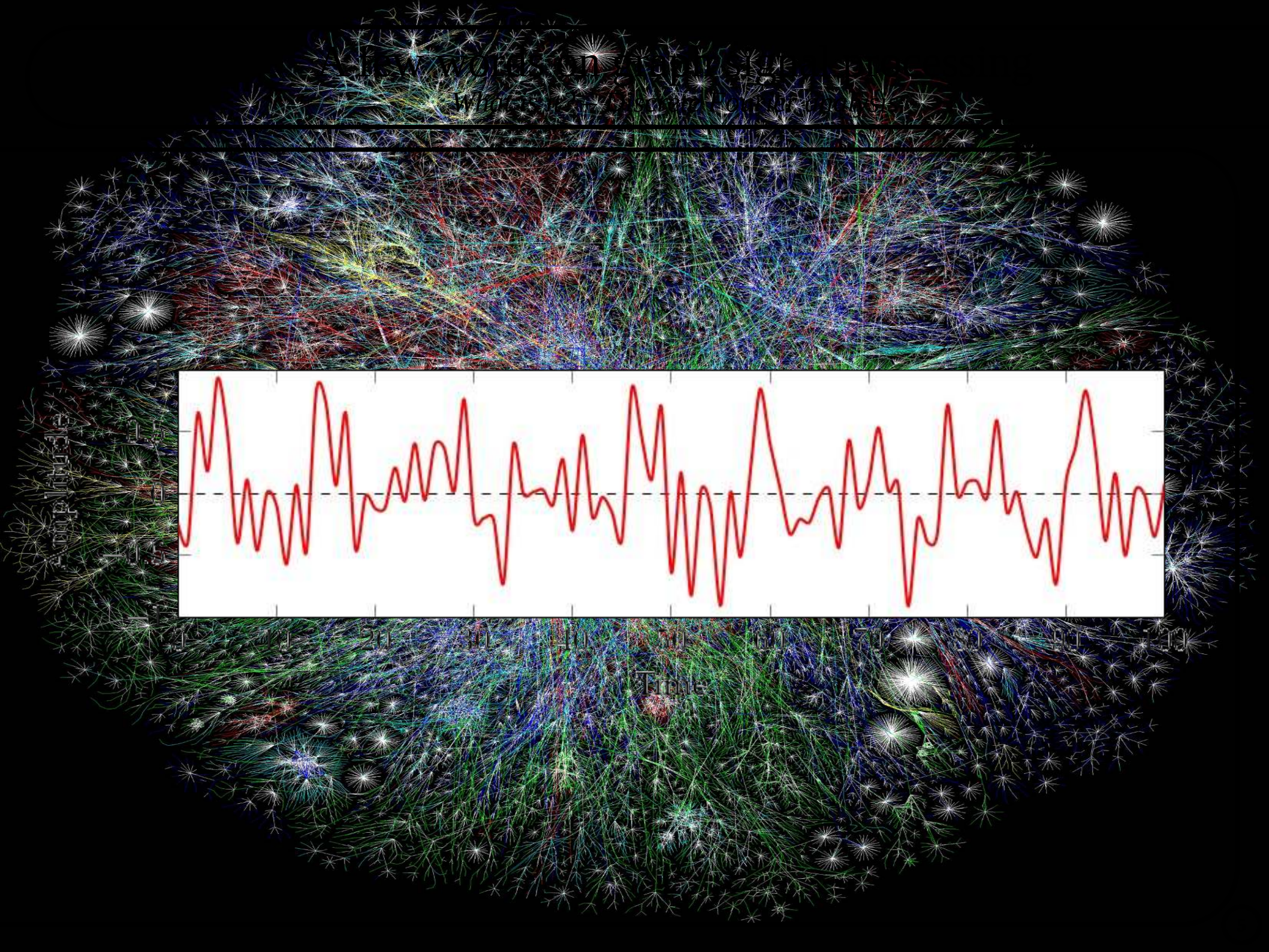
- * Why it may be necessary
- * The case of stationary signals

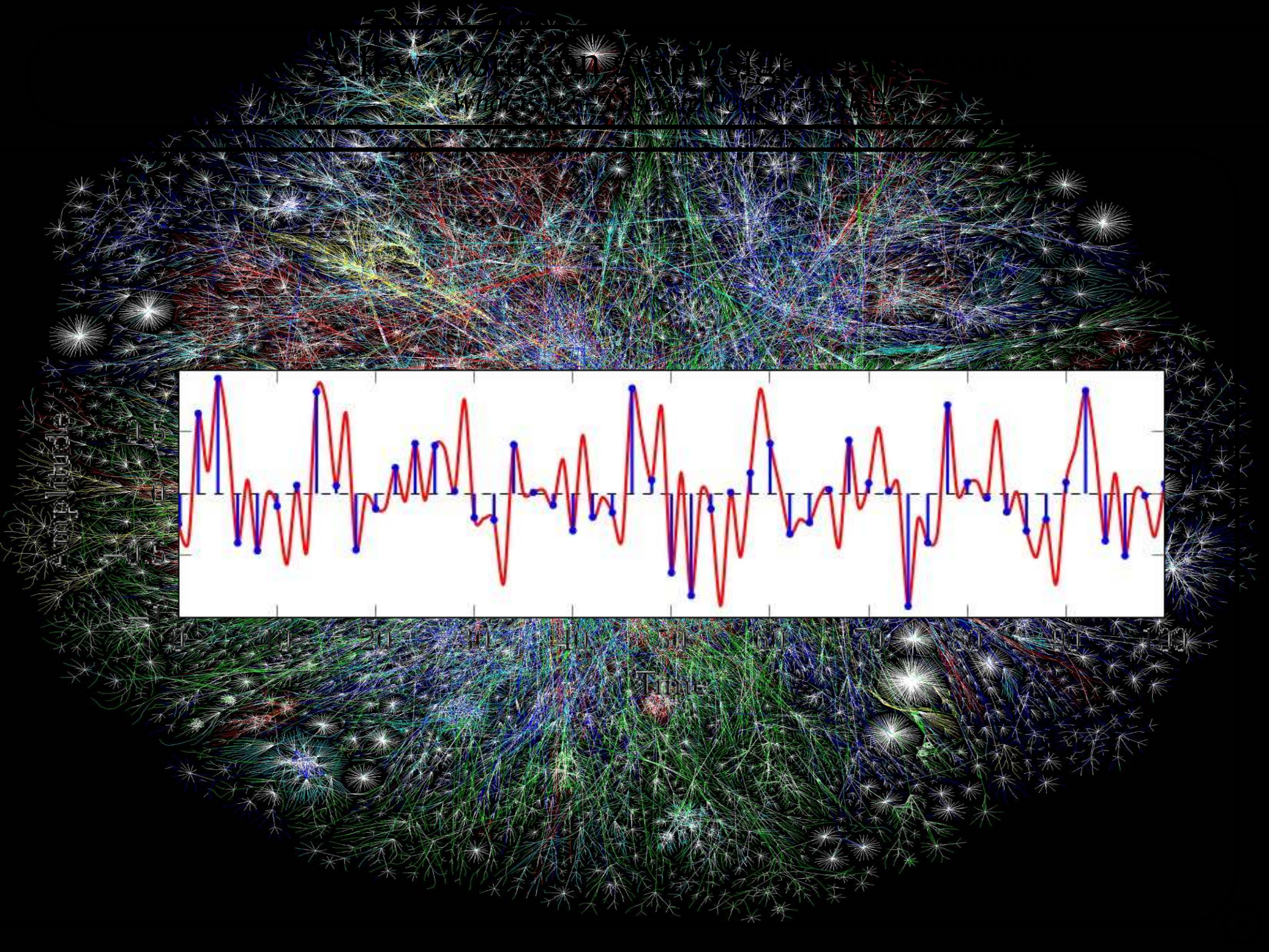
► Translations on irregular graphs

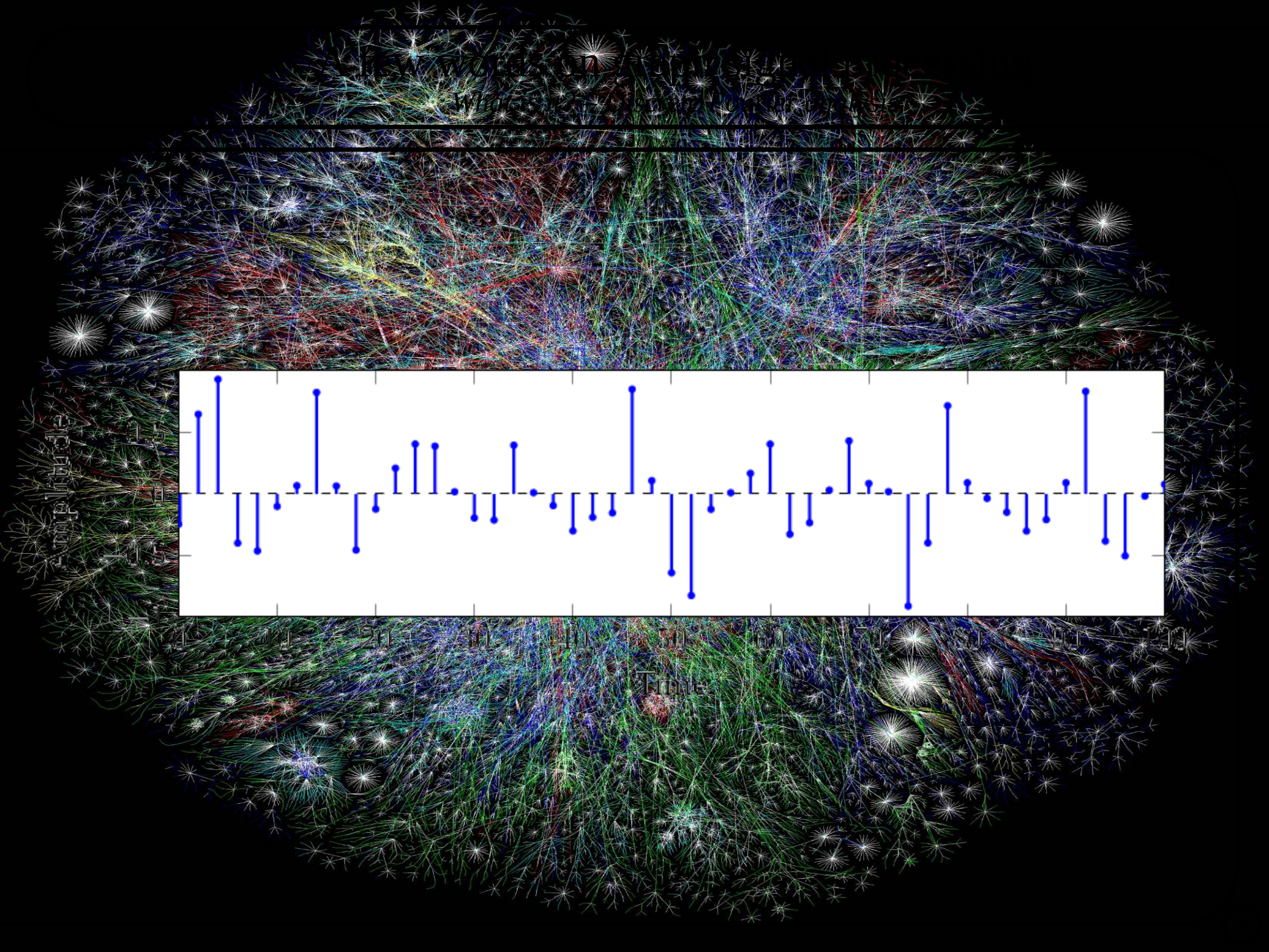
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► Extension of convolutional neural networks

- * Introduction of the method
- * Results

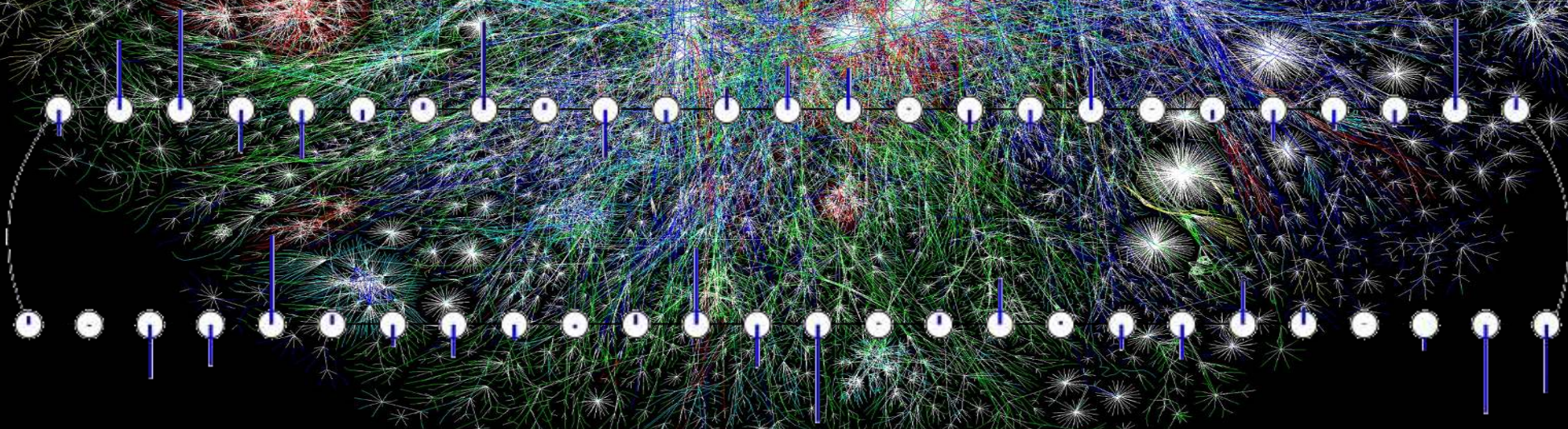
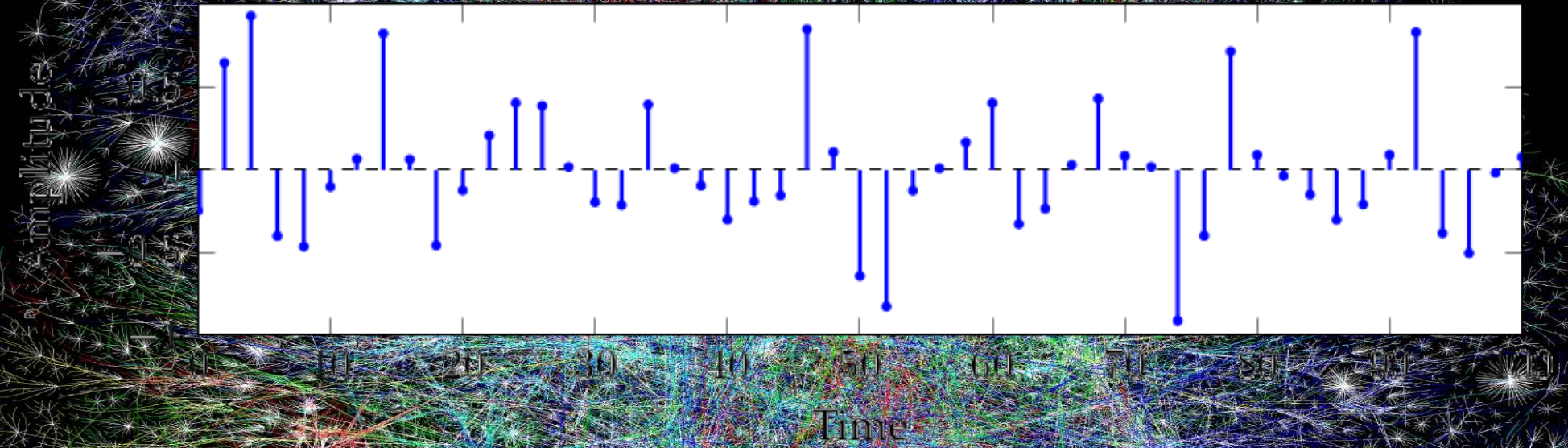




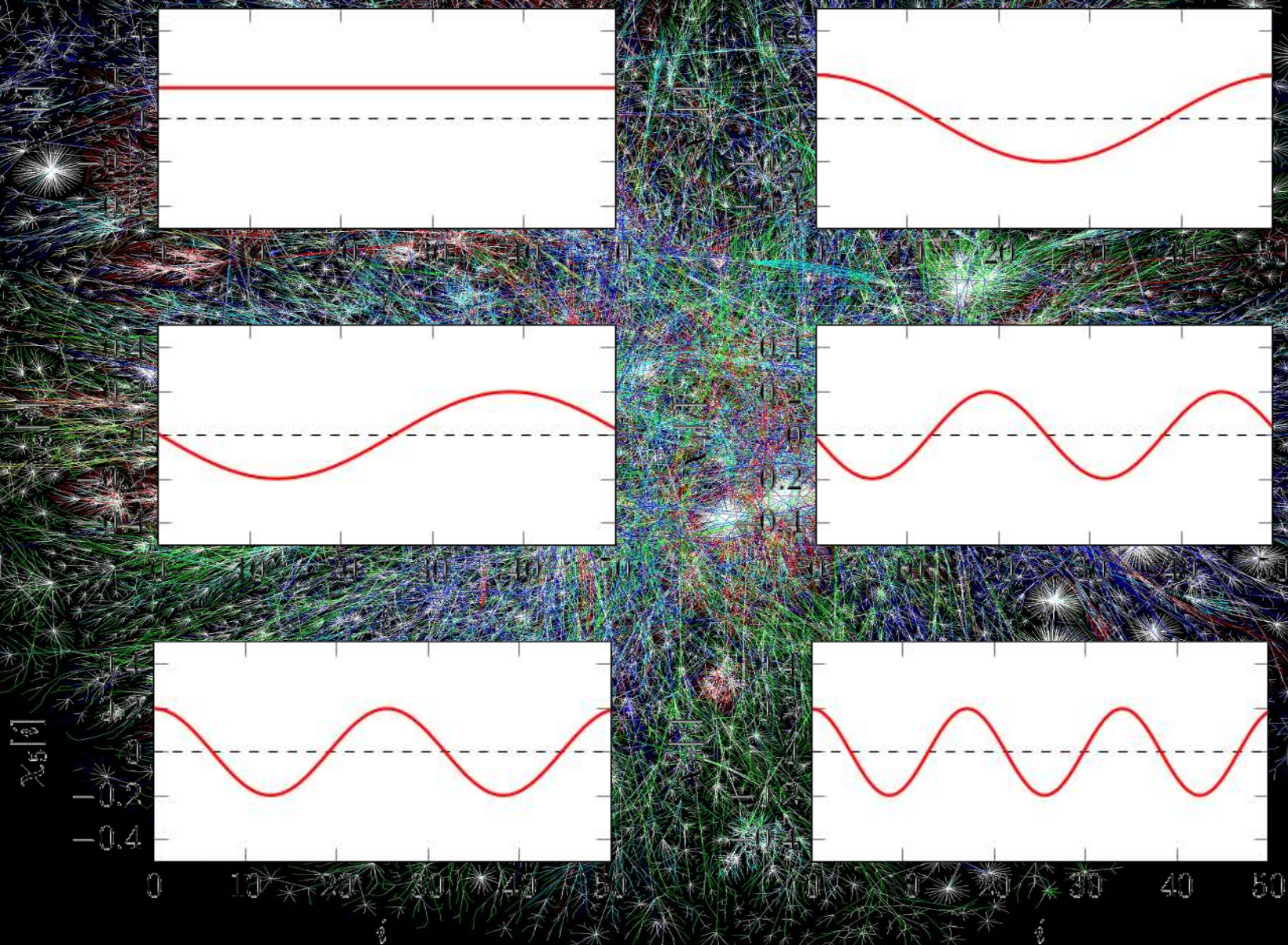


VIEW WORKS 10

What is it? A 100% natural product

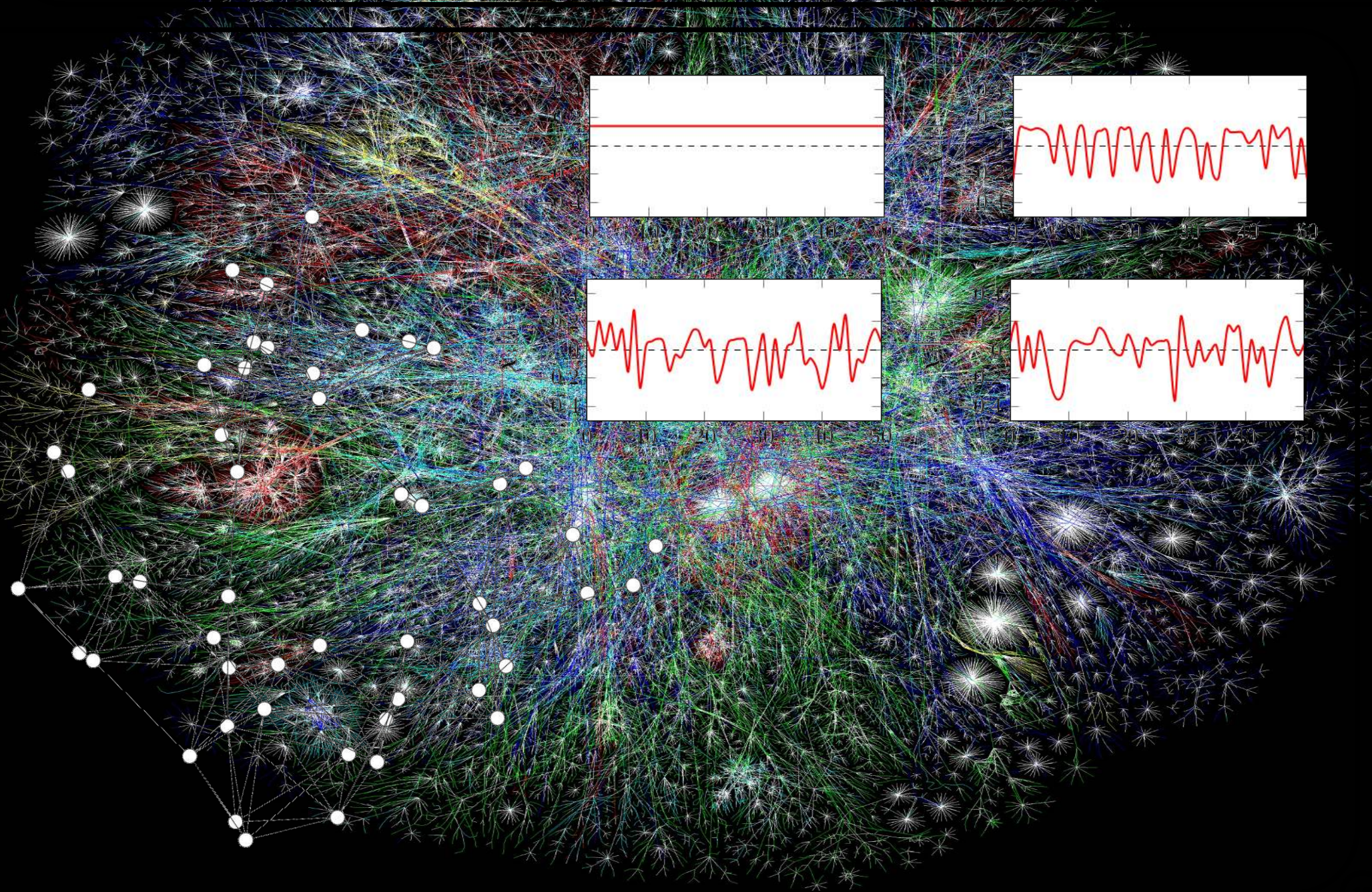


χ^2 and $\chi^2/\text{d.o.f.}$ The first eigenmode of the LSS



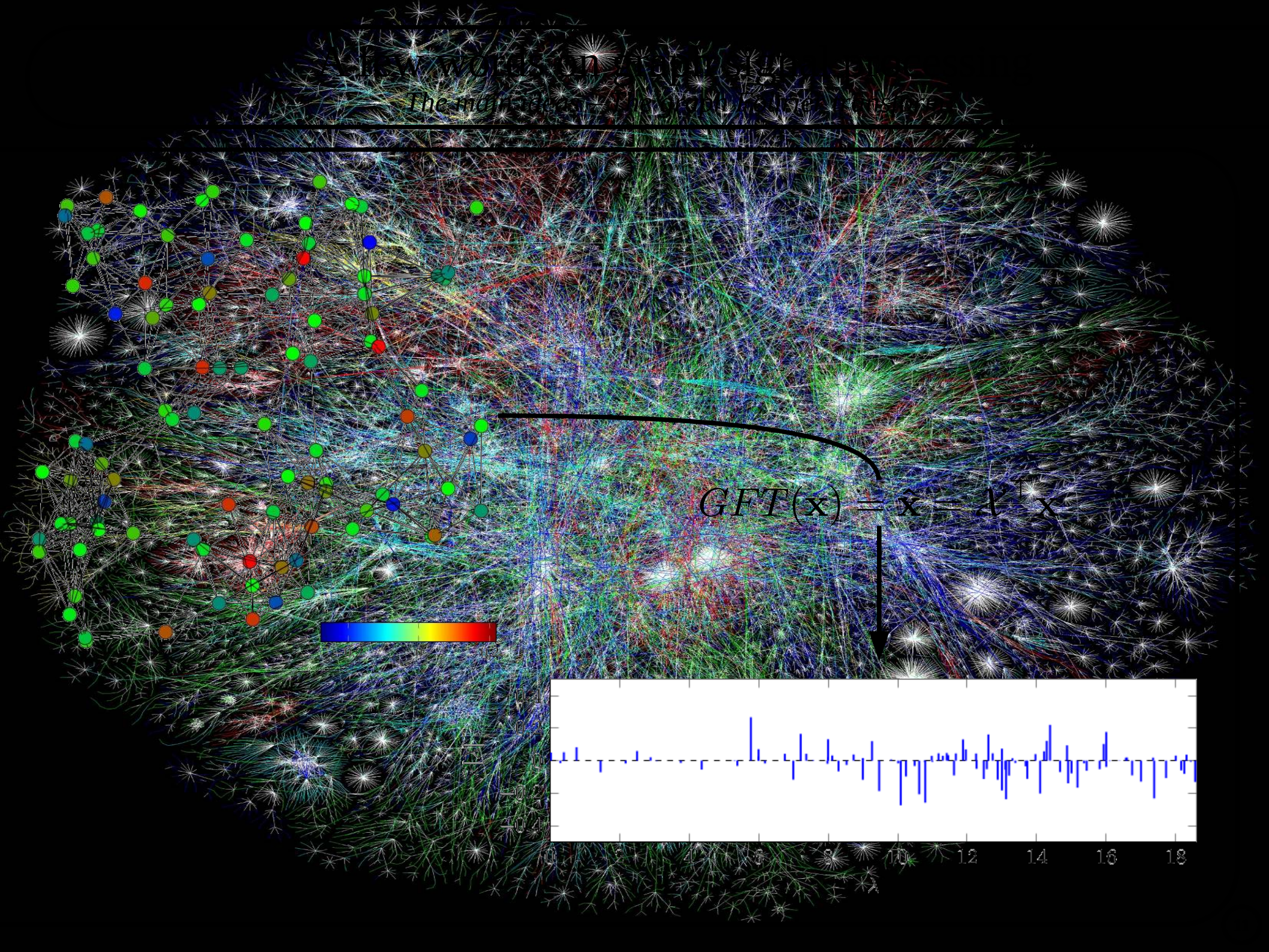
What's new in network science?

What is it? How does it work?



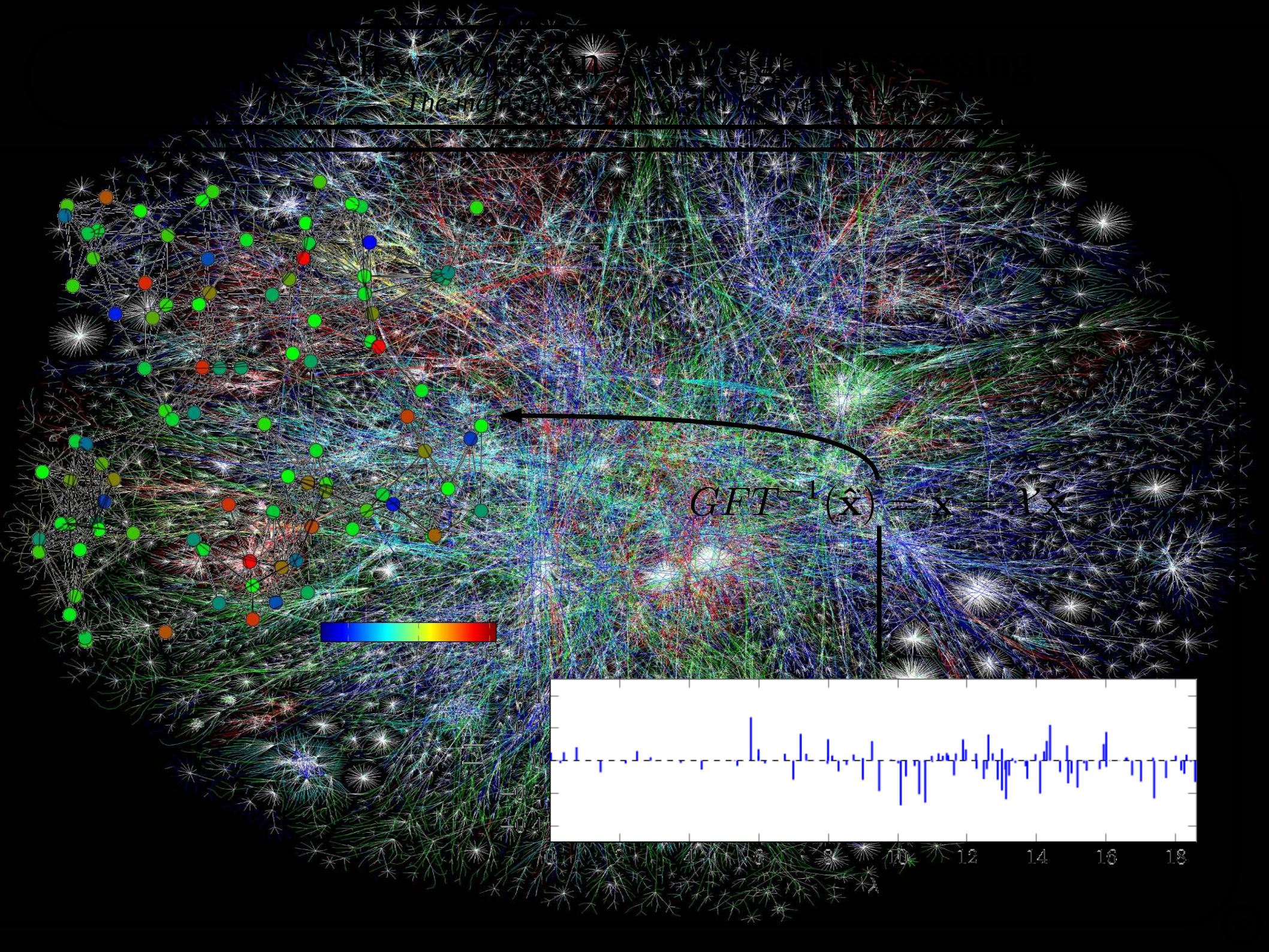
VIEW WORKS ON

The many faces of network science



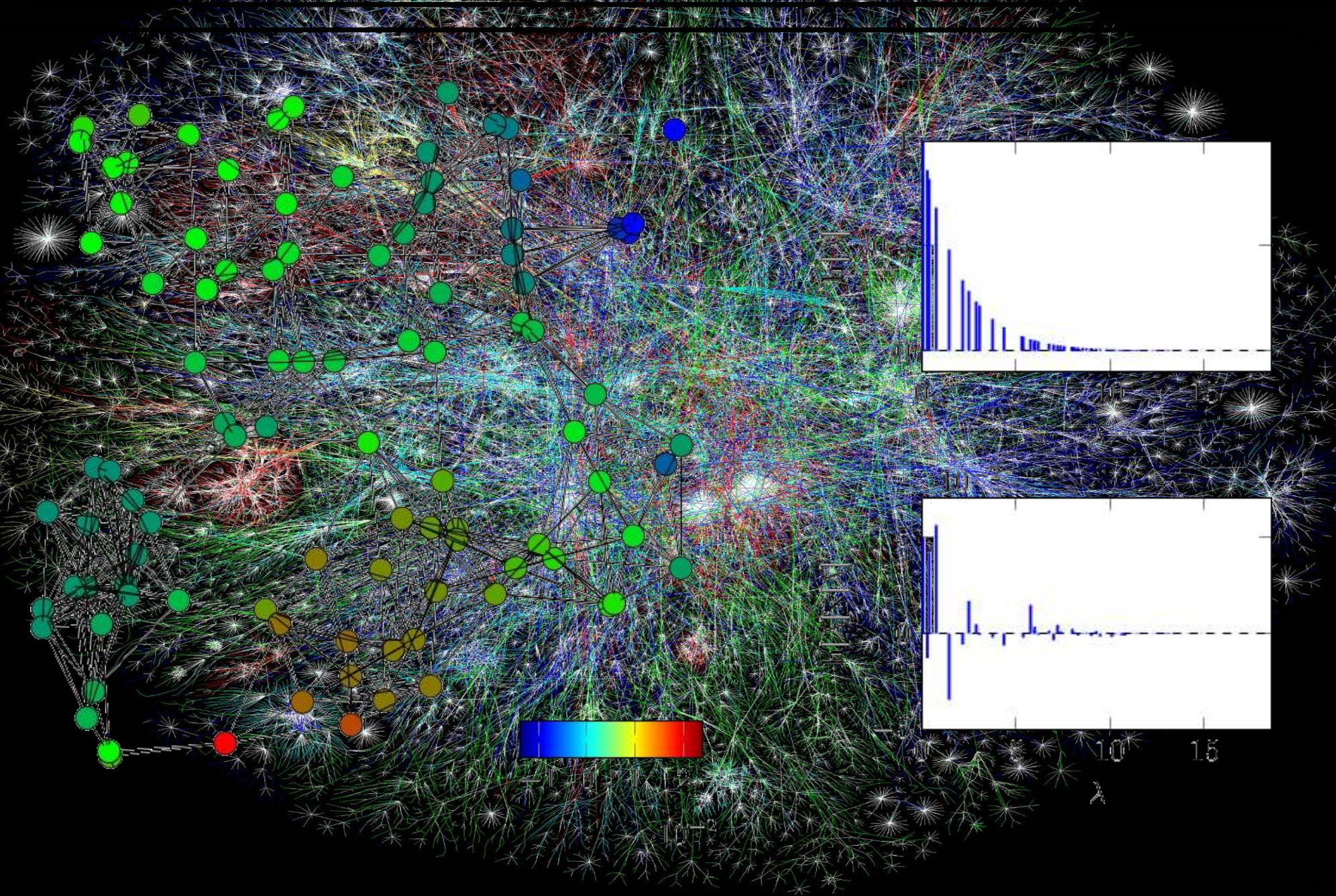
VIEW WORKS ON

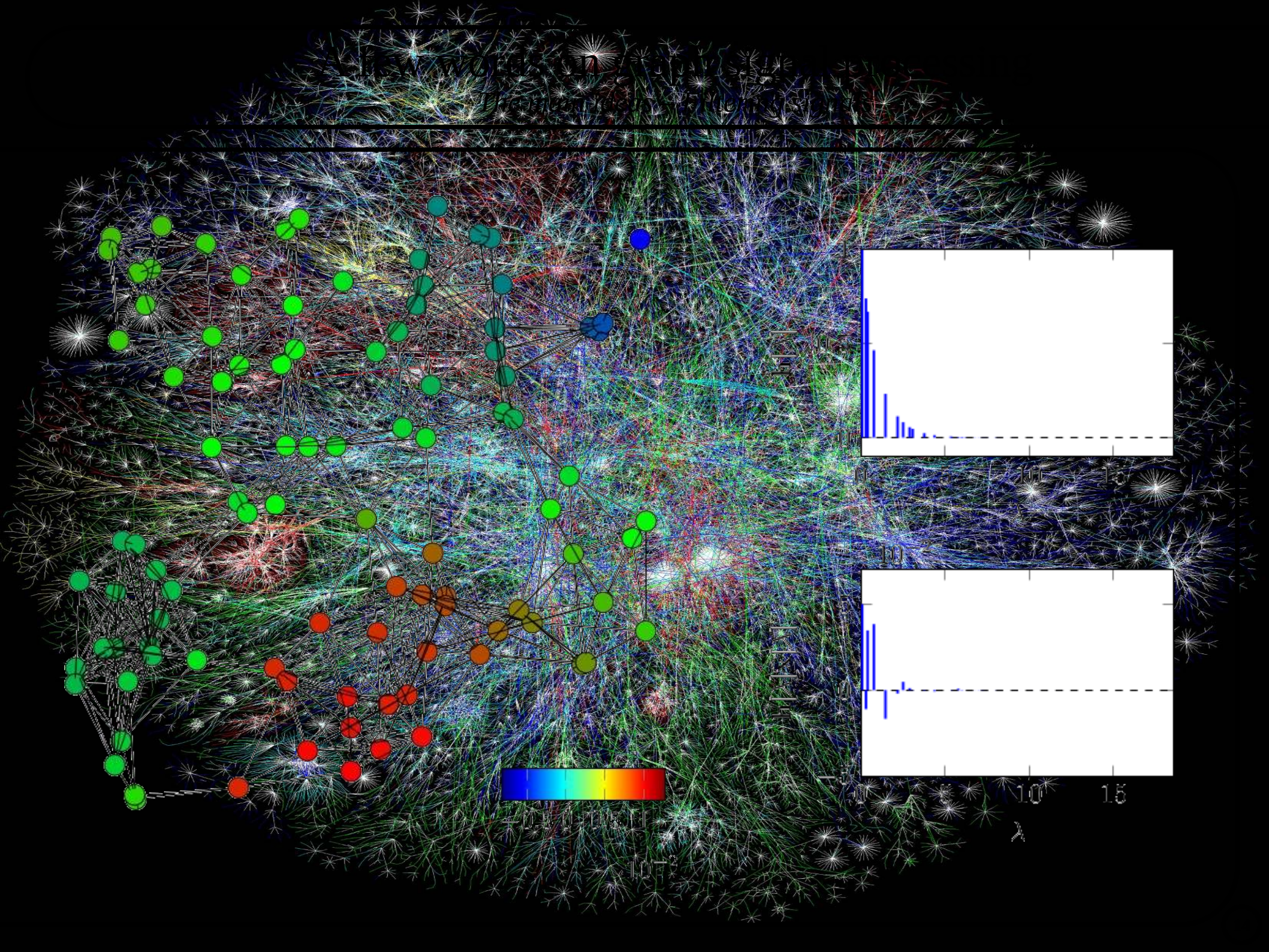
The mean-field limit of the Kuramoto model



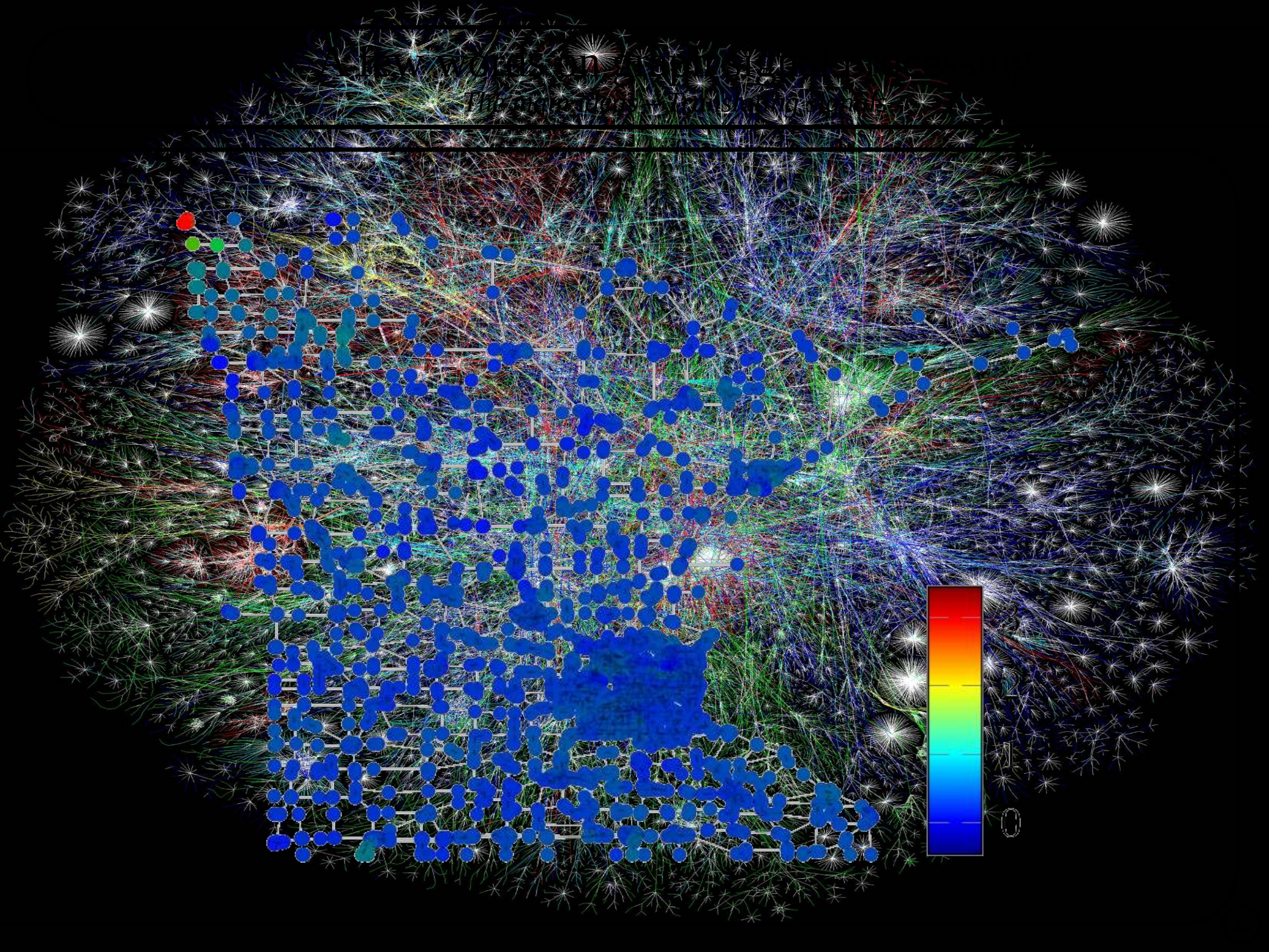
STANFORD UNIVERSITY

Department of Statistics

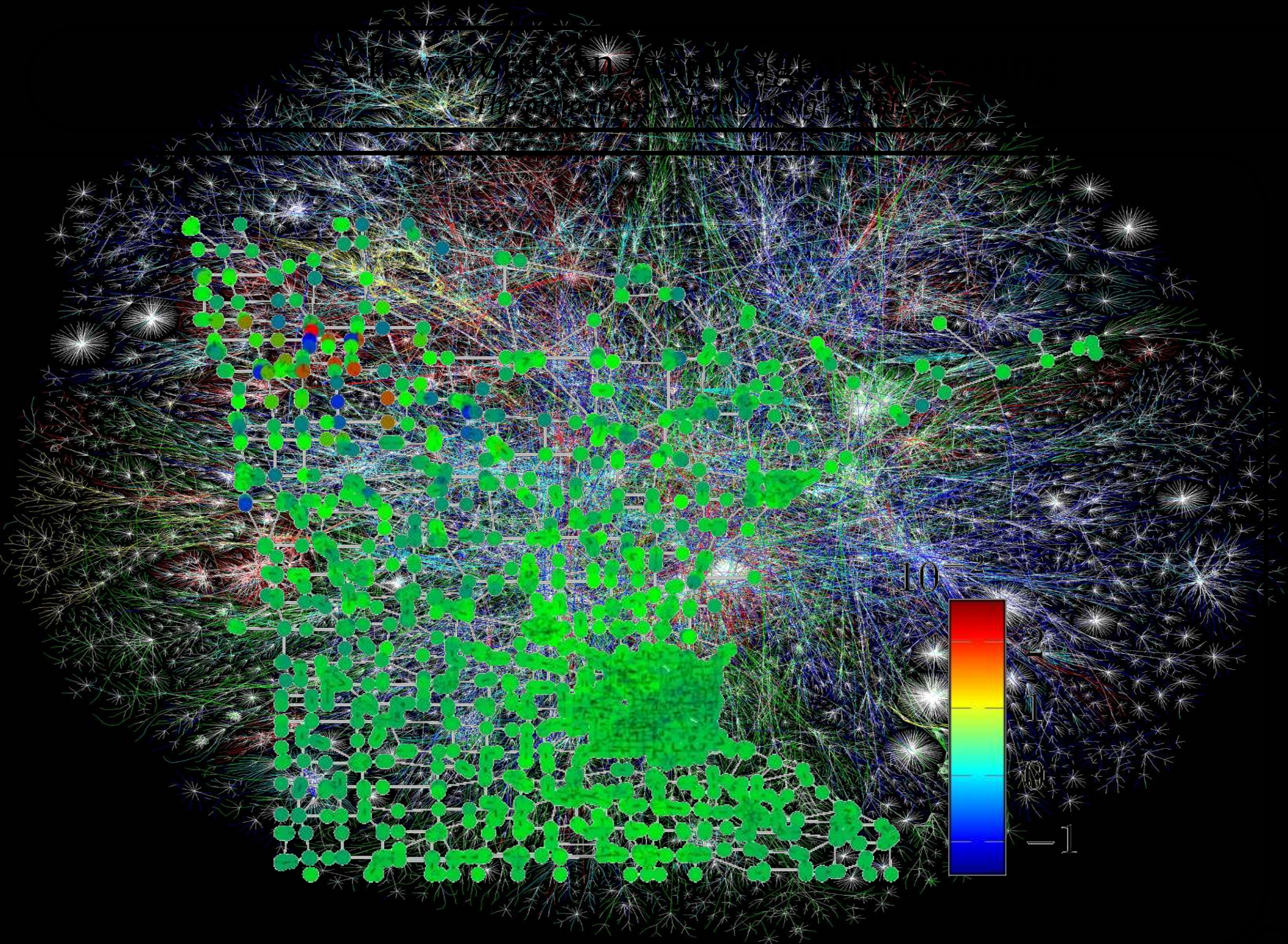




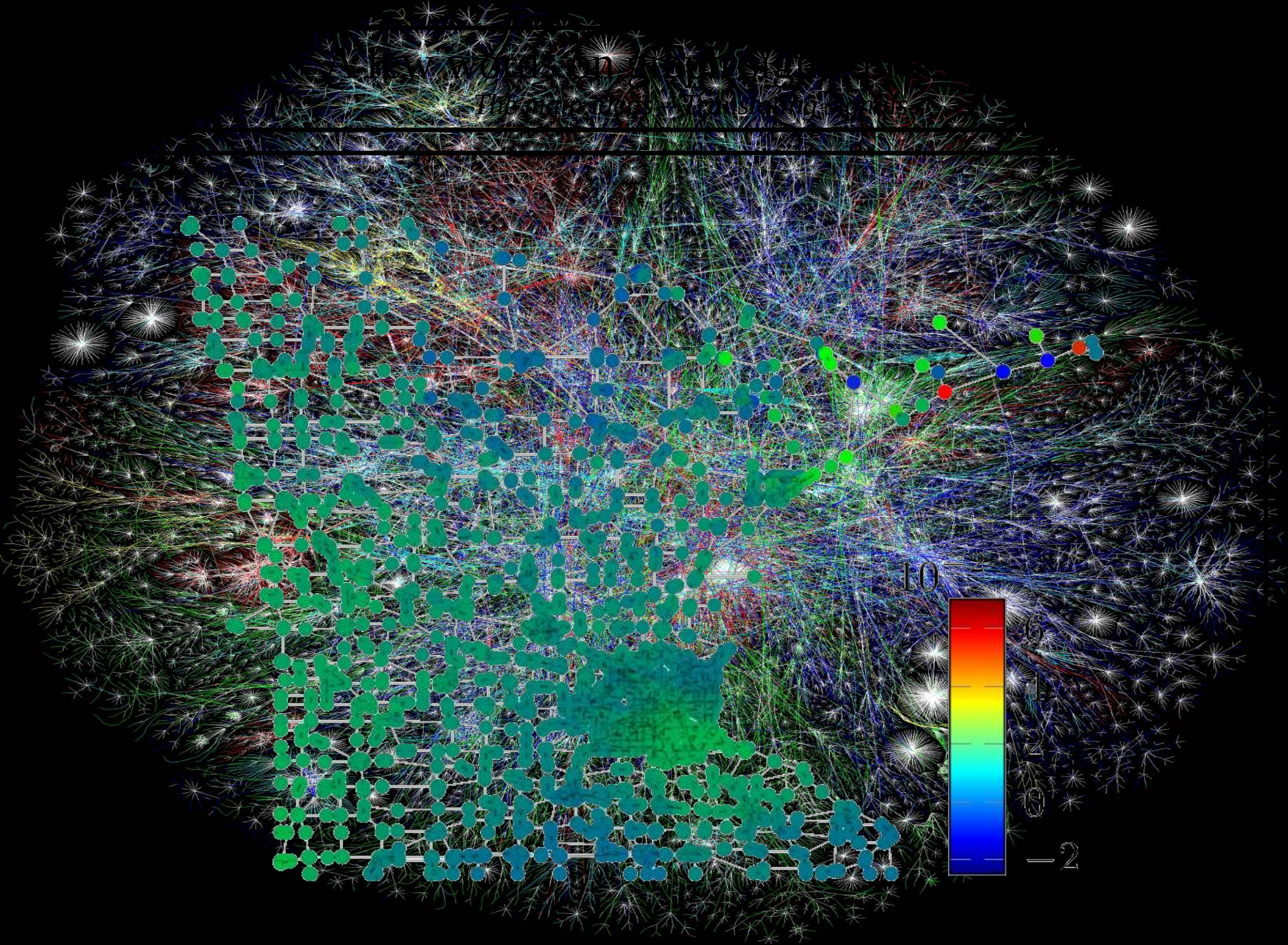
NEW WORK ON *The Geometry of the Universe*



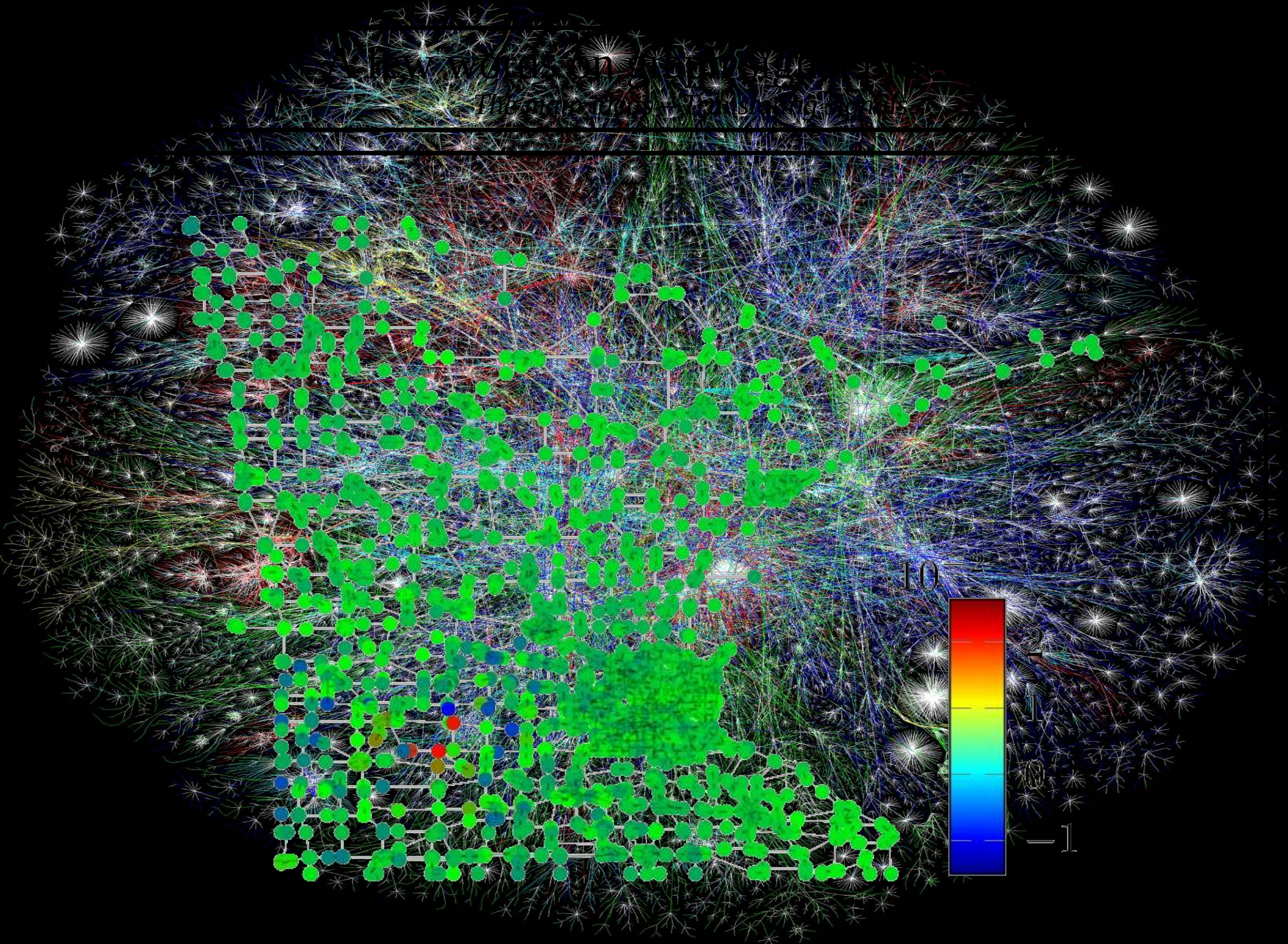
NEW WORK ON *The structure of the universe*

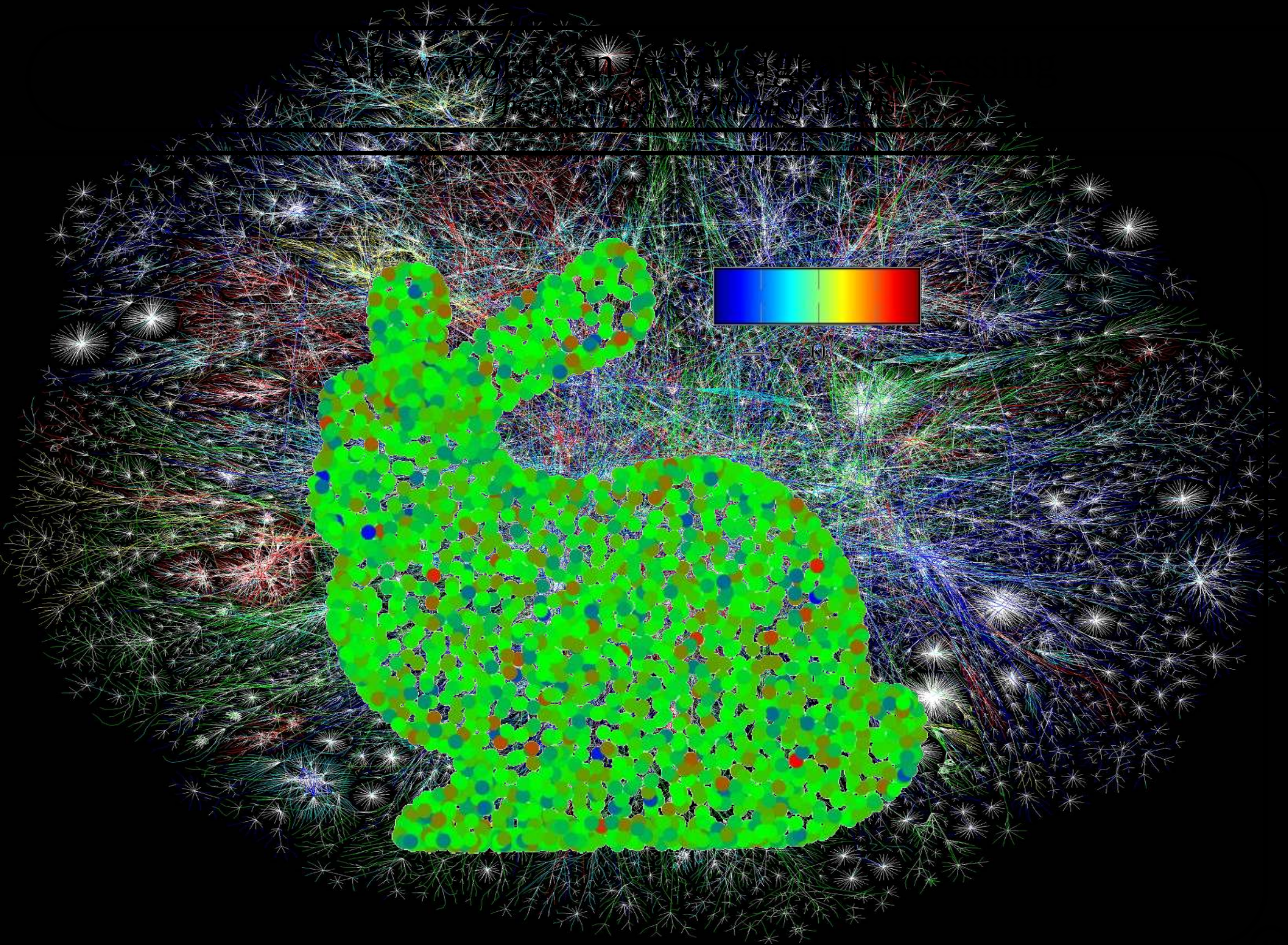


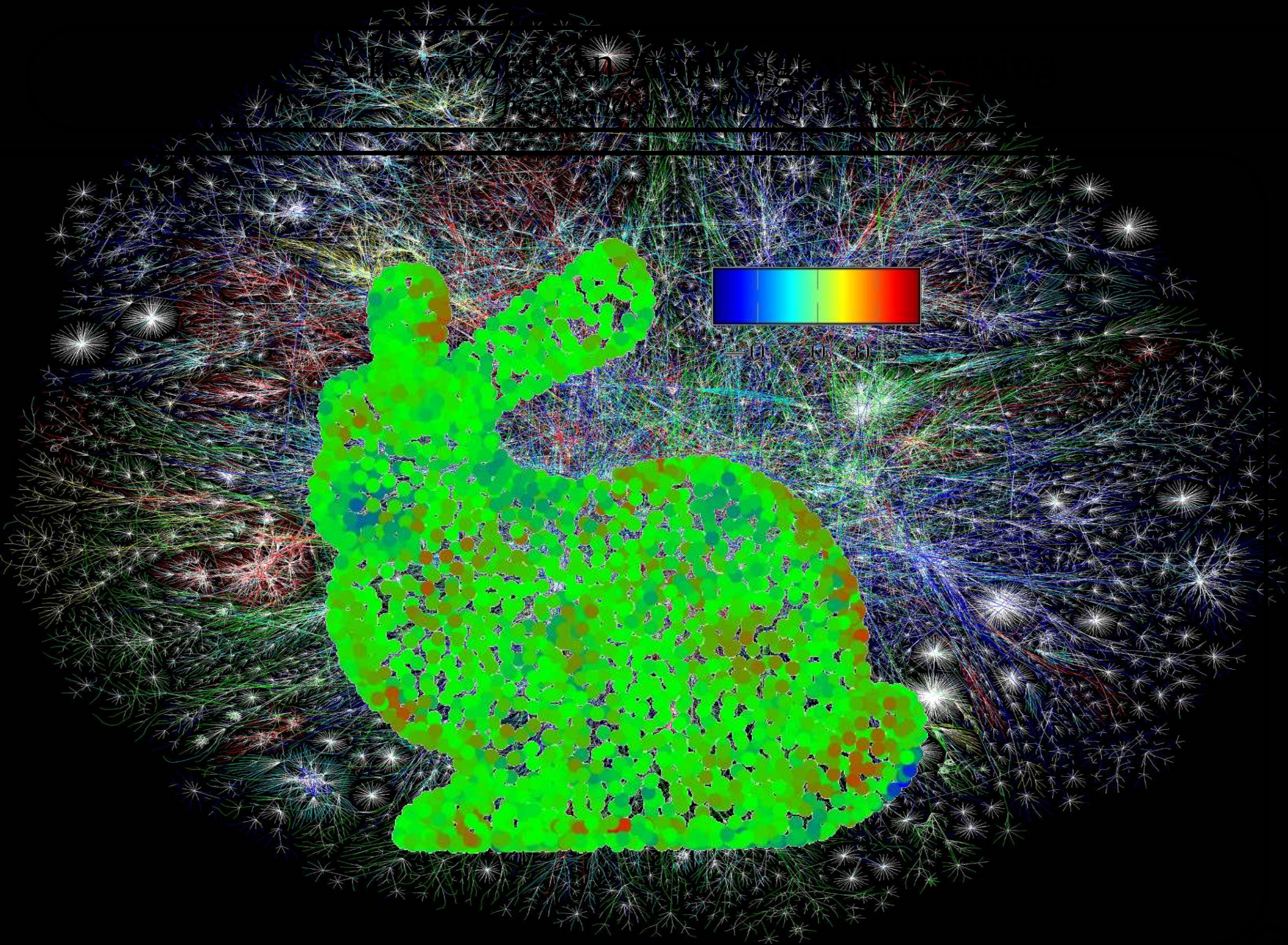
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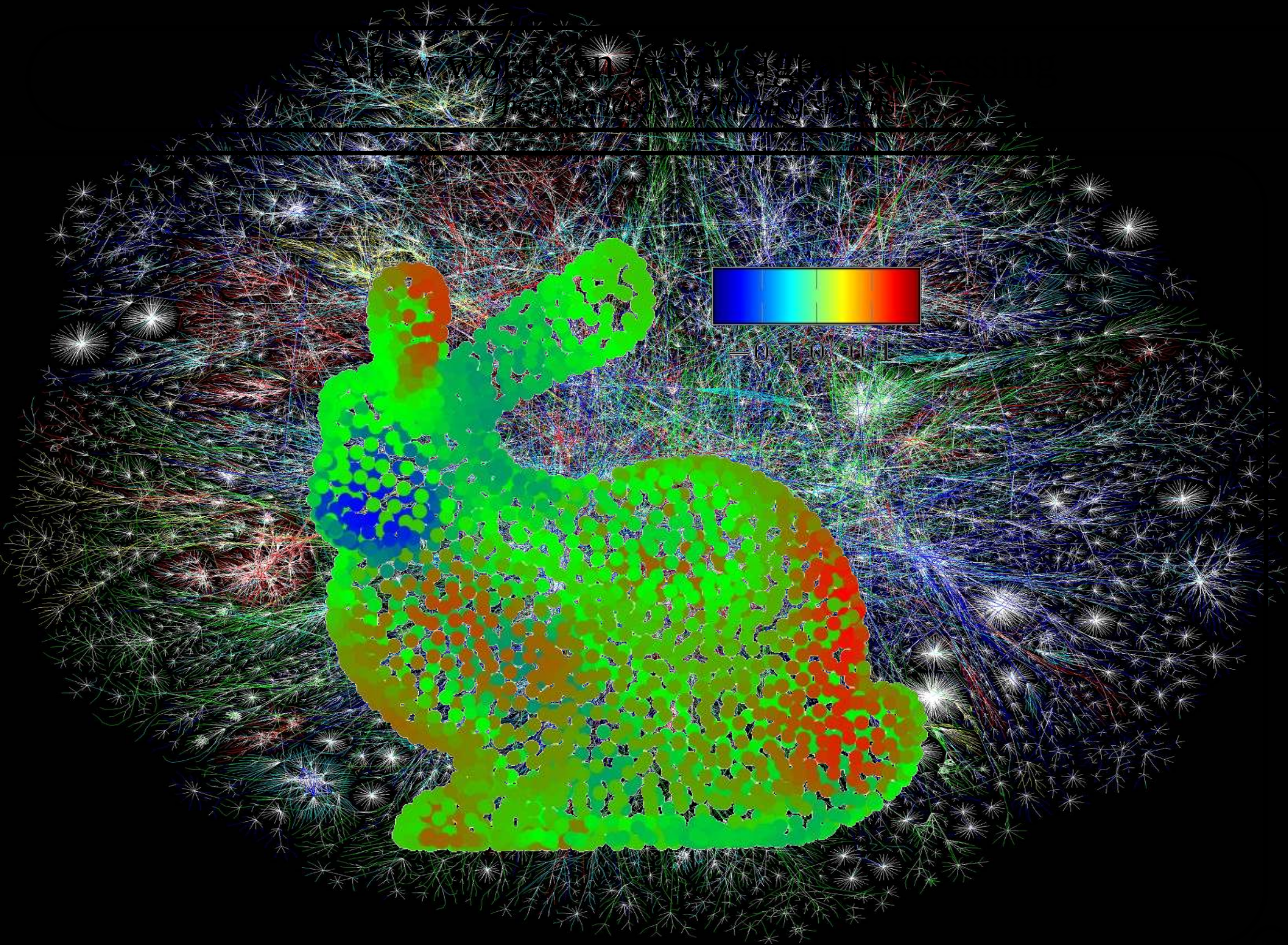


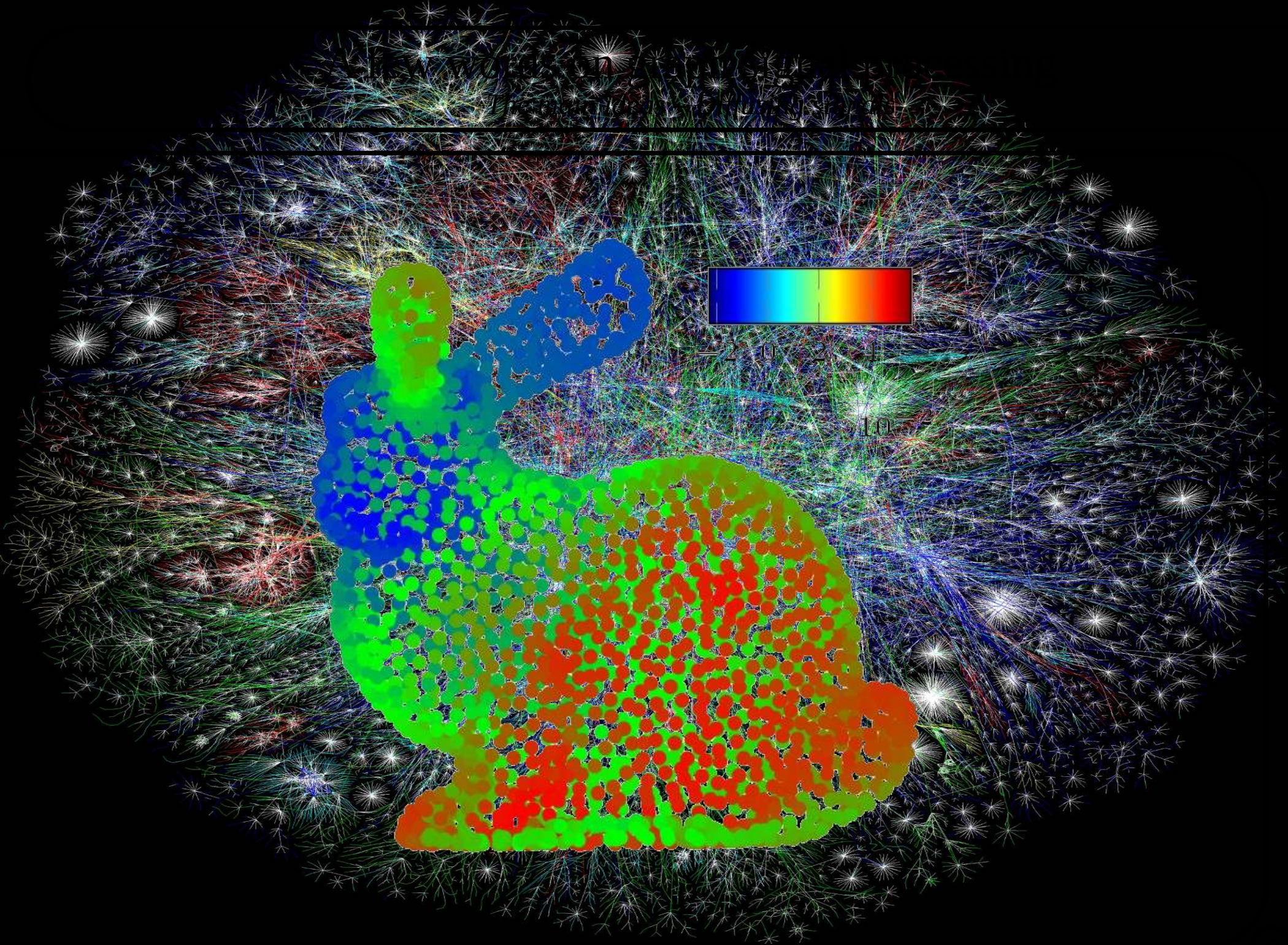
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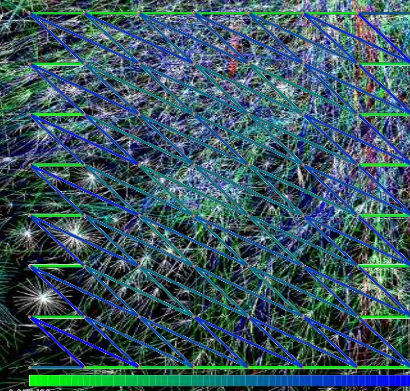
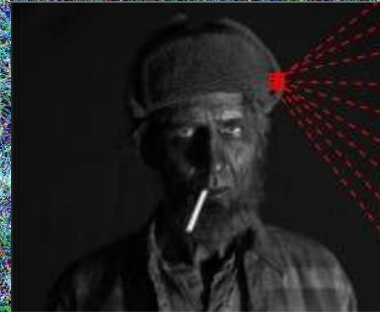




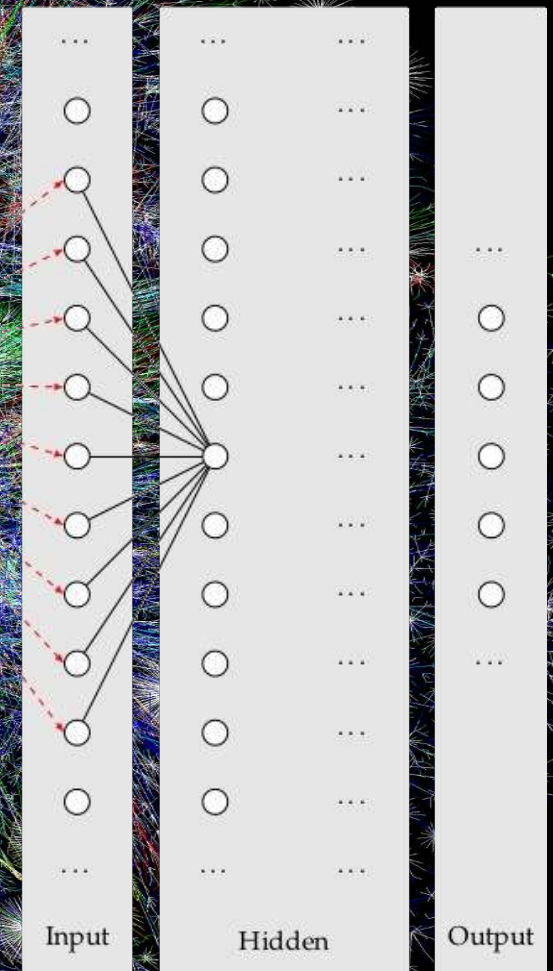
A NEW WORLD IN DATA

The new world of problems and solutions

- ▶ Denoising
- ▶ Downsampling
- ▶ Compression of signals
- ▶ Patterns recognition
- ▶ Machine learning
- ▶ Understanding the brain



0.953480



► A few words on graph signal processing

- * What is it?
- * The main ideas
- * Some problems and applications

► Graph inference from signals

- Why it may be necessary
- The case of stationary signals

► Translations on irregular graphs

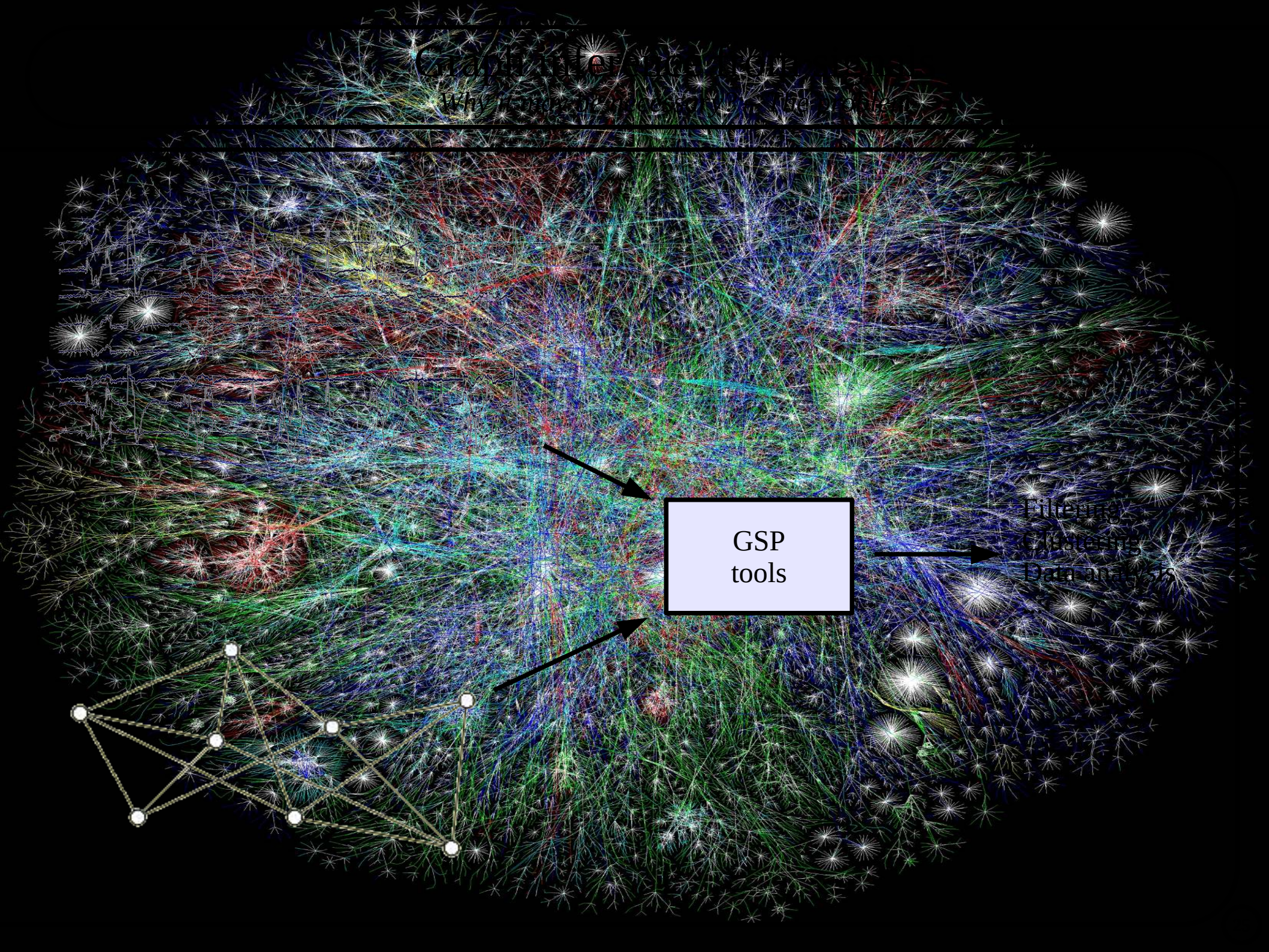
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► Extension of convolutional neural networks

- * Introduction of the method
- * Results

Graphs in the Wild

Why Random Graphs are not the Answer



GSP
tools

Filtering
Clustering
Data analysis

Graphs in the Wild

Why is this so hard to make sense of?

GSP
tools

Graphs in the Wild

Graphs in the Wild

Why it may be needed in the Wild

Graph
inference

GSP
tools

Filtering
Clustering
Data analysis



Graph Inference

Why it may be an essential building block

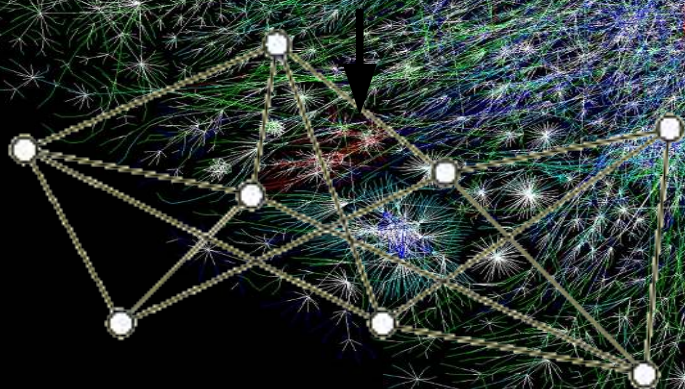
Signals:

Need for a model

Graph
inference

Graphs:

Need for assumptions



Graph Inference

Why it may be necessary for signals

Signals:

Need for a model

► Stationary processes

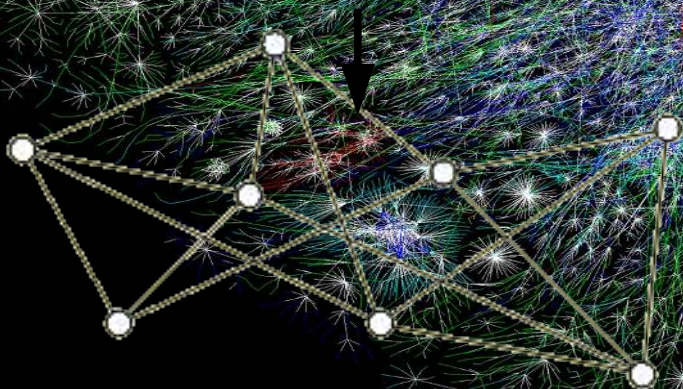
Graph
inference

Graphs:

Need for assumptions

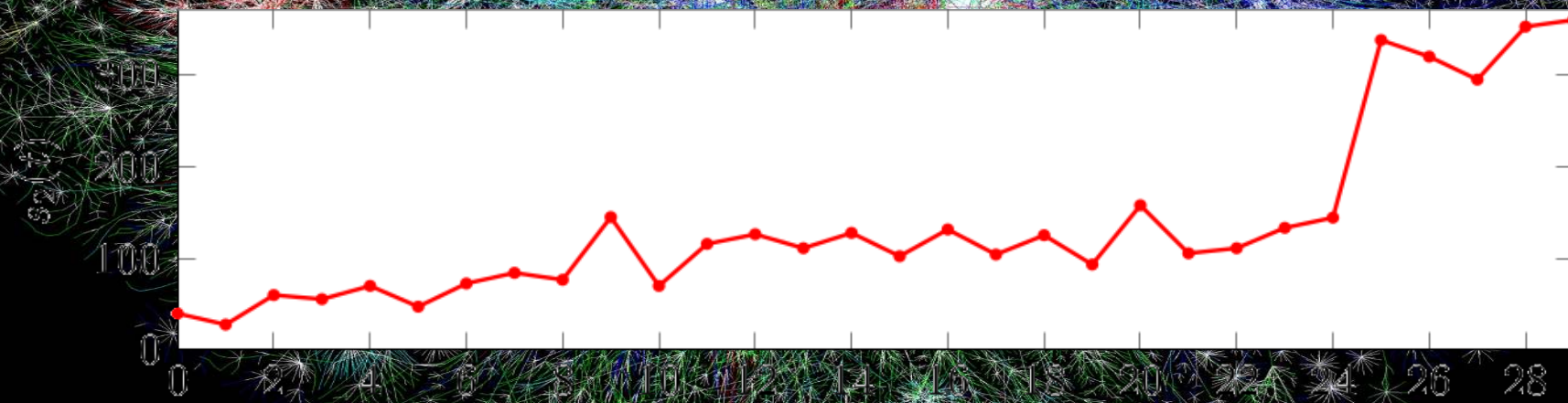
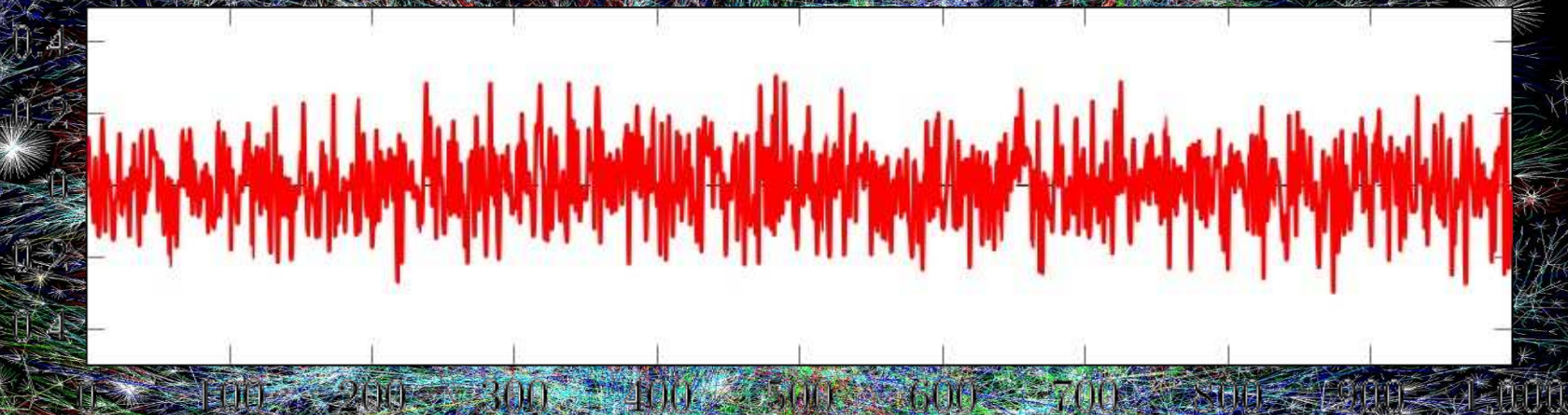
► Valid for stationary processes

► Desired features



Carapen River

Presence of stationary signals - stationary noise



Graph Filter with

The use of stationary signal conditions in graph filter

Definition

A stochastic graph signal $\mathbf{x}$ defined on the vertices of a graph G is called *graph wide (or second order) stationary* (GWSS), if and only if it satisfies the following properties:

Its first moment is constant over the vertex set: $m_{\mathbf{x}}[i] = \mathbb{E}\{\mathbf{x}[i]\} = c \in \mathbb{R}$

Its covariance is invariant with respect to the localization operator:

$$\gamma_{\mathbf{x}}[i, n] = \mathbb{E}\{(\mathbf{x}[i] - m_{\mathbf{x}})(\mathbf{x}[n] - m_{\mathbf{x}})\} = \mathcal{T}_i \gamma_{\mathbf{x}}[n]$$

Example

White noise is GWSS

If a graph filter g is applied to a GWSS signal, the result remains GWSS

Graphs and Neural Networks

Use of stochastic Hopfield Networks for associative memory

$$\forall i \in \{1, \dots, M\} : \mathbf{x}_i = \mathbf{T}^k \mathbf{y}_i$$

Graph Shift Operator

Use of stationary hypothesis: diffusion on a single graph

Number of signals

Observed signals

i.i.d signals

$$\forall i \in \{1, \dots, M\} : \mathbf{x}_i = \mathbf{T}^k \mathbf{y}_i$$

Graph shift operator used
for diffusion

Variable number of diffusions
(linear combination also ok)

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Graph Shift Operators

The case of stationary signals

Stationary
signals $\mathbf{X}$

Characterization

Space of valid
graph shift operators

Inference

$\tilde{\mathbf{T}}$

Stationary signals enforces coherence between the signal model and the

graph structure

Graph filter networks

The case of stationary signals – the data model

Characterization step enforces coherence between the signal model and the topology

Stationary
signals $\mathbf{X}$

Characterization

Space of valid
graph shift operators

Inference

$\tilde{\mathbf{T}}$

- $\tilde{\mathbf{T}} \triangleq \mathcal{X} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \mathcal{X}^T$ is symmetric

- $\forall i, j \in \{1, \dots, N\}; j \neq i: T(i, j) \geq 0$

- Let χ_1 be the constant sign eigenvector in $\mathcal{X}$, $\lambda_1 = 1$.

- $\forall i \in \{1, \dots, N\}; \lambda_i \in [-1, 1]$

Graph filter design

The case of stationary signals – the covariance matrix

Characterization step enforces coherence between the signal model and the topology

Stationary
signals $\mathbf{X}$

Characterization

Space of valid
graph shift operators

Inference

$\tilde{\mathbf{T}}$

Eigenvectors of the
covariance matrix

- $\tilde{\mathbf{T}} \triangleq \mathcal{X} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_N \end{pmatrix} \mathcal{X}^T$ is symmetric
 - $\forall i, j \in \{1, \dots, N\}: \lambda_i \geq 0, \mathbf{T}(i, j) \geq 0$
 - Let χ_1 be the constant sign eigenvector in $\mathcal{X}$: $\lambda_1 = 1$
 - $\forall i \in \{1, \dots, N\}: \lambda_i \in [-1, 1]$
- Diffusion matrix properties
- Scale factor

Graph filter networks

The case of stationary signals – for data sets of size

Characterization step enforces coherence between the signal model and the topology

Stationary
signals $\mathbf{X}$

Characterization

Space of valid
graph shift operators

Inference

$\tilde{\mathbf{T}}$

Eigenvectors of the
covariance matrix

Unknown

• $\tilde{\mathbf{T}} \triangleq \lambda^*$ is symmetric

• $\forall i, j \in \{1, \dots, N\} : \lambda_{ij} \geq 0, \mathbf{1}^T \lambda_{ij} \geq 0$

• Let χ_1 be the constant sign eigenvector in λ : $\lambda_1 = 1$

• $\forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1]$

Diffusion matrix
properties

Scale factor

Graphs

The case of stationary graphs

$$\forall i, j \in \{1, \dots, N\}; j \geq i : \tilde{\mathbf{T}}(i, j) \geq 0$$

$$\tilde{\mathbf{T}} = \mathbf{X} \begin{pmatrix} \widetilde{\lambda}_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \widetilde{\lambda}_N \end{pmatrix} \mathbf{X}^\top$$

$$\tilde{\mathbf{T}}(i, j) = \alpha_{ij1} \widetilde{\lambda}_1 + \dots + \alpha_{ijN} \widetilde{\lambda}_N$$

Graphs

The case of stationary graphs

$$\forall i, j \in \{1, \dots, N\}; j \geq i : \tilde{\mathbf{T}}(i, j) \geq 0$$

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Carsten Röhl

The case of symmetric $\mathfrak{sl}_3$ - $\mathfrak{sl}_3$

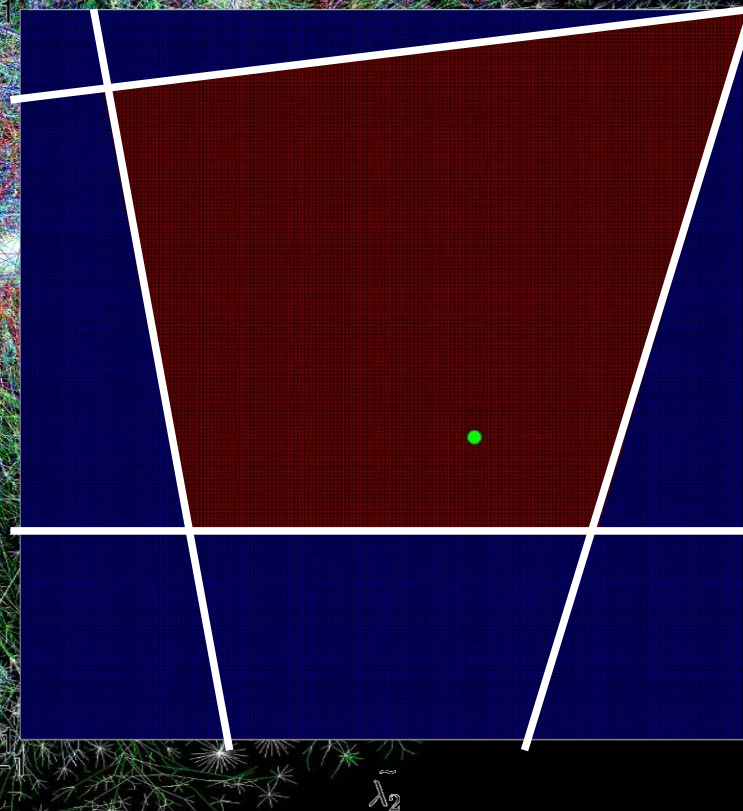
$$T(2,3) = \underbrace{\chi(2,1)\chi(3,1)}_{Q_{231}}\lambda_1 + \underbrace{\chi(2,2)\chi(3,2)}_{Q_{232}}\lambda_2 + \underbrace{\chi(2,3)\chi(3,3)}_{Q_{233}}\lambda_3$$

Graphs and Networks

The case of star graphs $S(3,1)$ and $S(3,2)$

$$T(2,3) = \underbrace{\chi(2,1)\chi(3,1)}_{Q_{231}}\lambda_1 + \underbrace{\chi(2,2)\chi(3,2)}_{Q_{232}}\lambda_2 + \underbrace{\chi(2,3)\chi(3,3)}_{Q_{233}}\lambda_3$$

$$W = \begin{pmatrix} 0.417 & 0.302 & 0.186 \\ 0.302 & 0.147 & 0.346 \\ 0.186 & 0.346 & 0.397 \end{pmatrix}$$

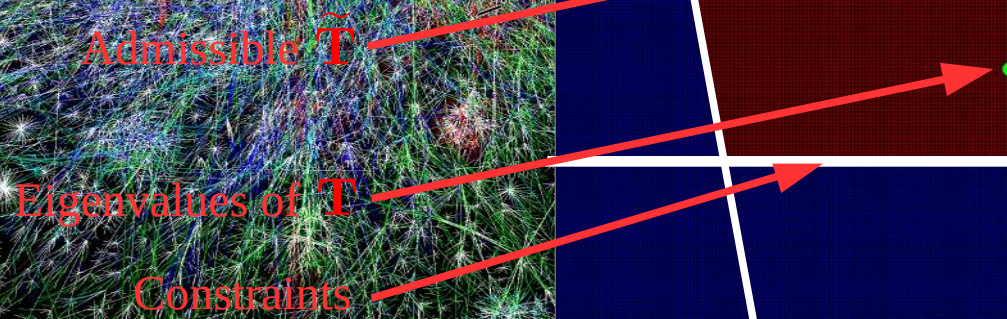


Chapter 10

The case of symmetric matrices

$$T(2,3) = \underbrace{\chi(2,1)\chi(3,1)}_{Q_{231}}\lambda_1 + \underbrace{\chi(2,2)\chi(3,2)}_{Q_{232}}\lambda_2 + \underbrace{\chi(2,3)\chi(3,3)}_{Q_{233}}\lambda_3$$

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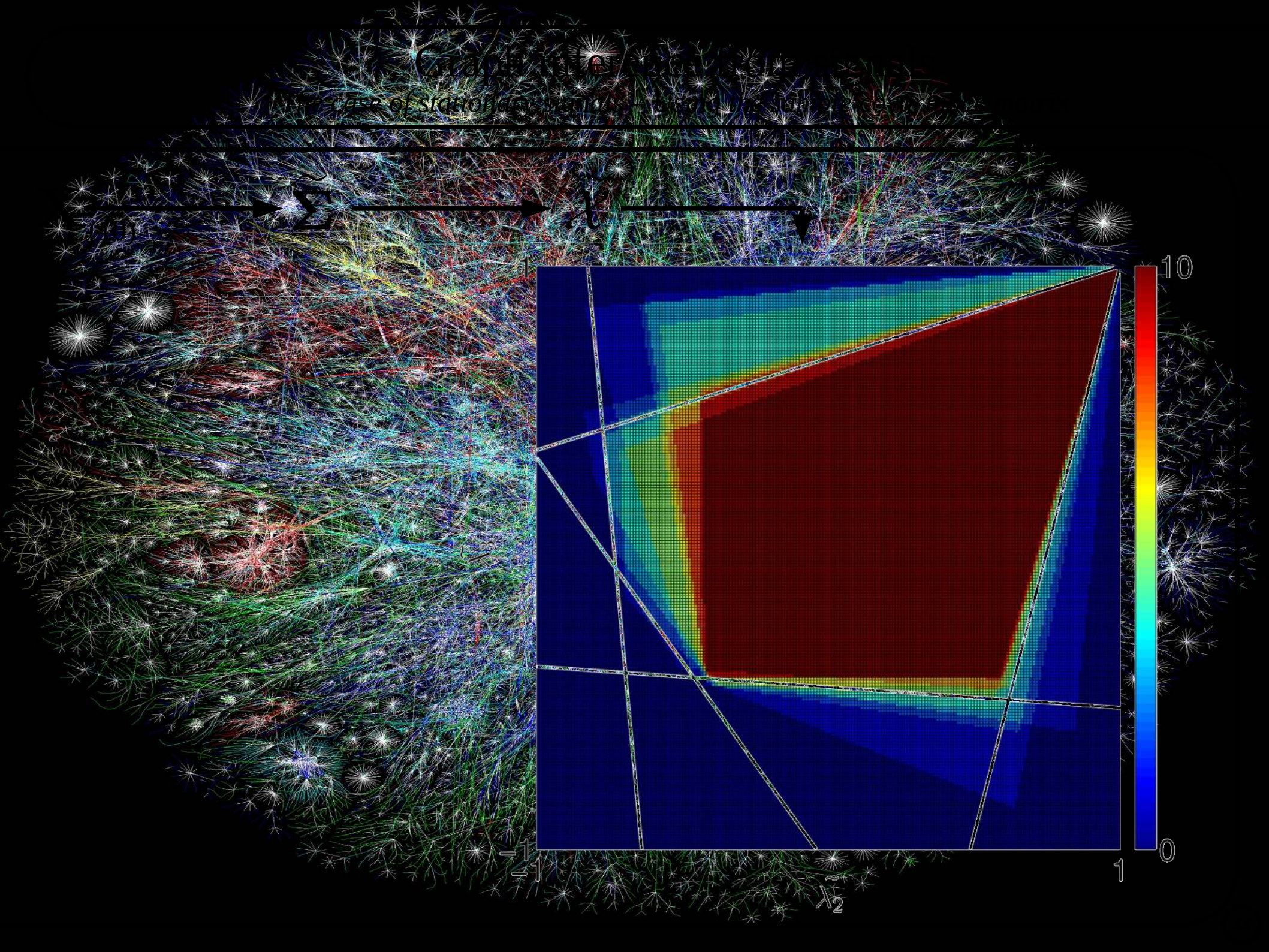


λ_2

1

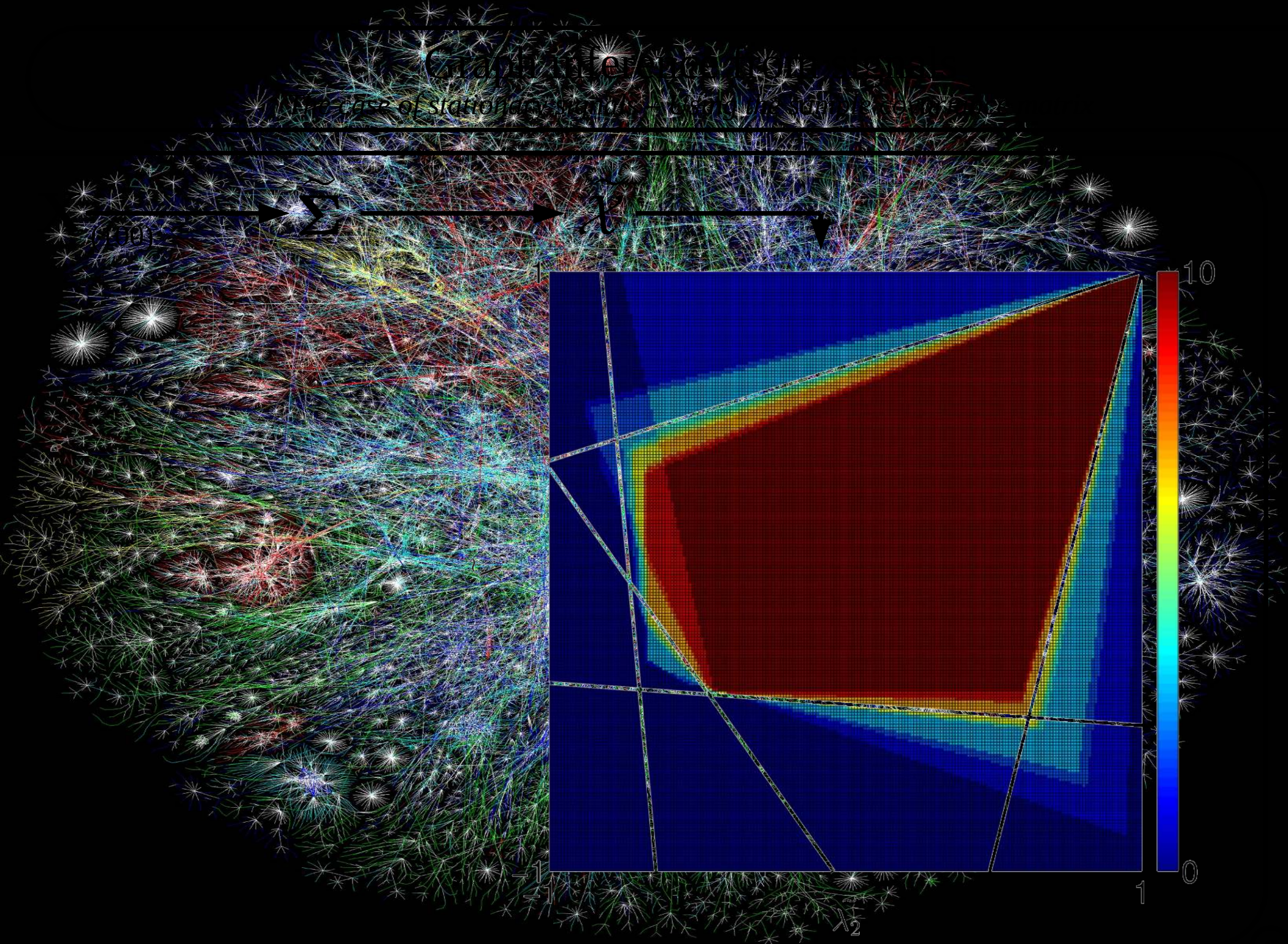
Carapin et al.

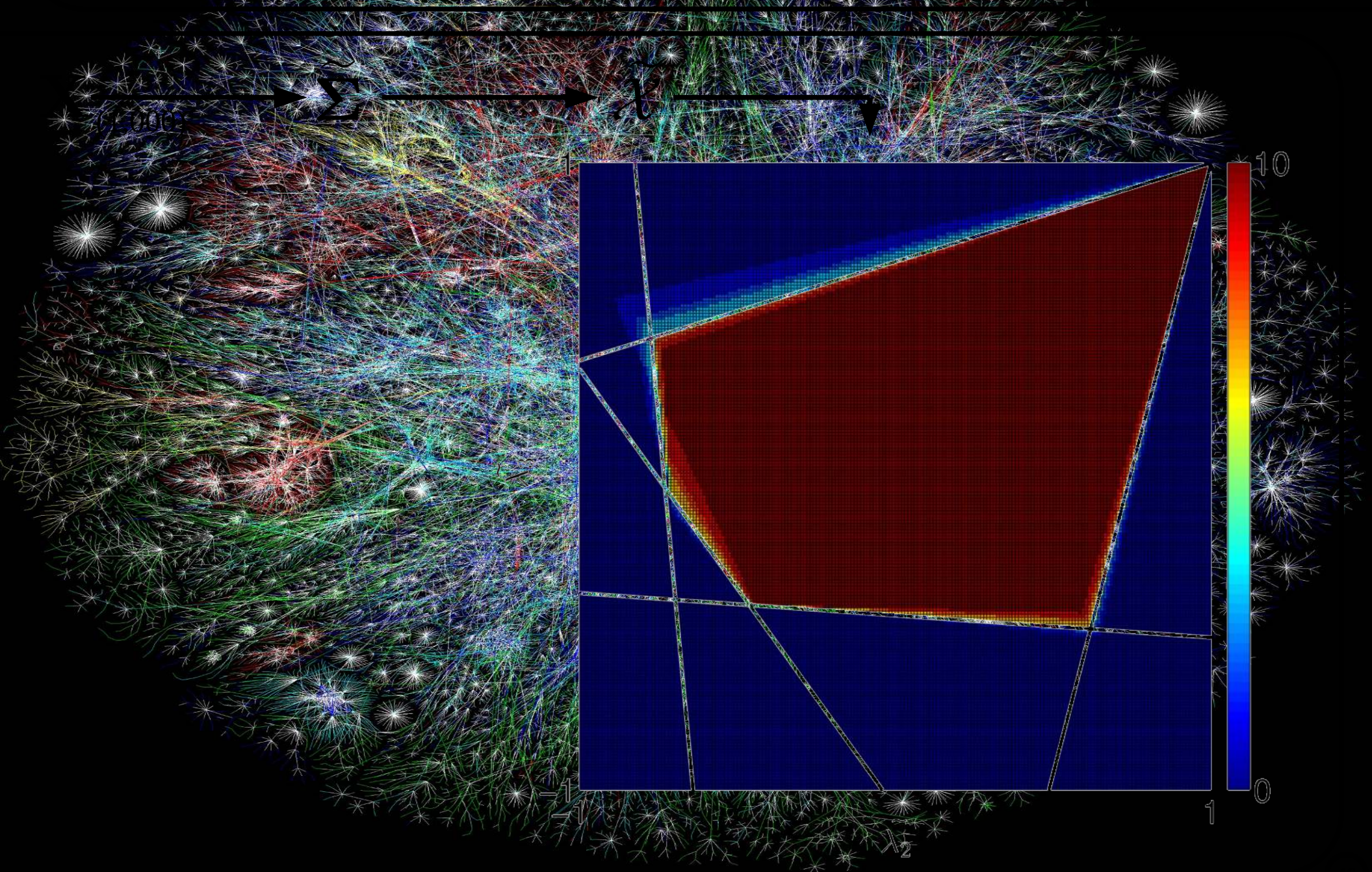
The case of stationary wavelets - the case of stationary wavelets



Carapin et al.

The case of stationary wavelets - time 1.5 s, $\sigma = 0.1$





Graph Inference

The case of stationary signals on networks

Inference requires selection criteria

Stationary
signals $\mathbf{X}$

Characterization

Space of valid
graph shift operators

Inference

$\tilde{\mathbf{T}}$

Knowledge of the underlying topology
Knowledge of the diffusion matrix
Knowledge of the signals on the graph
Knowledge of $\mathbf{A}$ (close to) binary matrix

Graph Inference

The case of stationary signals on networks

Inference requires selection criteria

Stationary
signals $\mathbf{X}$

Characterization

Space of valid
graph shift operators

Inference

$\tilde{\mathbf{T}}$

Our proposed methods

Kalotolias?

Knowledge of the underlying topology
Knowledge of the diffusion matrix
Knowledge of the signals on the graph
Knowledge of $\mathbf{A}$ (close to) binary matrix

Graph Partitioning

The case of stationary nodes and edges

$$\lambda_1, \dots, \lambda_N = \arg \min_{\lambda_1, \dots, \lambda_N} \sum_{i=1}^N \lambda_i$$

$$\text{s. t. } \begin{cases} \text{Polytope constraints} \\ \forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

Graphs and Networks

The case of stationary graphs and networks

Trace of the matrix

$$\lambda_1, \dots, \lambda_N = \arg \min_{\lambda_1, \dots, \lambda_N}$$

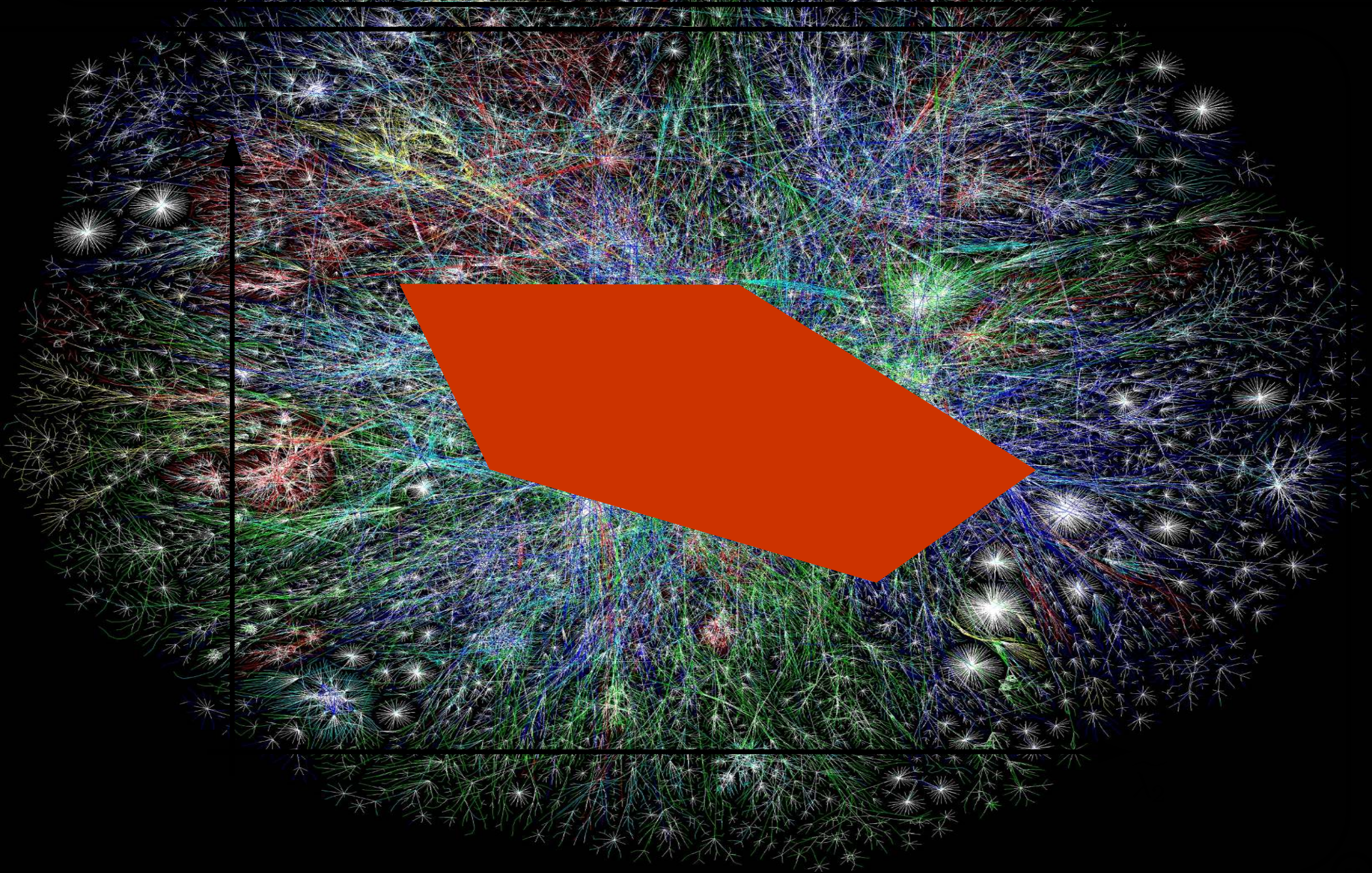
$$\sum_{i=1}^N \lambda_i$$

Polytope constraints

$$\text{s.t.} \begin{cases} \forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

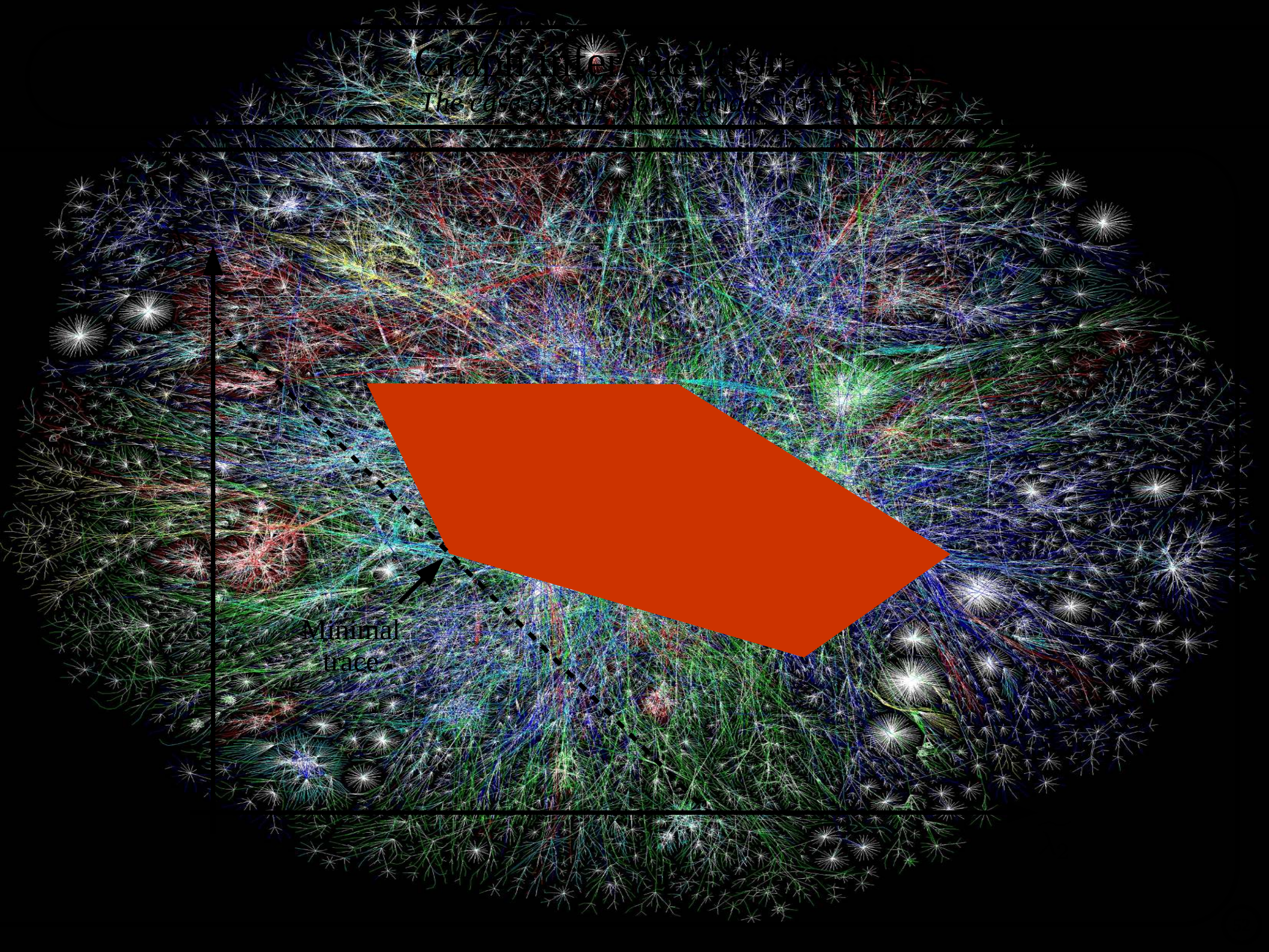
Carapen R. U. V.

The case of an open system



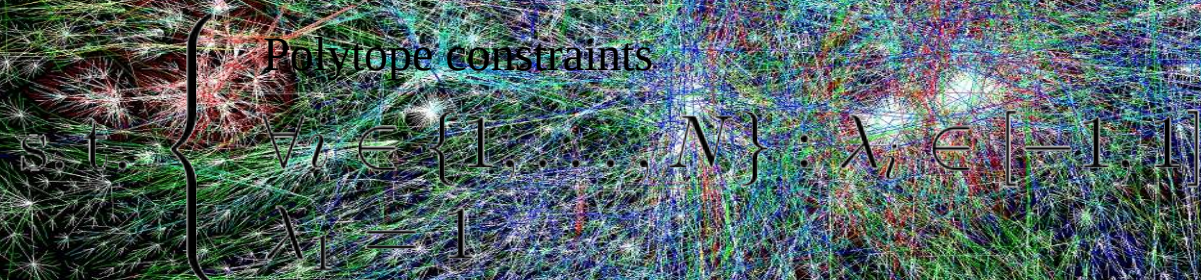
Graphs in the Wild

The case of an open system



The case of stationarity

$$\lambda_1, \dots, \lambda_N = \arg \min_{\lambda_1, \dots, \lambda_N} \mathbf{1}_N^\top \mathcal{X} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_N \end{pmatrix} \mathcal{X}^\top \mathbf{1}_N$$



$$\text{s.t.} \left\{ \begin{array}{l} \text{Polytope constraints} \\ \forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{array} \right.$$

Graph Partitioning

The case of stationary nodes (e.g. [1], [2], [3], [4], [5])

ℓ_1 norm

$$-\lambda_1, \dots, -\lambda_N = \arg \min_{\lambda_1, \dots, \lambda_N}$$

$$\mathbf{1}_N^\top \mathbf{x} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \lambda_N \end{pmatrix} \mathbf{x}^\top \mathbf{1}_N$$

$$\text{s.t.} \begin{cases} \text{Polytope constraints} \\ v_i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

Graph Embedding

The case of stationary and/or random walk

$\mathbf{X}$

Inference method
(eg. Kalofolias)

$$\hat{\mathbf{T}}_K = \hat{\mathbf{X}}_K \hat{\mathbf{\Lambda}}_K \hat{\mathbf{X}}_K^T$$

Graphs and Networks

The case of stationary and/or ergodic graphs

$\mathbf{X}$

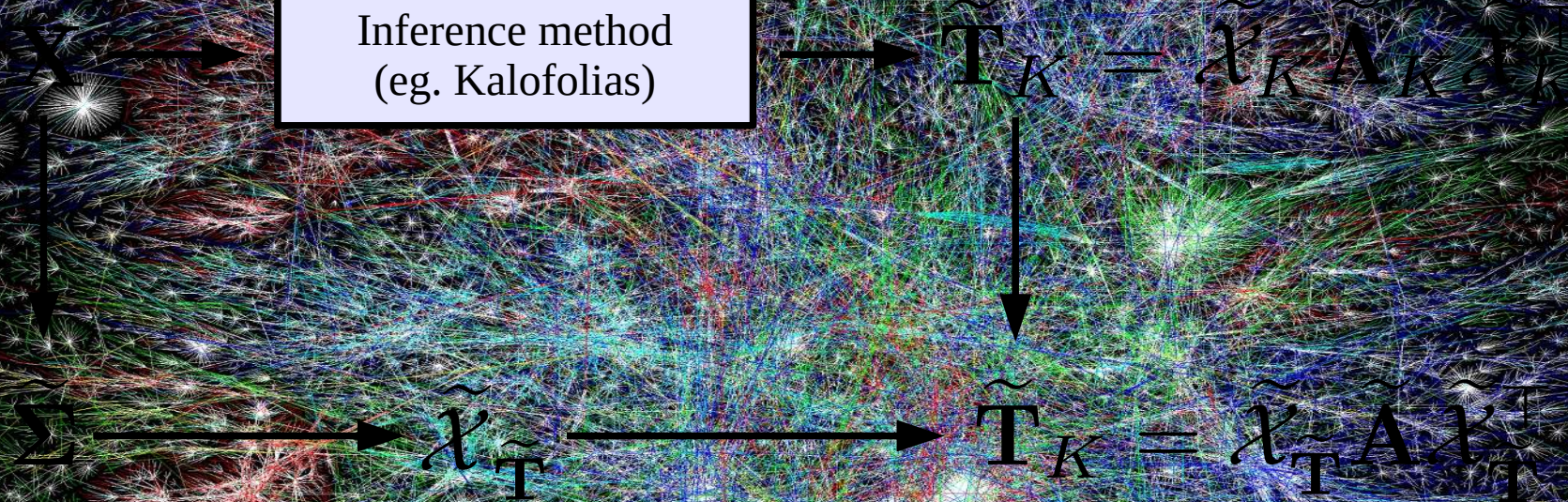
Inference method
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Not those of the
covariance matrix

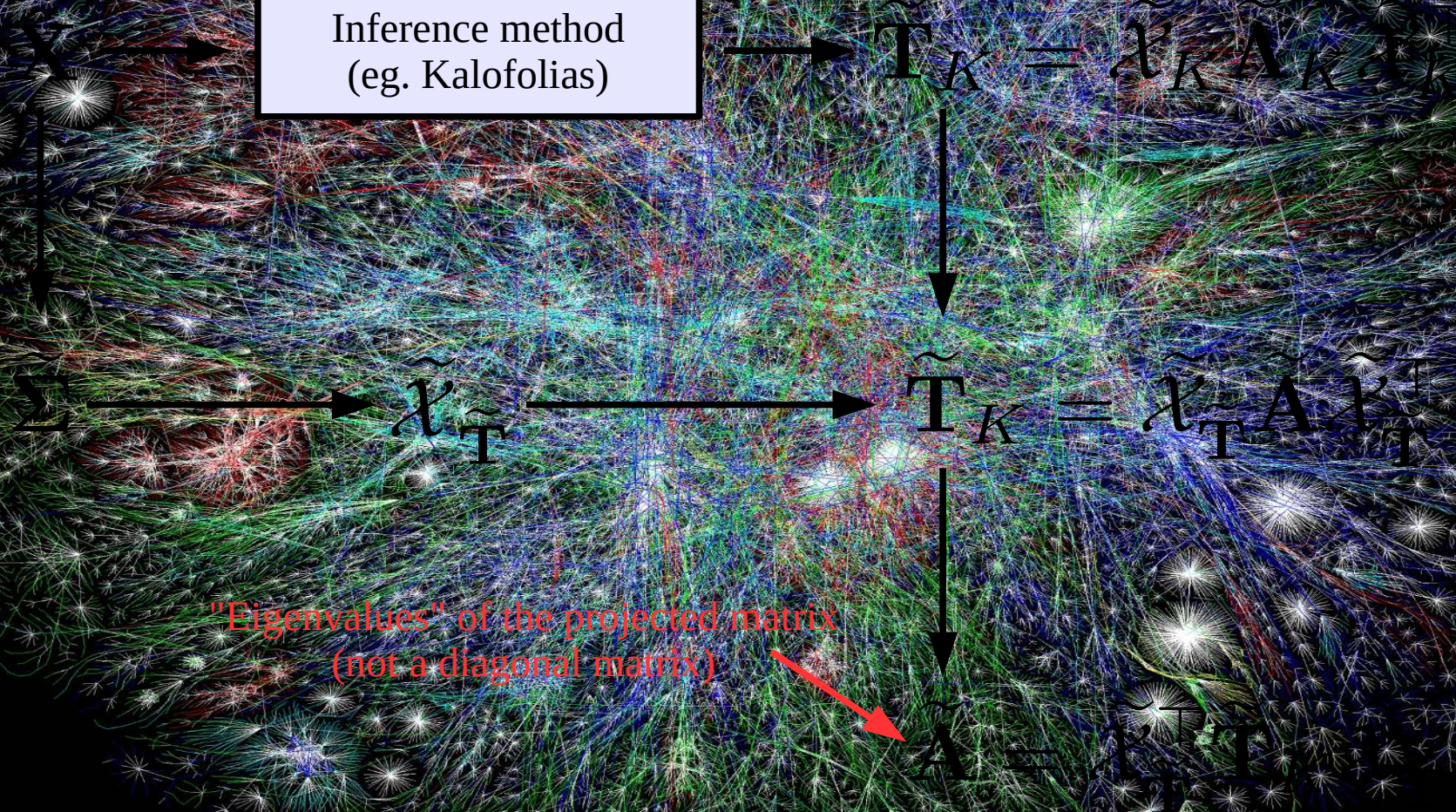
Graph Embedding

The case of stationary and/or random walk



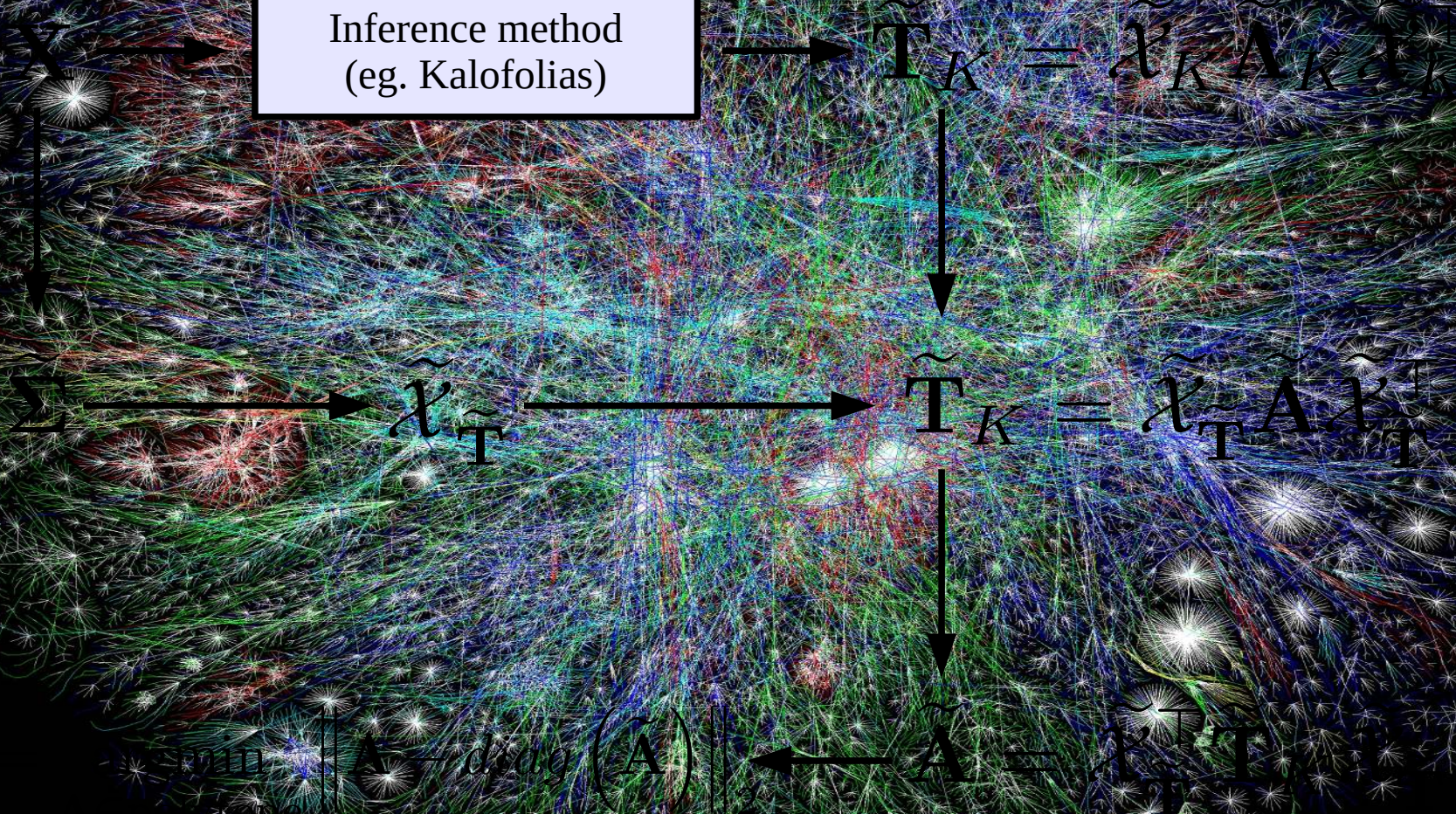
Graph Embedding

The case of stationary and/or random walks



Graph Embedding

The case of stationary and/or random walks



Graph Inference

The case of stationary and/or random walks

Inference method
(eg. Kalofolias)

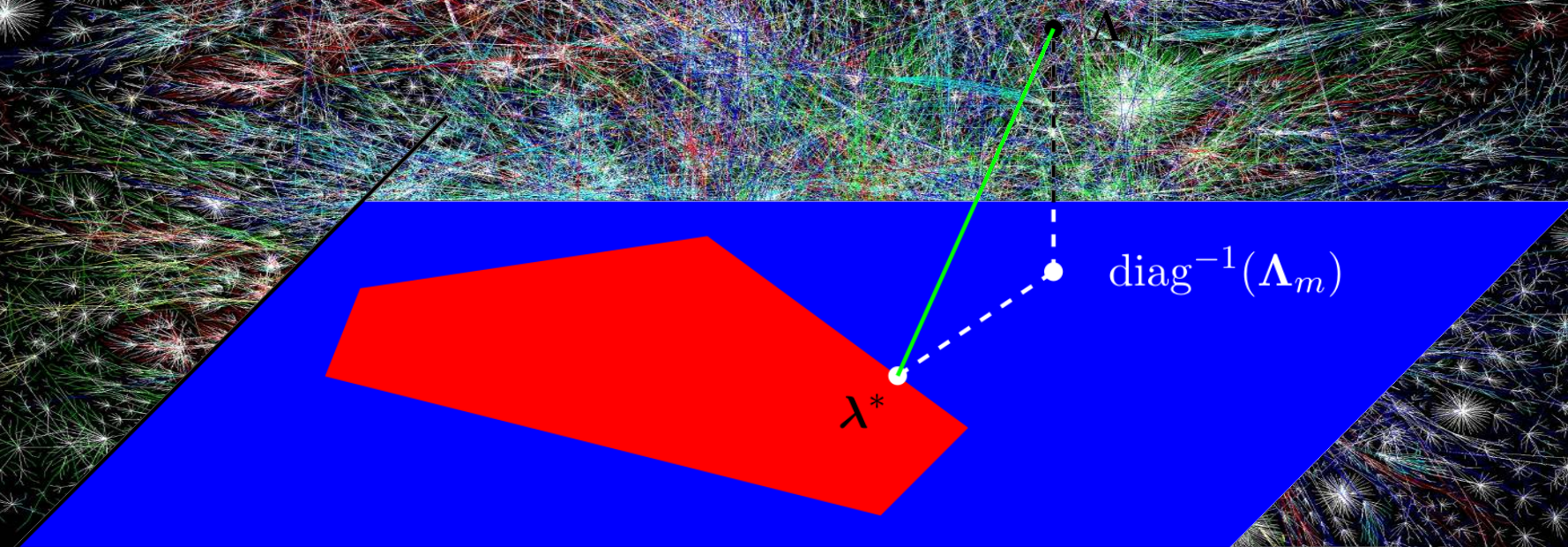
$$\hat{\mathbf{T}}_K = \hat{\mathcal{X}}_K \hat{\mathbf{A}}_K \hat{\mathcal{X}}_K^T$$

$$\Sigma \longrightarrow \tilde{\mathcal{X}}_{\tilde{\mathbf{T}}} \longrightarrow \tilde{\mathbf{T}}_K = \tilde{\mathcal{X}}_{\tilde{\mathbf{T}}} \tilde{\mathbf{A}} \tilde{\mathcal{X}}_{\tilde{\mathbf{T}}}^T$$

$$\min_{\tilde{\mathbf{A}}} \|\tilde{\mathbf{A}} - \text{diag}(\tilde{\mathbf{A}})\| \longleftarrow \tilde{\mathbf{A}} = \tilde{\mathcal{X}}_{\tilde{\mathbf{T}}}^T \tilde{\mathbf{T}}_K \tilde{\mathcal{X}}_{\tilde{\mathbf{T}}}$$

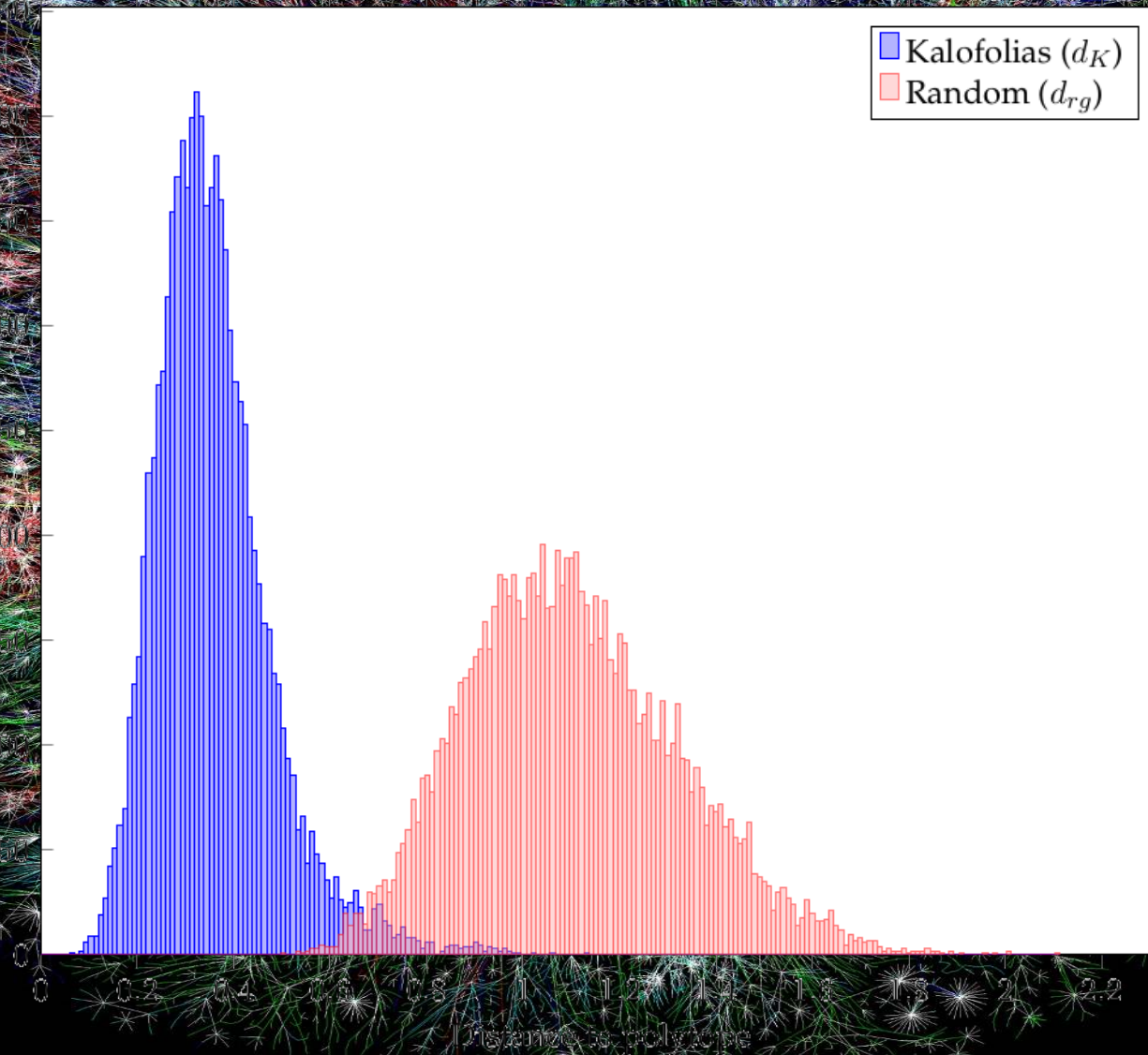
Carapin et al. (2017)

The case of an open system



Kalofolias

The case of Salicornia



Graph of the network of the case of stations in Brittany

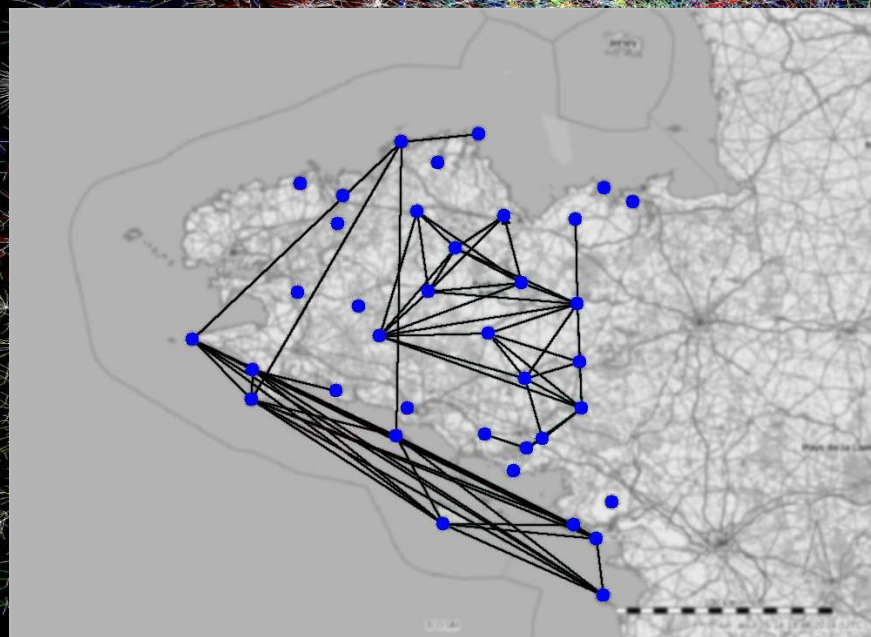


37 stations after preprocessing

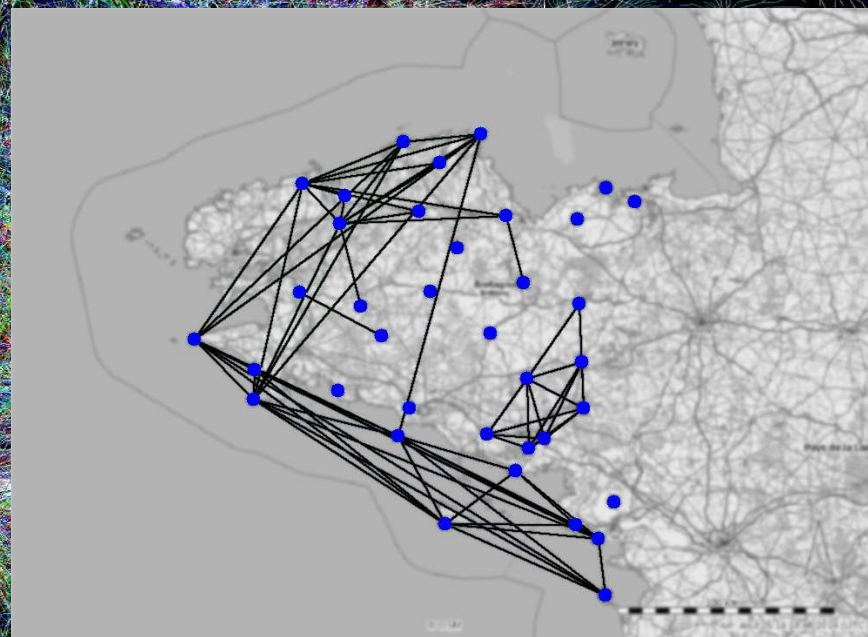
744 observations per station

Graphs

Use of stationary and non-stationary



Kalman



Regularized Kalman

Graphical Lasso

Choice of stationary variables according to $S(\mathbf{X})$

	polytope	$L_{1,1}$	Tr	$S(\mathbf{X})$
Simple	✓	36.9974	0.0013	0.0551
Sparse	✓	36.9971	0.9093	0.0585
Kalofolias	0.0313	36.9979	0	0.0751
Kalofolias closest	✓	36.9974	0.0298	0.0548
Segarra <i>et al.</i>	0.0062	36.9993	1.97×10^{-5}	0.0245
Segarra <i>et al.</i> closest	✓	36.9974	0.0046	0.0551
Graphical lasso	1.3730	35.3539	13.4977	32.8421
Graphical lasso closest	✓	36.9984	13.3584	0.0335

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- * The case of stationary signals

► Translations on irregular graphs

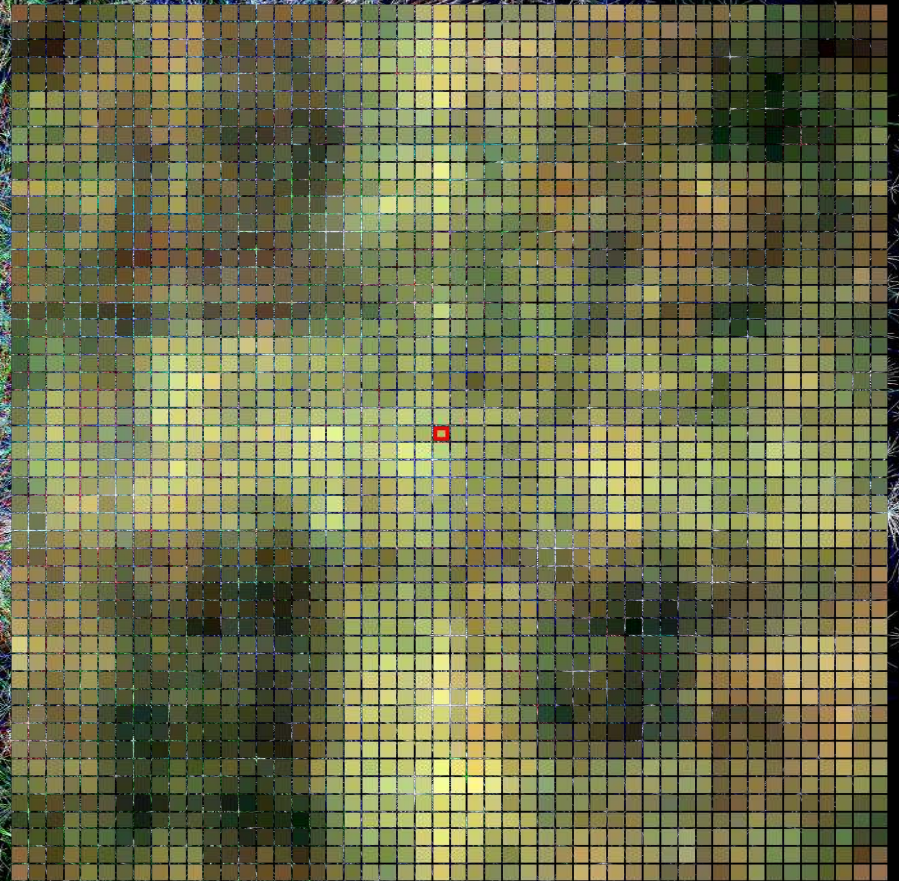
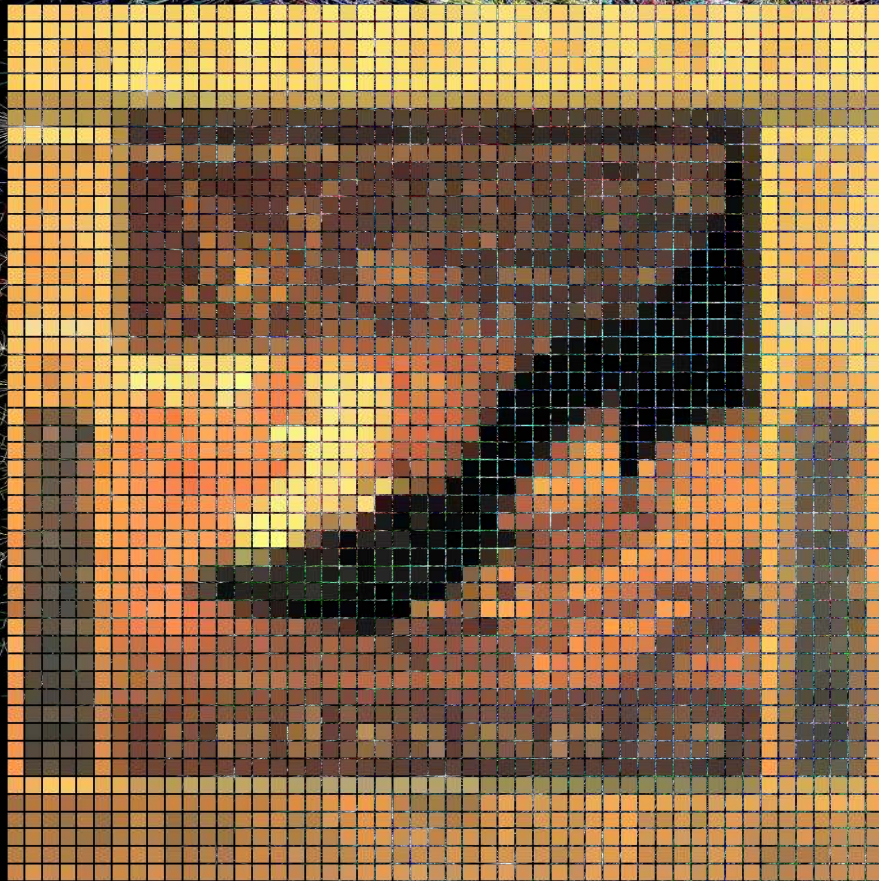
- * How we define them
- * How we identify them
- * Example on a random graph

► Extension of convolutional neural networks

- * Introduction of the method
- * Results

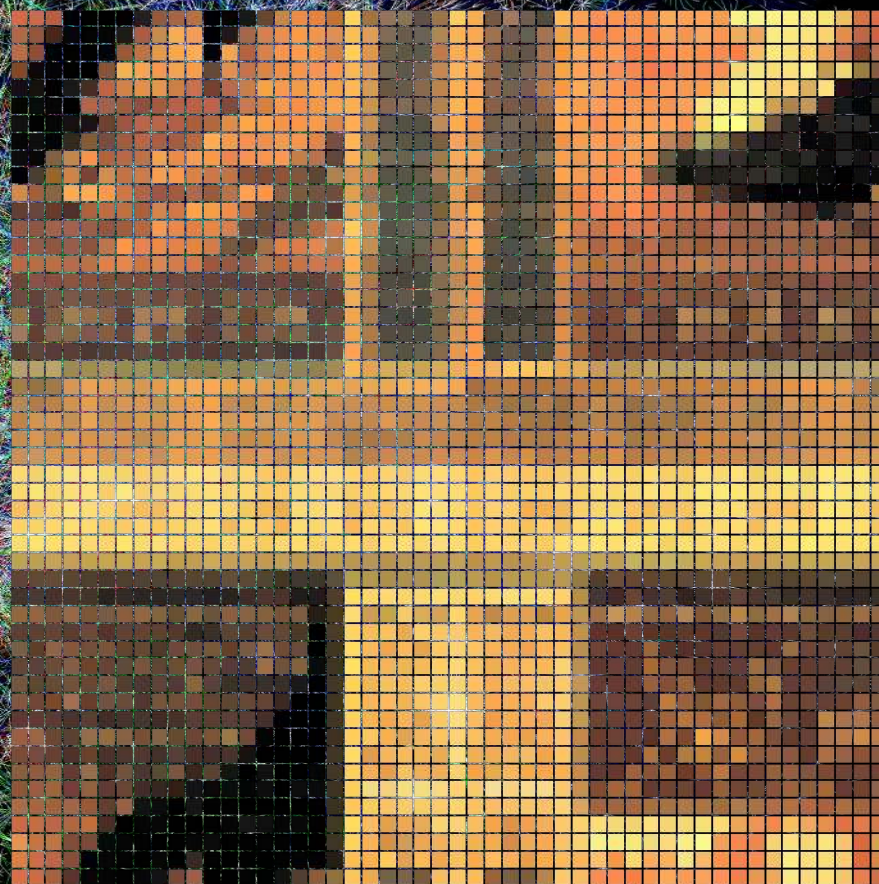
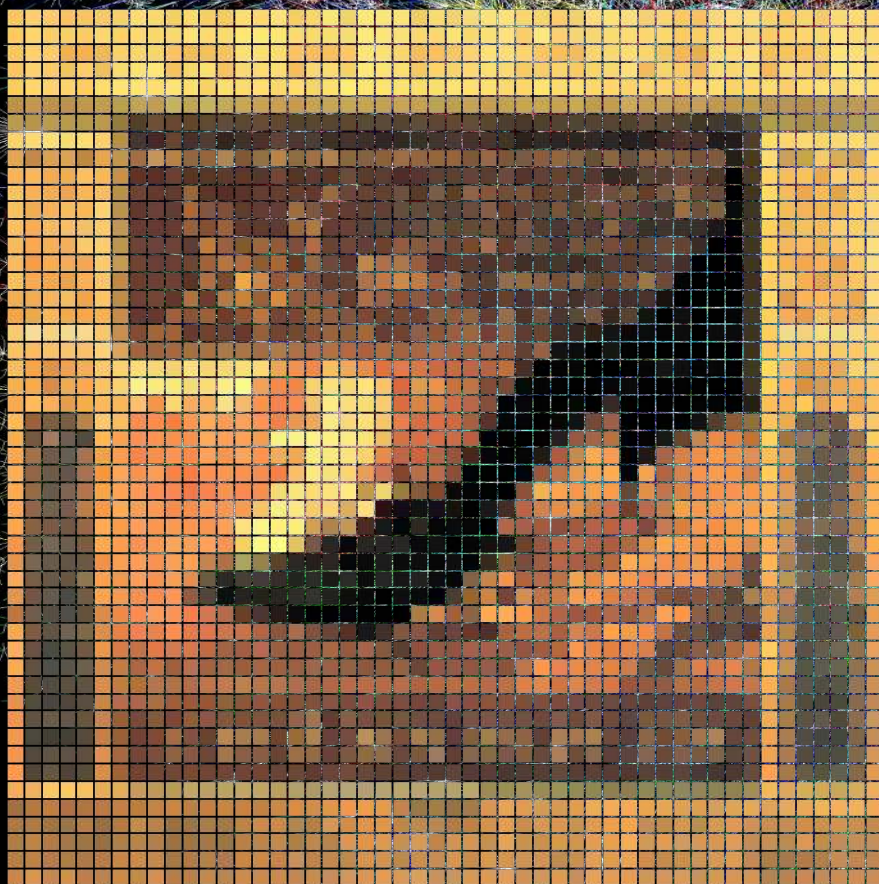
Fractal Images

How to define fractal images on a grid



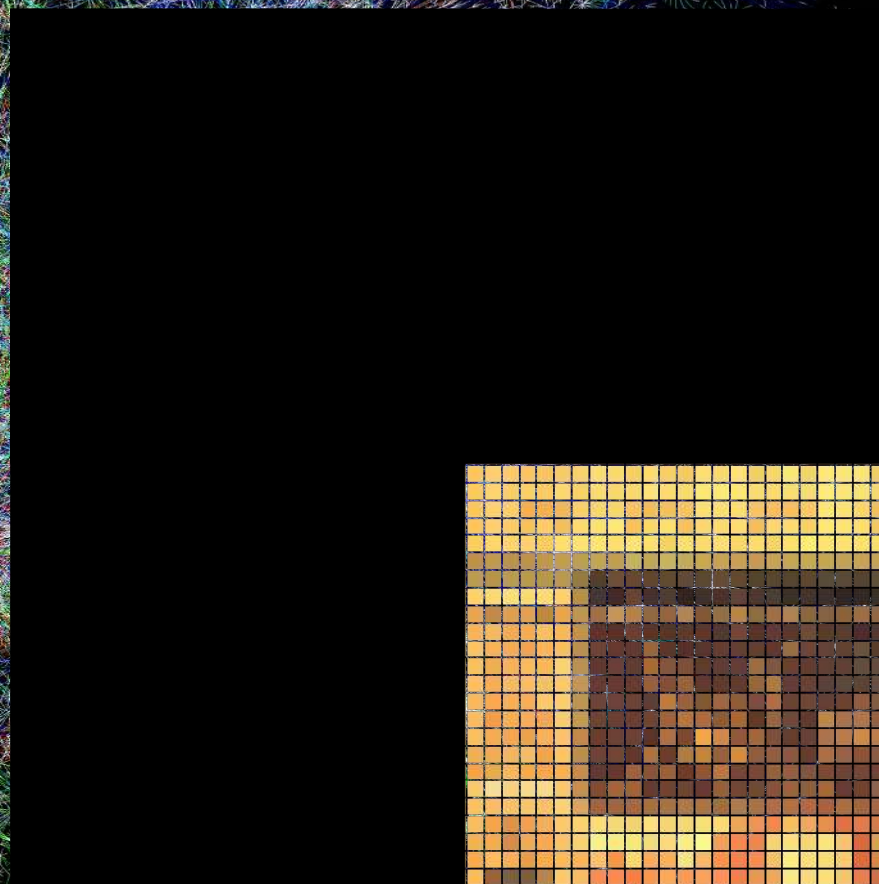
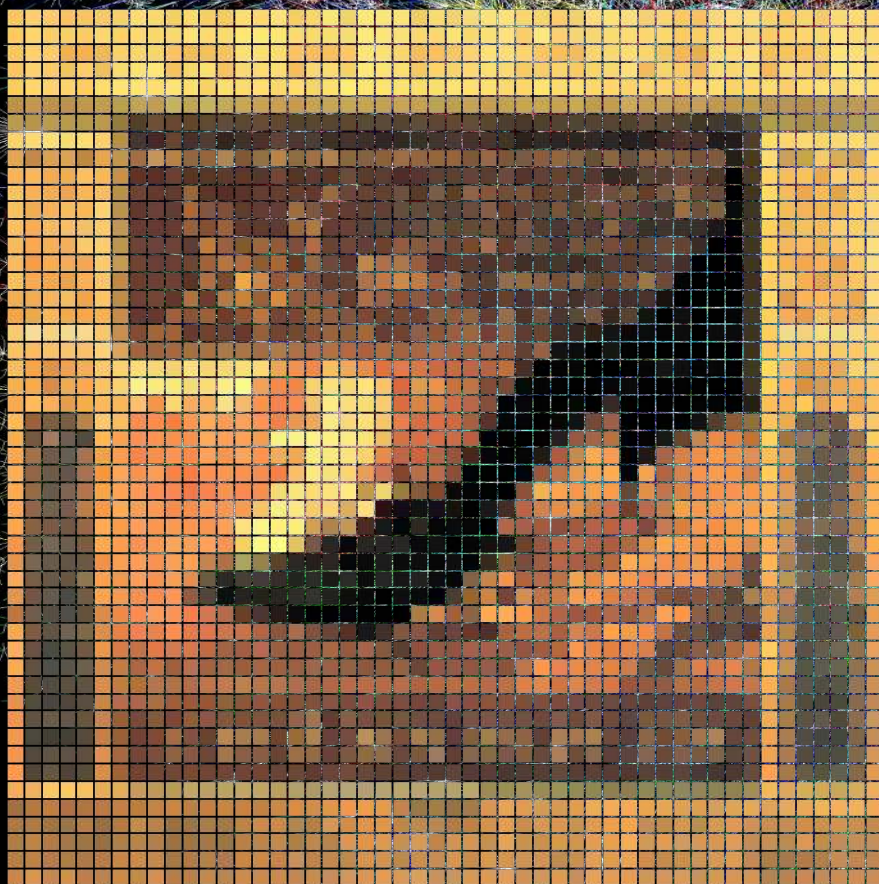
11.1115 - 11.1115

How to design a presentation



Fractal Images

How to define fractal images



Translations

How to formalize a "nice" mapping

Injectivity

$$\forall v_1, v_2 \in V : (\psi(v_1) = \psi(v_2) \neq \perp) \Rightarrow (v_1 = v_2)$$

Restriction to the edges

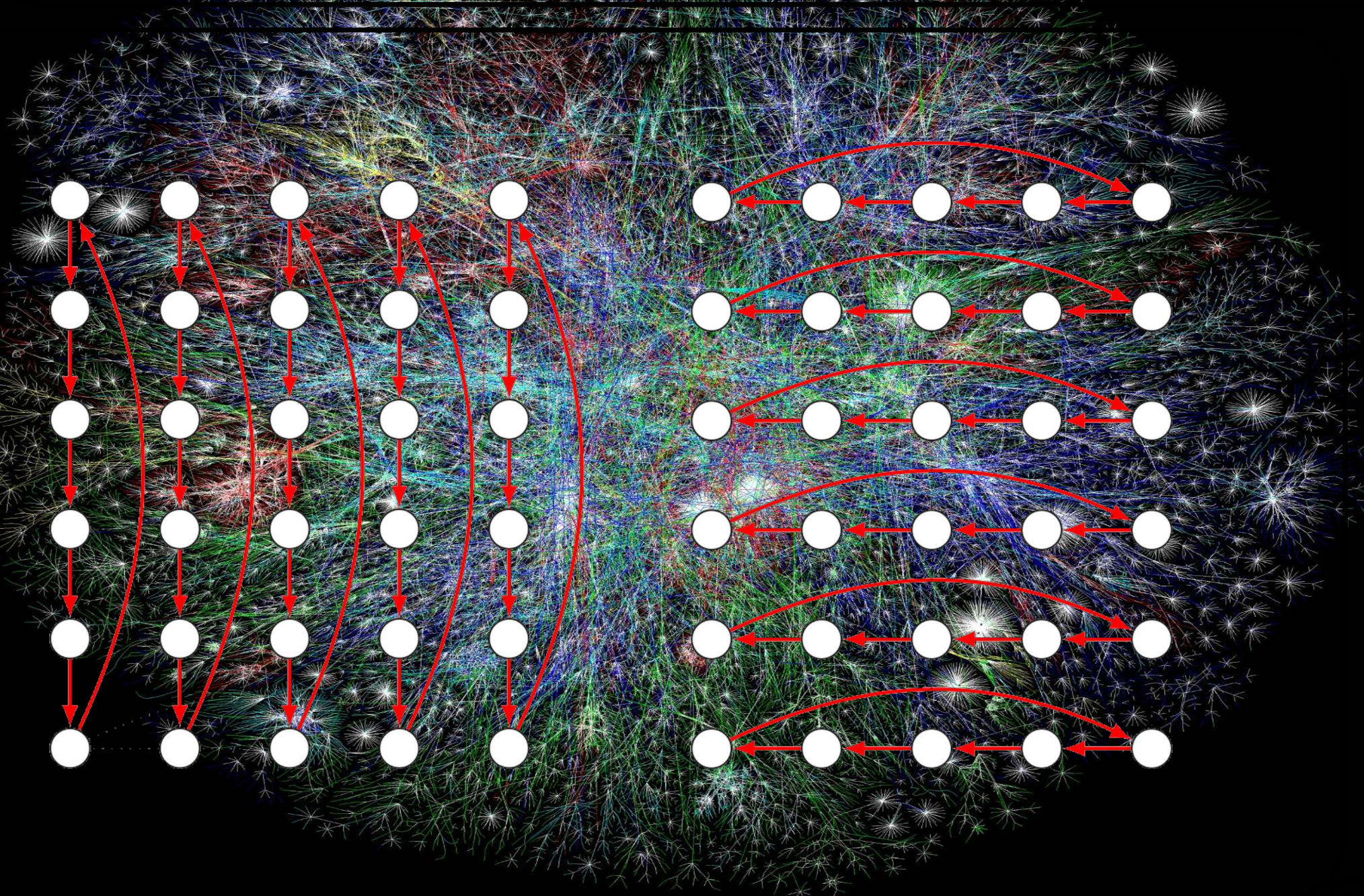
$$\forall v \in V : (\{v, \psi(v)\} \in \mathcal{E}) \vee (\psi(v) = \perp)$$

Preservation of the neighborhood

$$\forall v_1, v_2 \in V : (\{v_1, v_2\} \in \mathcal{E} \Leftrightarrow \{\psi(v_1), \psi(v_2)\} \in \mathcal{E} \vee (\psi(v_1) = \perp) \vee (\psi(v_2) = \perp))$$

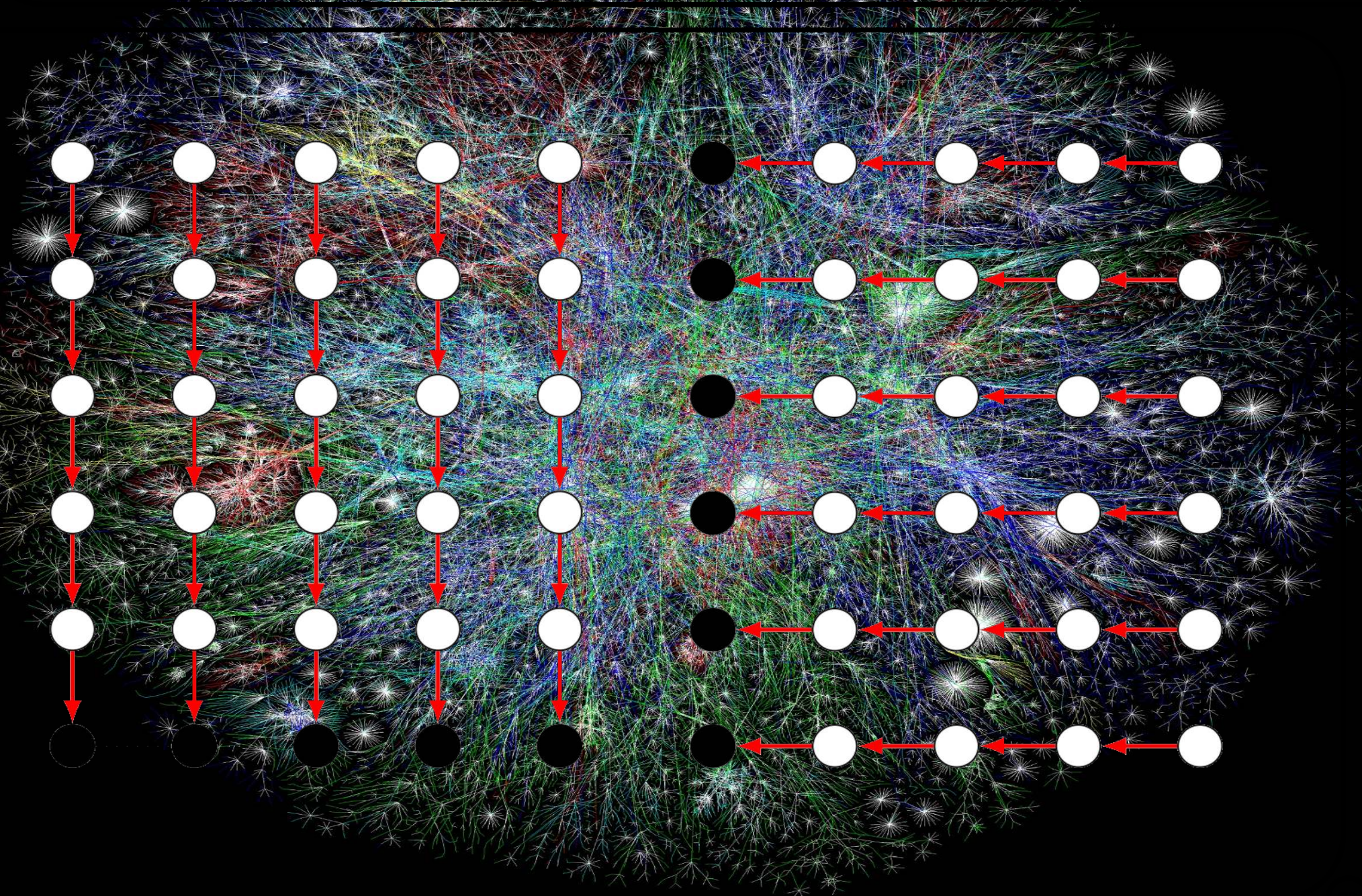
Transitions

How to identify them? How to calculate them?



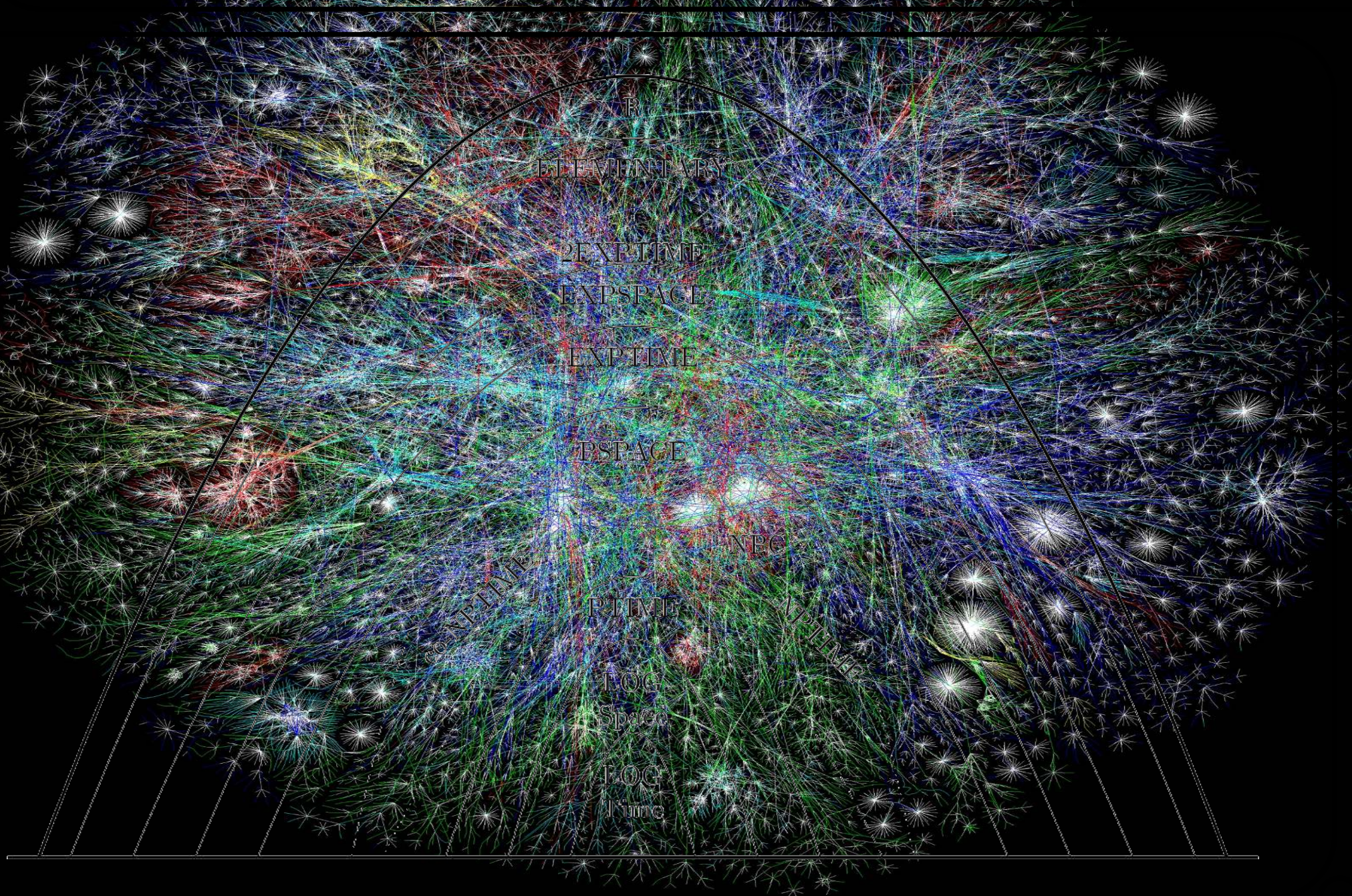
Transitions

How to identify them - I will show you a few ways



Transcendental

How to identify them - identifying the individual



Translocation

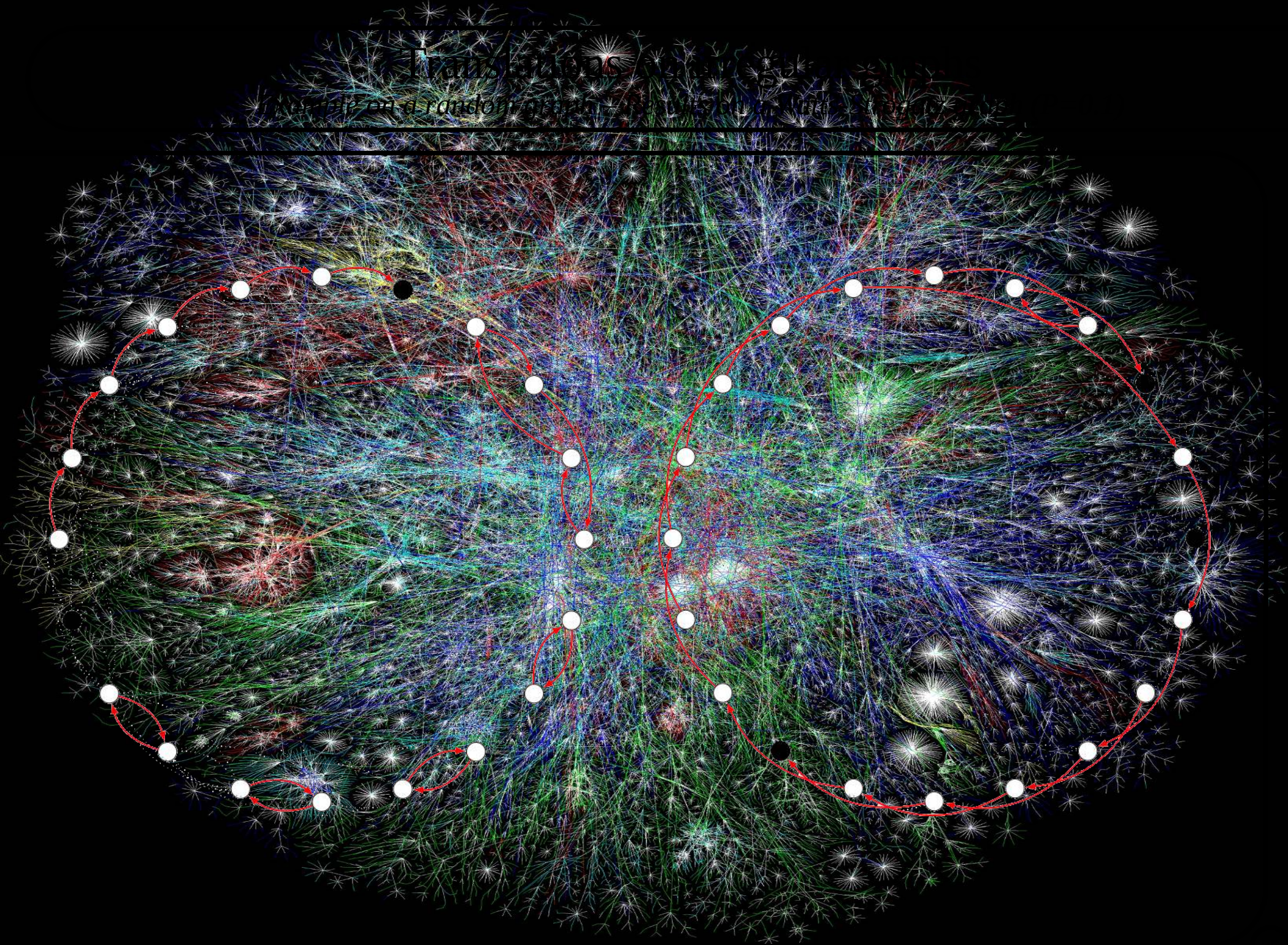
How to identify translocation events in a cell

$$\mathbf{A} \rightarrow \arg \min_{\mathbf{A}_\psi \in \{0,1\}^{N \times N}} \|\mathbf{A} \cdot \mathbf{A}_\psi - \mathbf{A}_\psi \cdot \mathbf{A}\|_1 \quad \text{Minimize deformation}$$

$$\left\{ \begin{array}{l} \forall i, j \in [1, N] : \mathbf{A}_\psi[i, j] \leq \mathbf{A}[i, j] \\ \forall i \in [1, N] : \sum_{j=1}^N \mathbf{A}_\psi[i, j] \leq 1 \\ \forall j \in [1, N] : \sum_{i=1}^N \mathbf{A}_\psi[i, j] \leq 1 \\ \|\mathbf{A}_\psi\|_1 \leq C \end{array} \right. \quad \left. \begin{array}{l} \text{Restriction to edges} \\ \text{Injectivity} \\ \text{Avoid empty solution} \end{array} \right\}$$

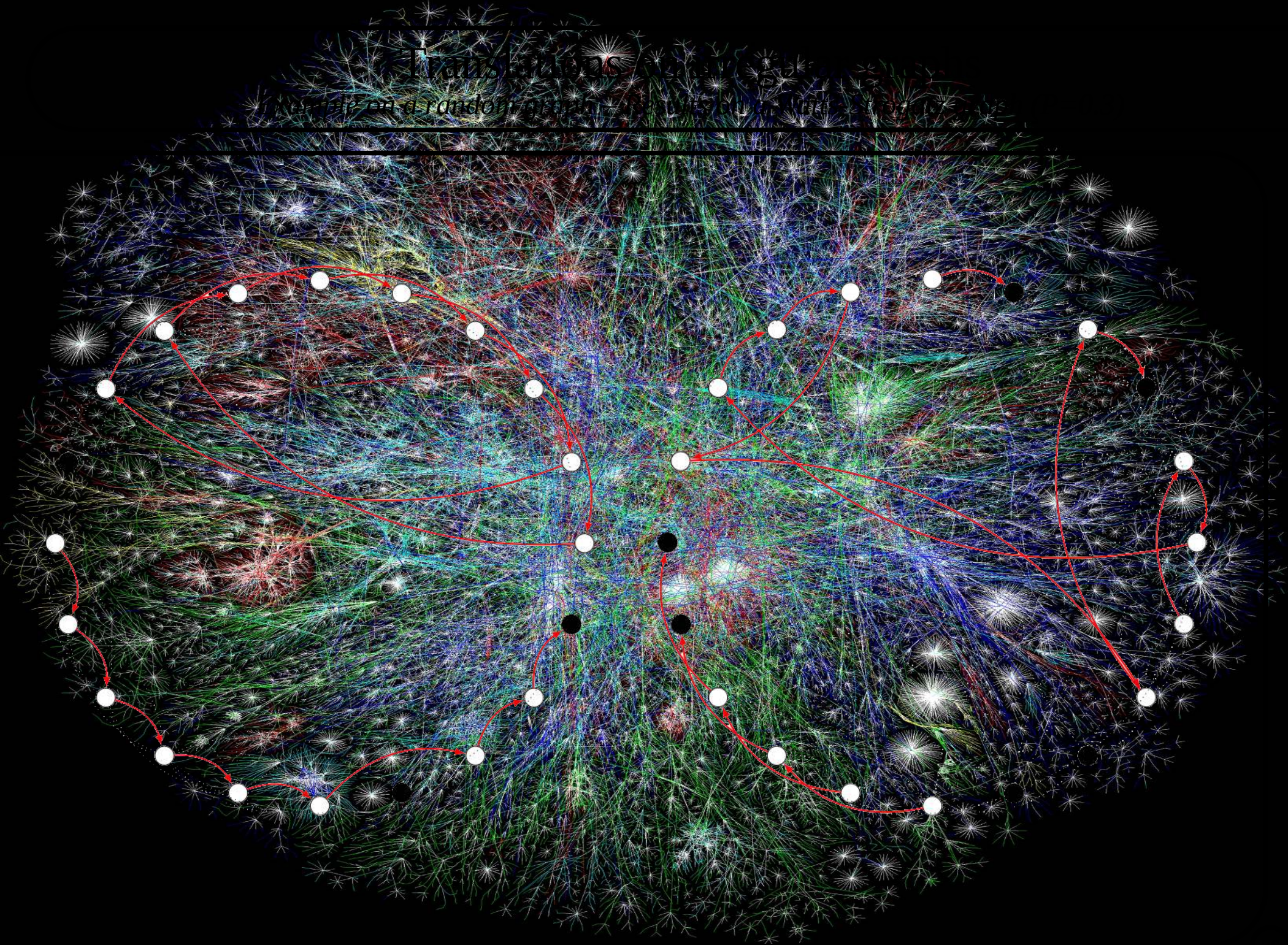
Transients

Model on a random graph - $\beta = 0.5$ and $\alpha = 0.1$



Transients

Number of a random graph - $\log_{10}(n) \approx 10$



► A few words on graph signal processing

- * What is it?
- * The main ideas
- * Some problems and applications

► Graph inference from signals

- * Why it may be necessary
- * The case of stationary signals

► Translations on irregular graphs

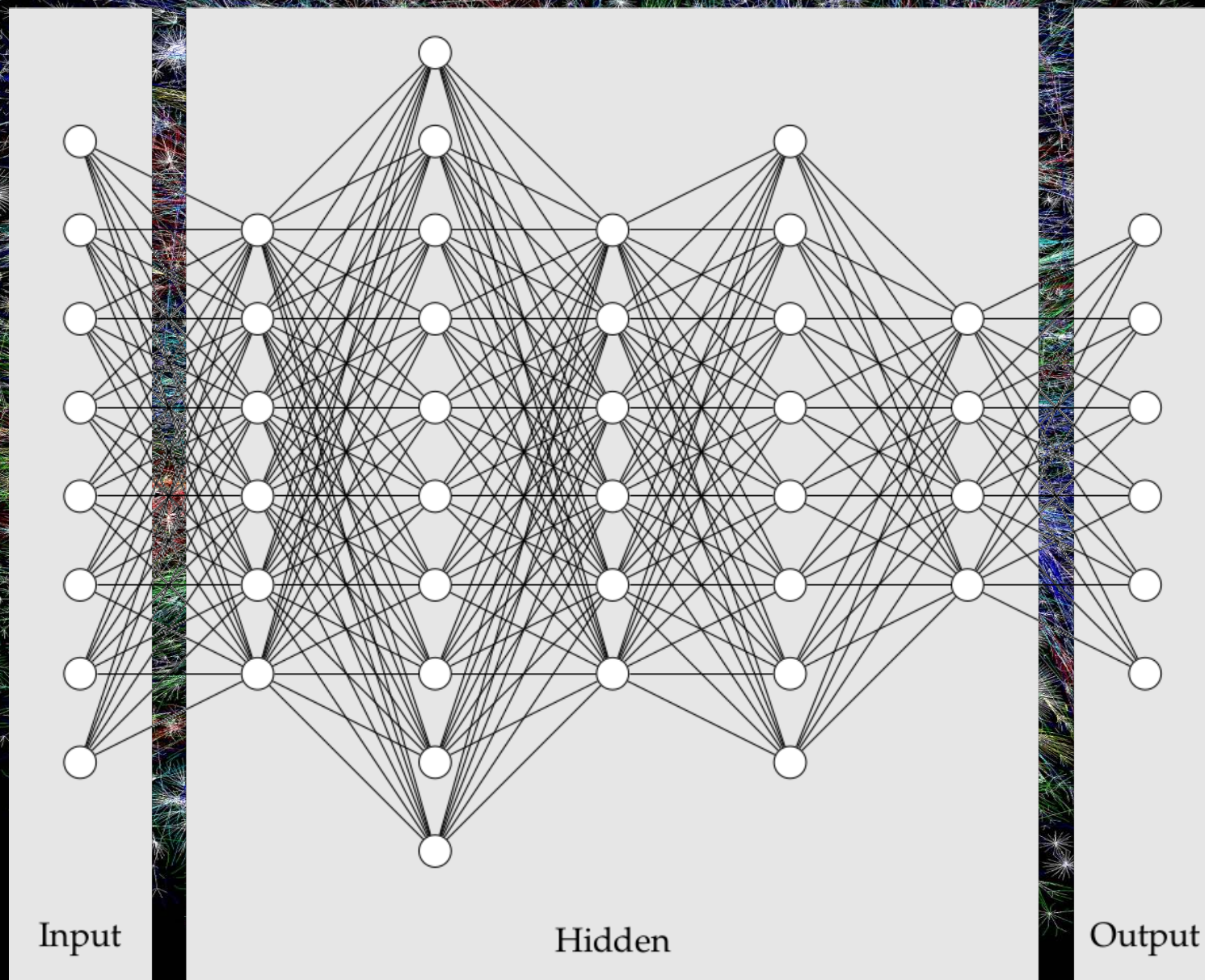
- * How we define them
- * How we identify them
- * Examples and applications

Extension of convolutional neural network

Design of the method

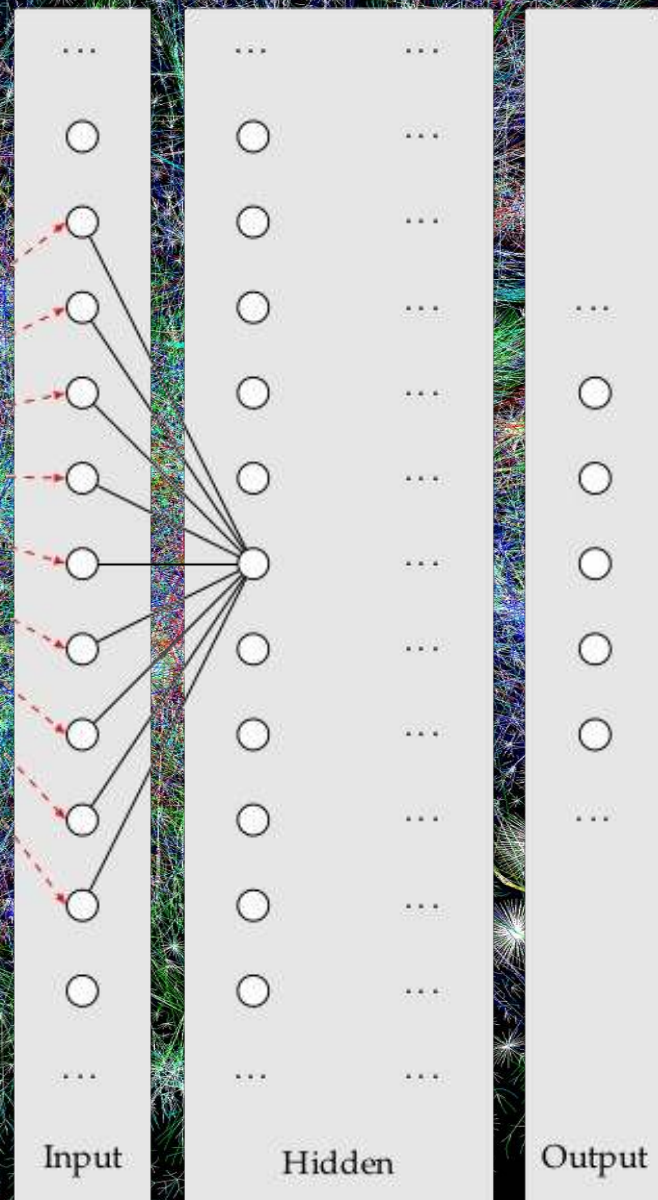
Neural Networks

Introduction to the Neural Network



Extension of the ANN

Introduction of the Path to the Output Layer



Extension of GSPW [13]

Introduction and motivation – Graphs – Graphs

Convolution in the spectral domain

Bruna *et al.* [2013], Henaff *et al.* [2015]
Defferrard *et al.* [2016]

— GSP equivalents
— Polynomial filters

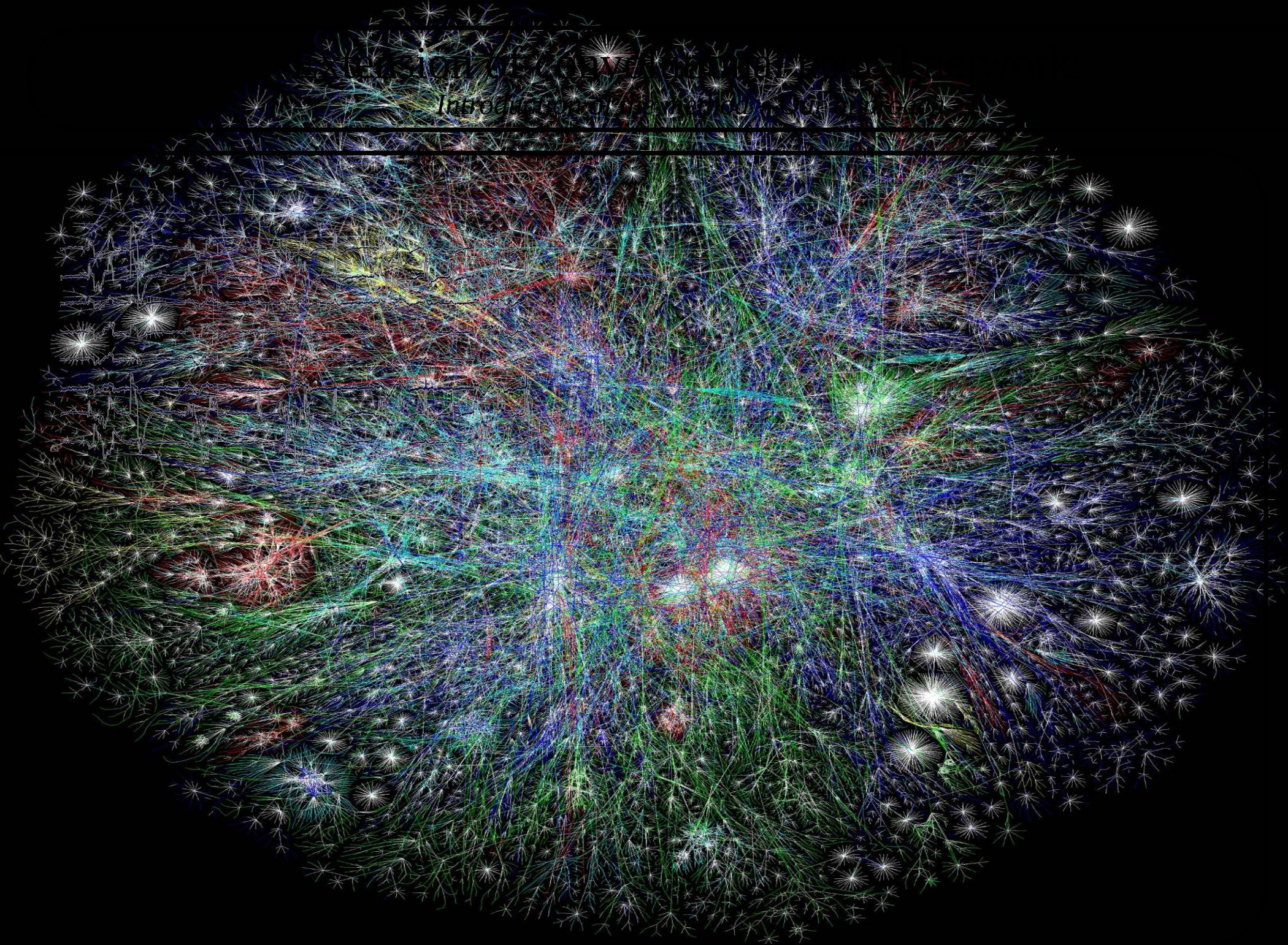
Convolution in the graph domain

Atwood & Towsley [2015]
Pan *et al.* [2017]
Hechtlinger *et al.* [2017]
Mallat *et al.* [2017]

— Arbitrary mapping
— Mapping as a function of weights
— Mapping using random walks
— Mapping learned simultaneously

Section 10.10

Introduction of the new model



Introduction of Graph Neural Network



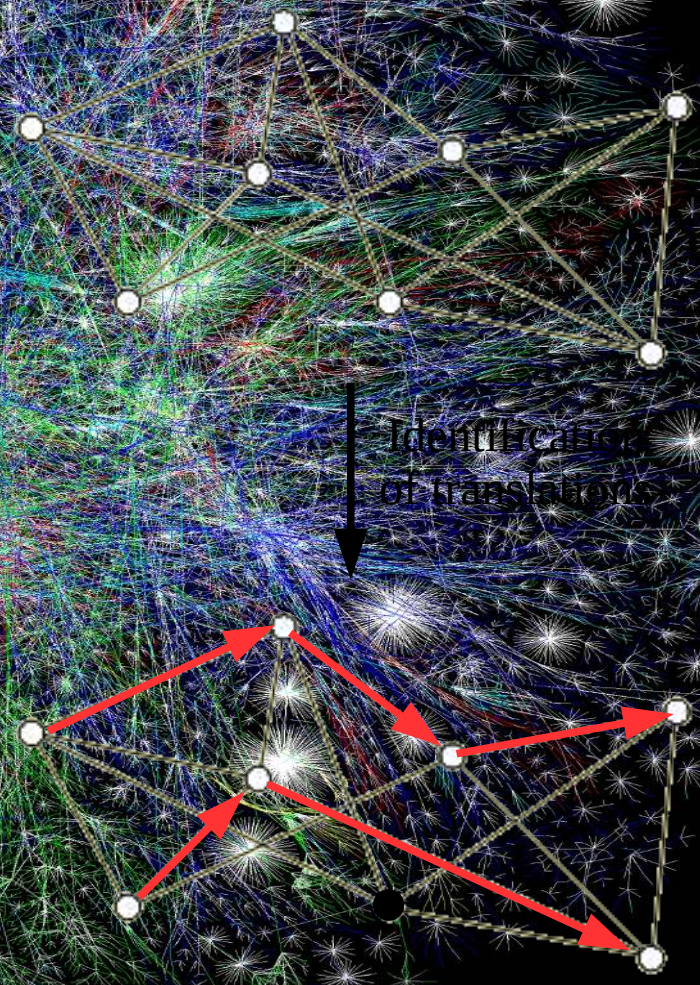
Graph

inference

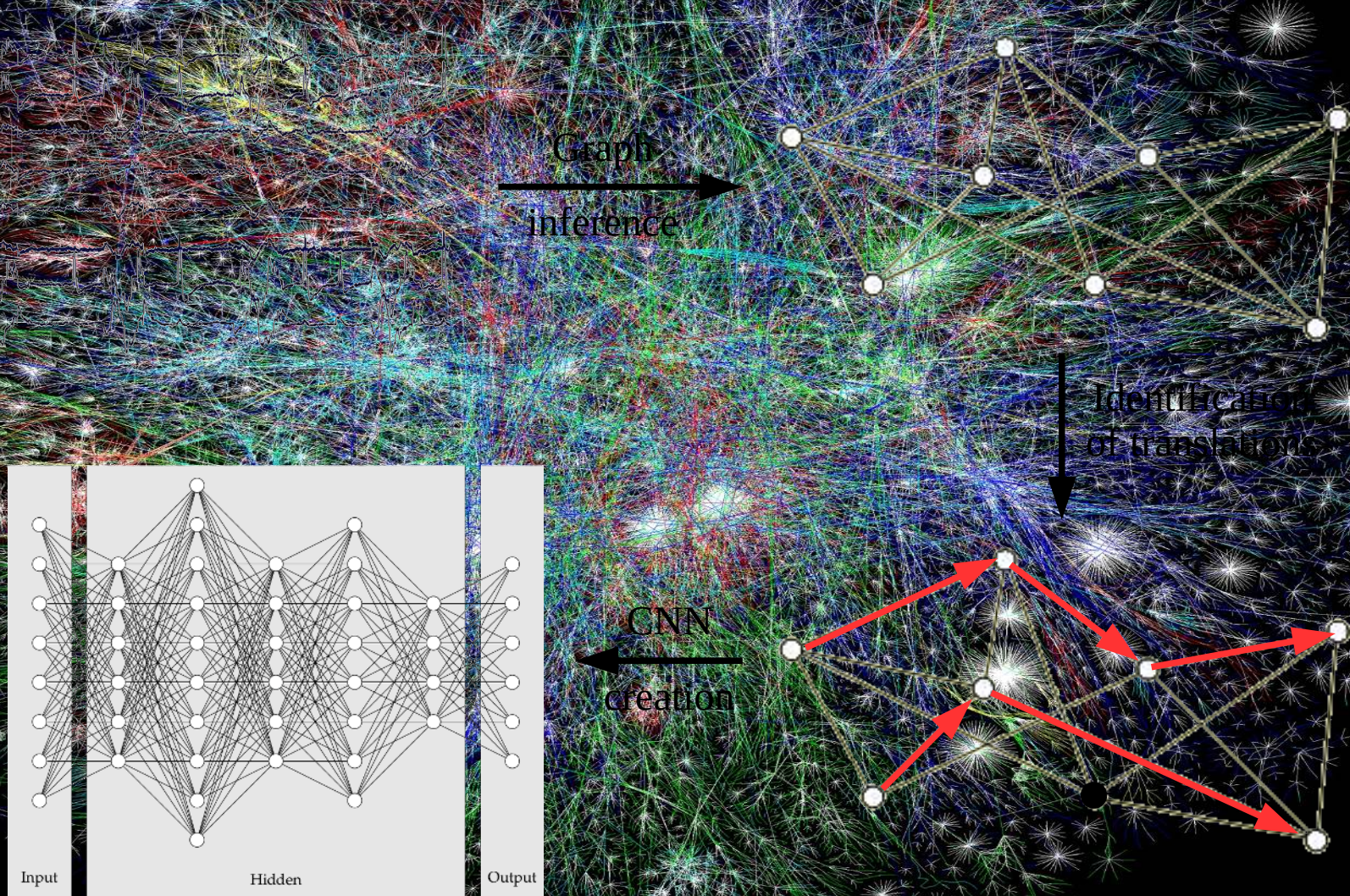
Extension of GEMMA to Incorporation of observed network data

Graph
inference

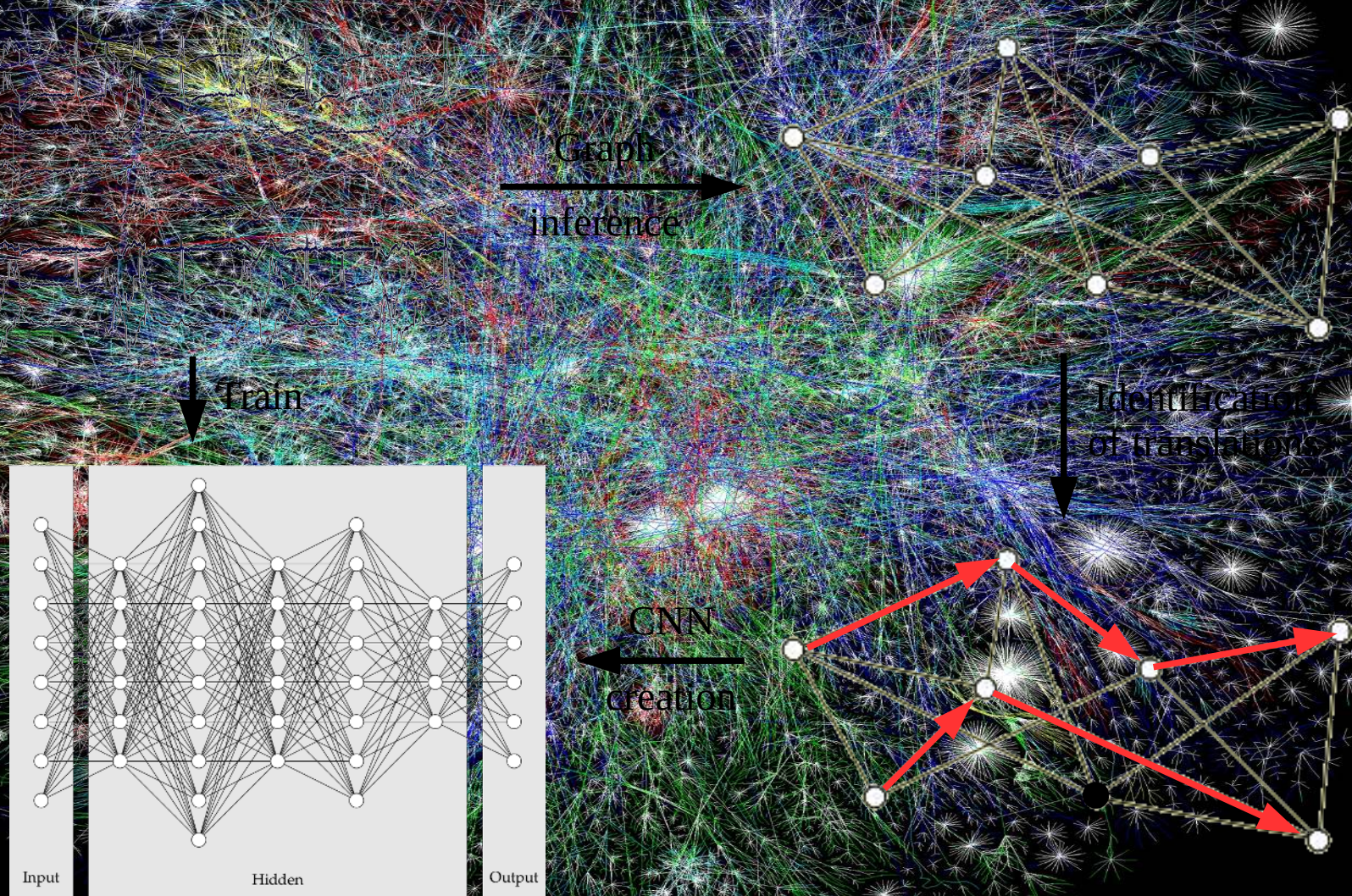
Identification
of translations



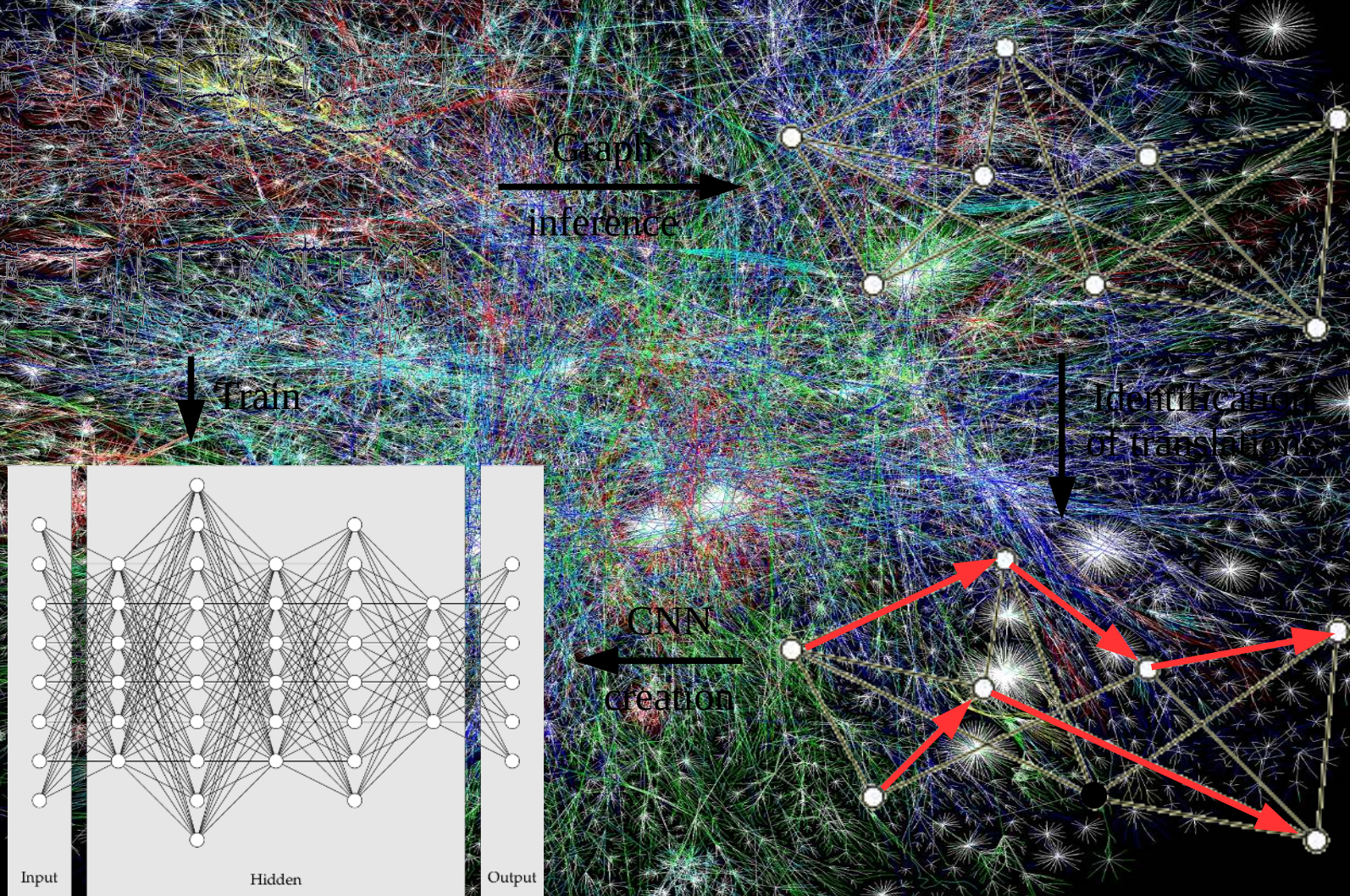
Generation of CNN with Introduction of Graphical Structure



Generation of CNN with Introduction of Graphical Structure



Generation of CNN with Introduction of Graphical Structure



Results



- ▶ 7 convolutional layers
- ▶ 1 kernel size (2 hops)
- ▶ + avoid overfitting
- ▶ max pooling + softmax

Extension of CoMVA [1]

Results: 100,000 nodes, 100,000 edges

► Covariance + max K per row: $\mathcal{G}_S$

► Kalofolias + max K per row: $\mathcal{G}_K$

► Regularized Kalofolias + max K per row: $\mathcal{G}_K^*$

► Sparse + max K per row: $\mathcal{G}_{\text{sparse}}$

Baseline (3x3 sliding kernel): $\mathcal{G}_0$

Extension of GMM to 2D

Results: *Low-dimensional space* (1000)

► Covariance + max K per row: $\mathcal{G}_S$

► Kalofolias + max K per row: $\mathcal{G}_K$

► Regularized Kalofolias + max K per row: $\mathcal{G}_K^*$

► Sparse + max K per row: $\mathcal{G}_{\text{sparse}}$

Smoothness

Stationarity

Baseline (3x3 sliding kernel): $\mathcal{G}_0$

Selection of $\mathcal{G}$ via $\mathcal{G}_g$

Results of $\mathcal{G}_g$ on $\mathcal{G}$ for $\mathcal{G}_g$

$\mathcal{G}_g$	$\mathcal{G}_{\tilde{\Sigma}}$	$\mathcal{G}_K$	$\mathcal{G}_K^*$	$\mathcal{G}_{\text{sparse}}$
85.98% (193)	85.03% (246)	85.33% (195)	85.81% (277)	77.54% (169)

Selection of the GWN model

Results of the comparison of the GWN models

$\mathcal{G}_g$	$\mathcal{G}_{\tilde{\Sigma}}$	$\mathcal{G}_K$	$\mathcal{G}_K^*$	$\mathcal{G}_{\text{sparse}}$
85.98% (193)	85.03% (246)	85.33% (195)	85.81% (277)	77.54% (169)

Smoothness

Stationarity

SELECTION OF KERNEL SIZE

Results of the proposed method

$\mathcal{G}_g$	$\mathcal{G}_{\tilde{\Sigma}}$	$\mathcal{G}_K$	$\mathcal{G}_K^*$	$\mathcal{G}_{\text{sparse}}$
85.98% (193)	85.03% (246)	85.33% (195)	85.81% (277)	77.54% (169)

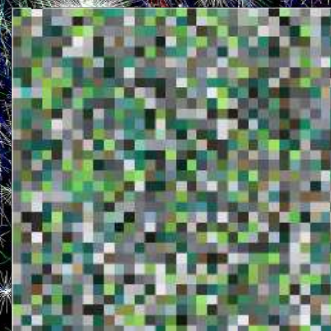
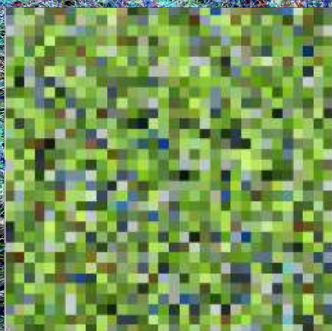
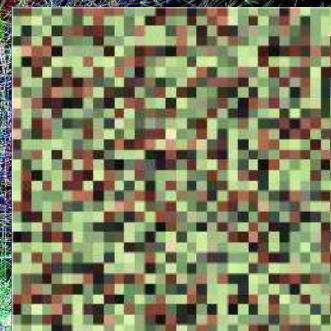
Smoothness

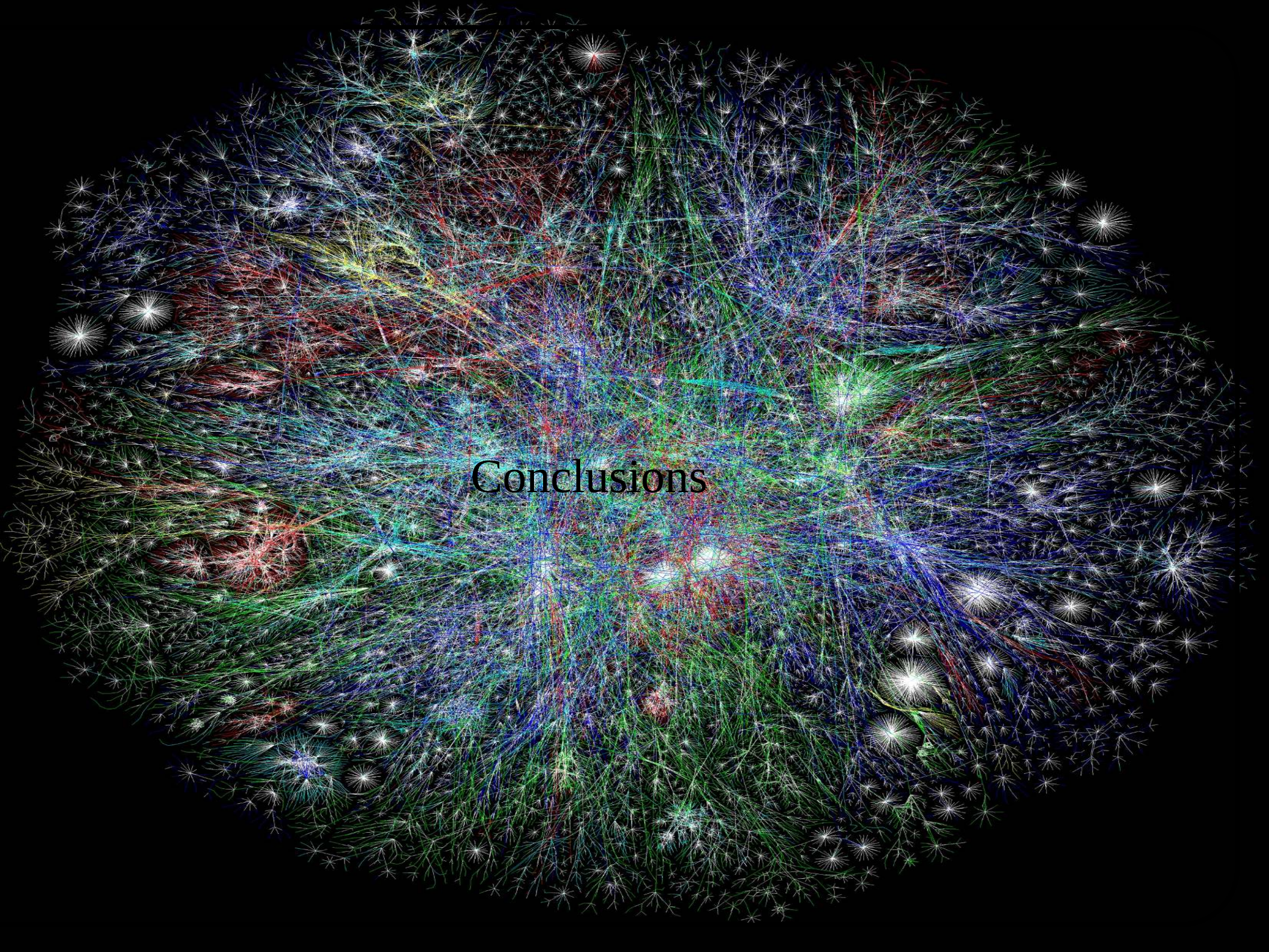
Stationarity

	$\mathcal{G}_g$	$\mathcal{G}_{\tilde{\Sigma}}$	$\mathcal{G}_K$	$\mathcal{G}_K^*$	$\mathcal{G}_{\text{sparse}}$
Initial kernel size	13 (495)	13 (527)	13 (528)	13 (495)	13 (530)
Mean kernel size	12.433	11.523	11.27	12.194	8.137

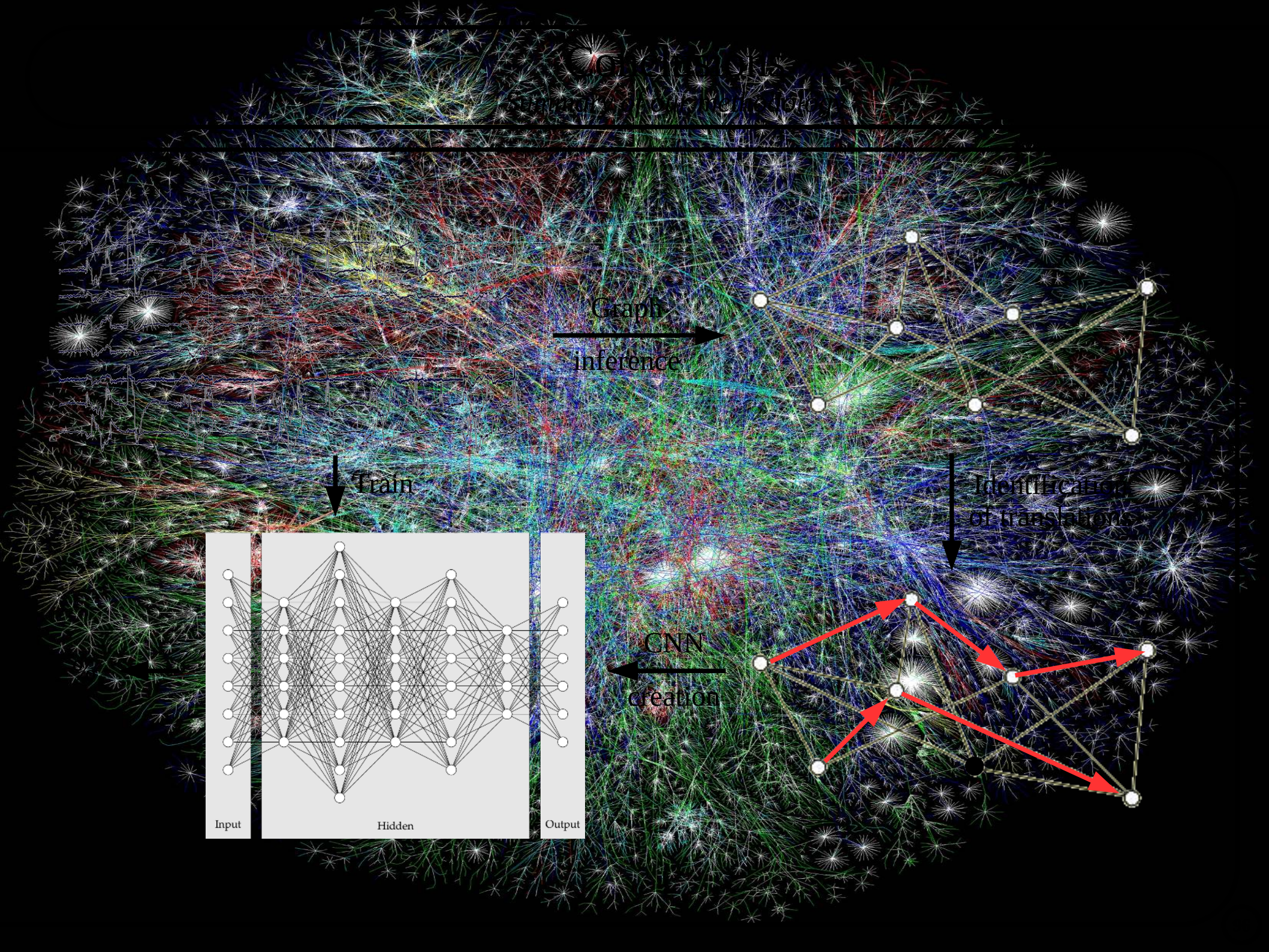
Restoration of Low Resolution

Results from the proposed method



The image features a dense, circular network of fine, multi-colored lines (red, green, blue, yellow) radiating from a central point. These lines connect to numerous small, bright, starburst-like nodes. The overall shape is roughly circular, with the lines and nodes filling the space against a black background. The word "Conclusions" is centered within this network.

Conclusions



Graph

inference

Train

Identification
of translations

CNN
creation

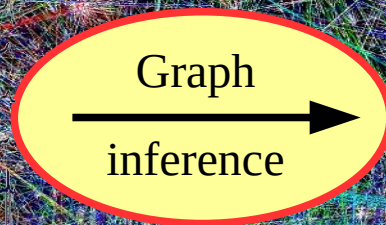
Input

Hidden

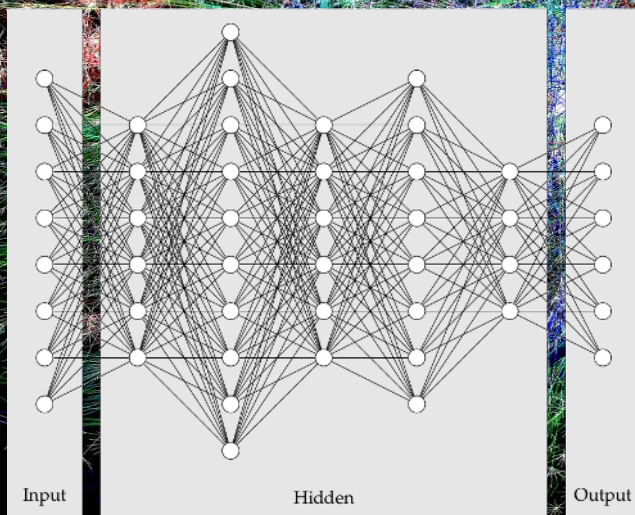
Output

Complexity Reduction

New strategies
Redundant constraints
Best graph selection

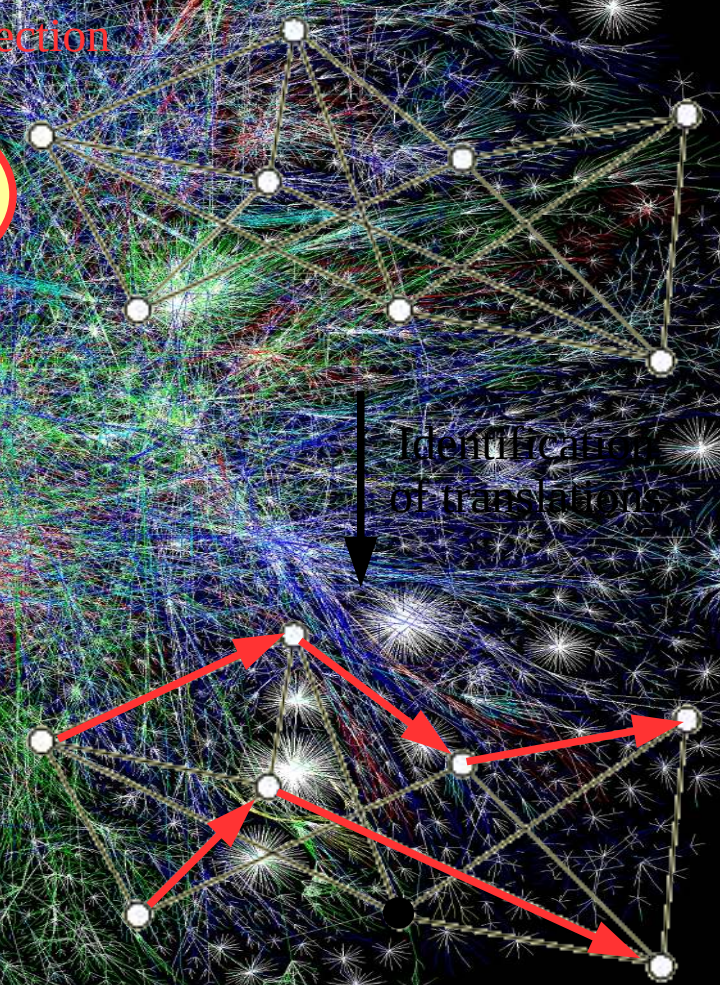


Train



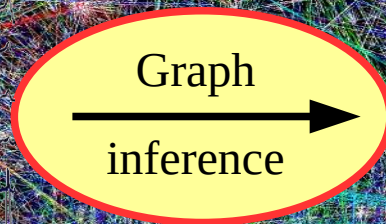
Identification of translations

CNN creation

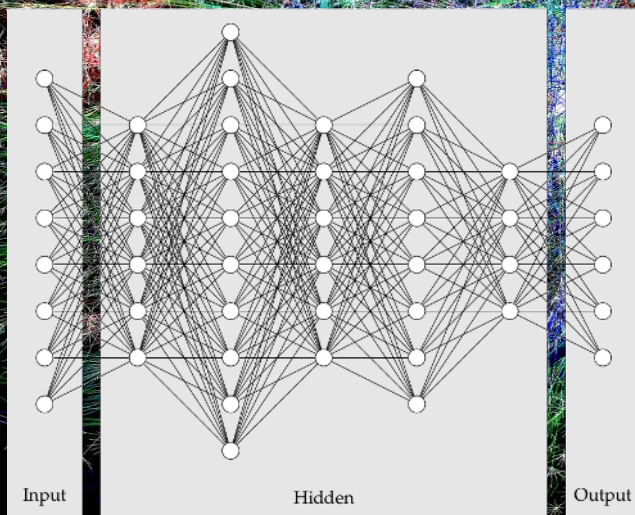


Graph Inference

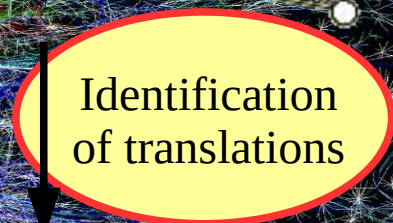
New strategies
Redundant constraints
Best graph selection



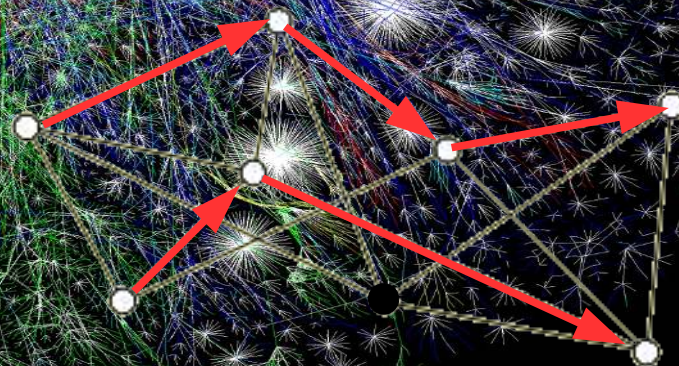
Train



Other relaxations
Next slide :)

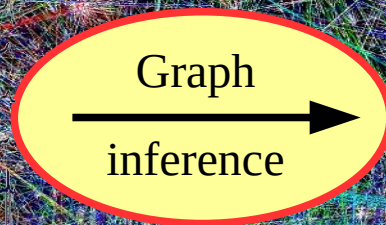


CNN
creation



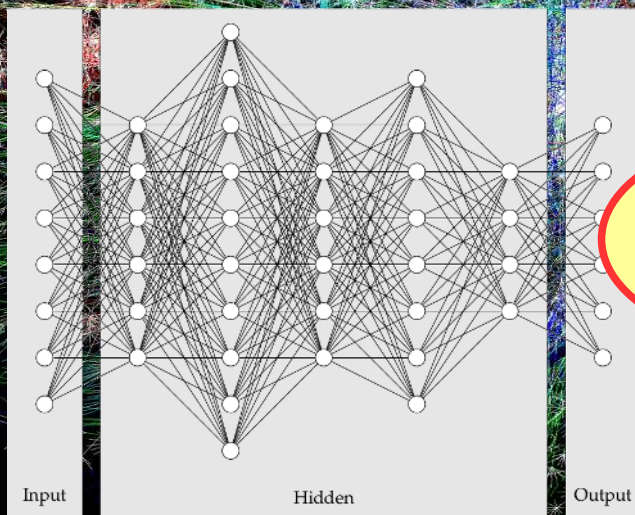
Complexity Reduction

New strategies
Redundant constraints
Best graph selection

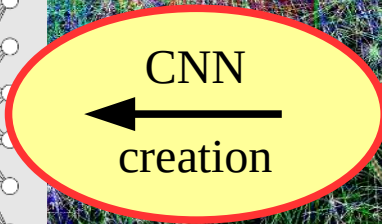
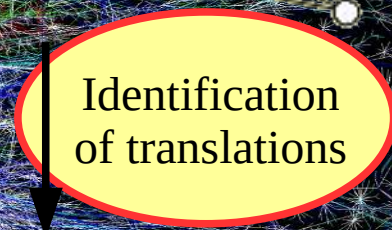


Train

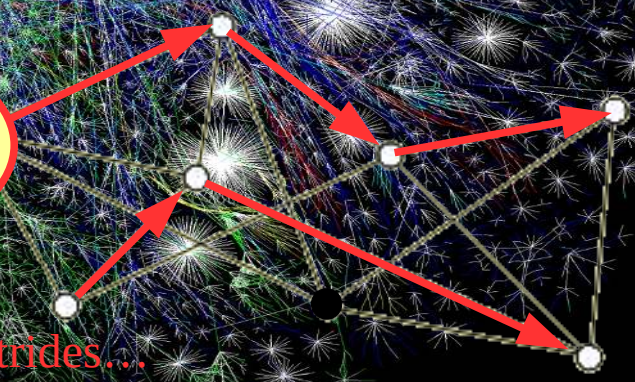
A black arrow pointing downwards from the main graph to the CNN diagram.



Other relaxations
Next slide :)

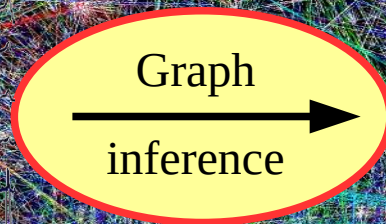


Grid search
Max-pooling, strides...
Various kernel sizes

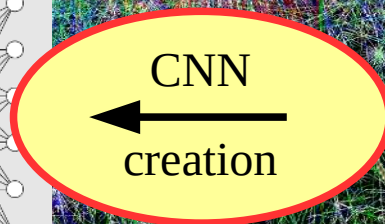
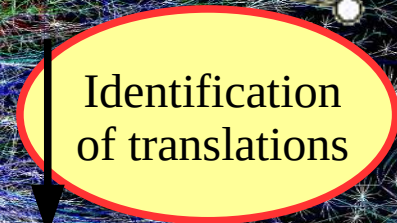


Complexity Reduction

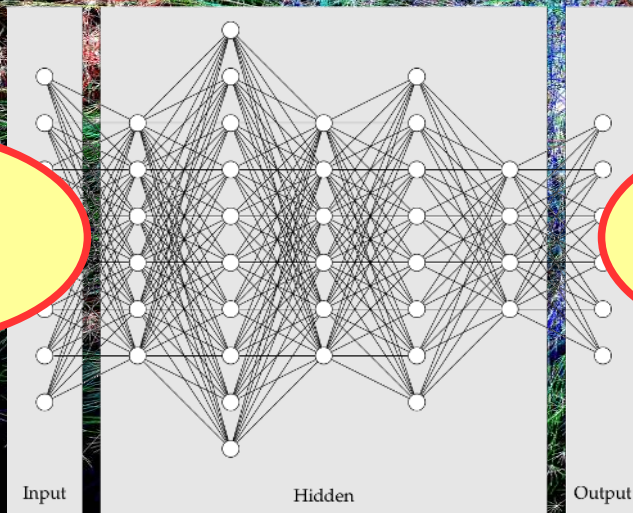
New strategies
Redundant constraints
Best graph selection



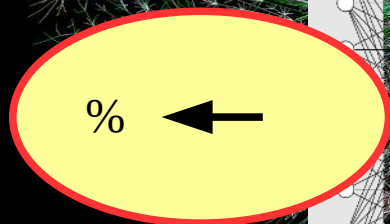
Other relaxations
Next slide :)



Grid search
Max-pooling, strides...
Various kernel sizes

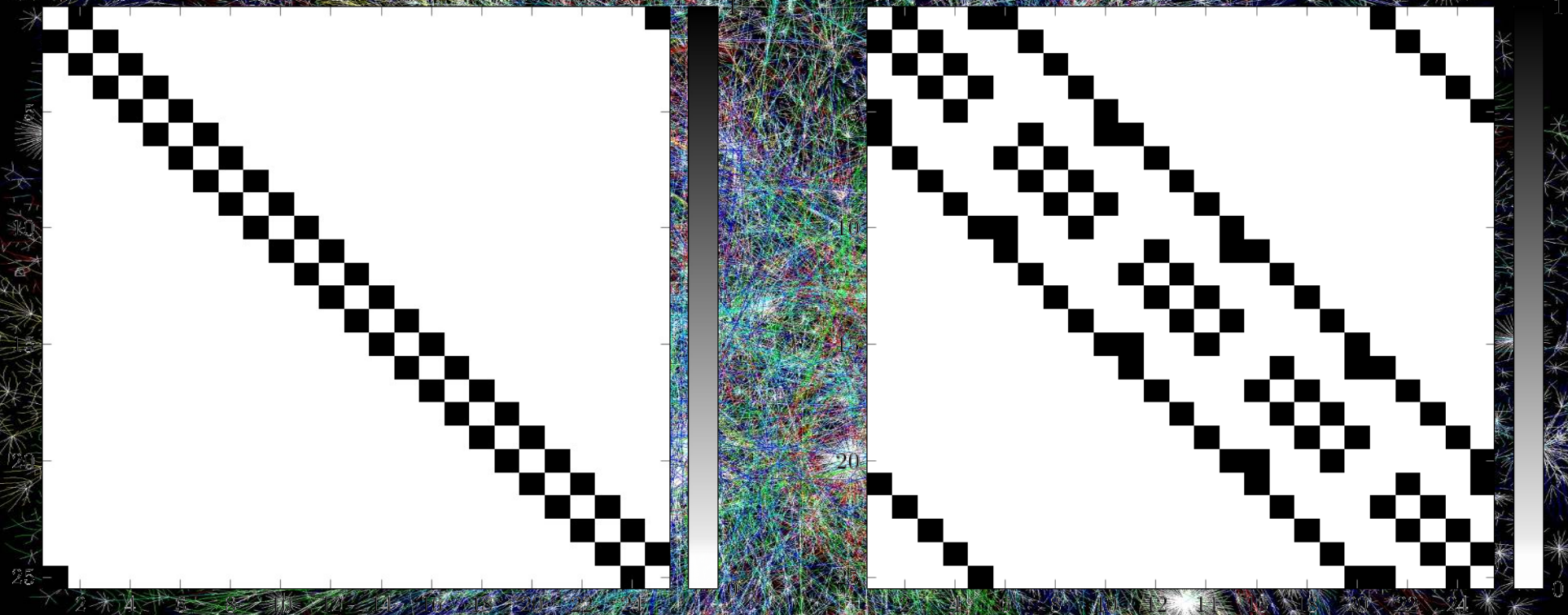


Train



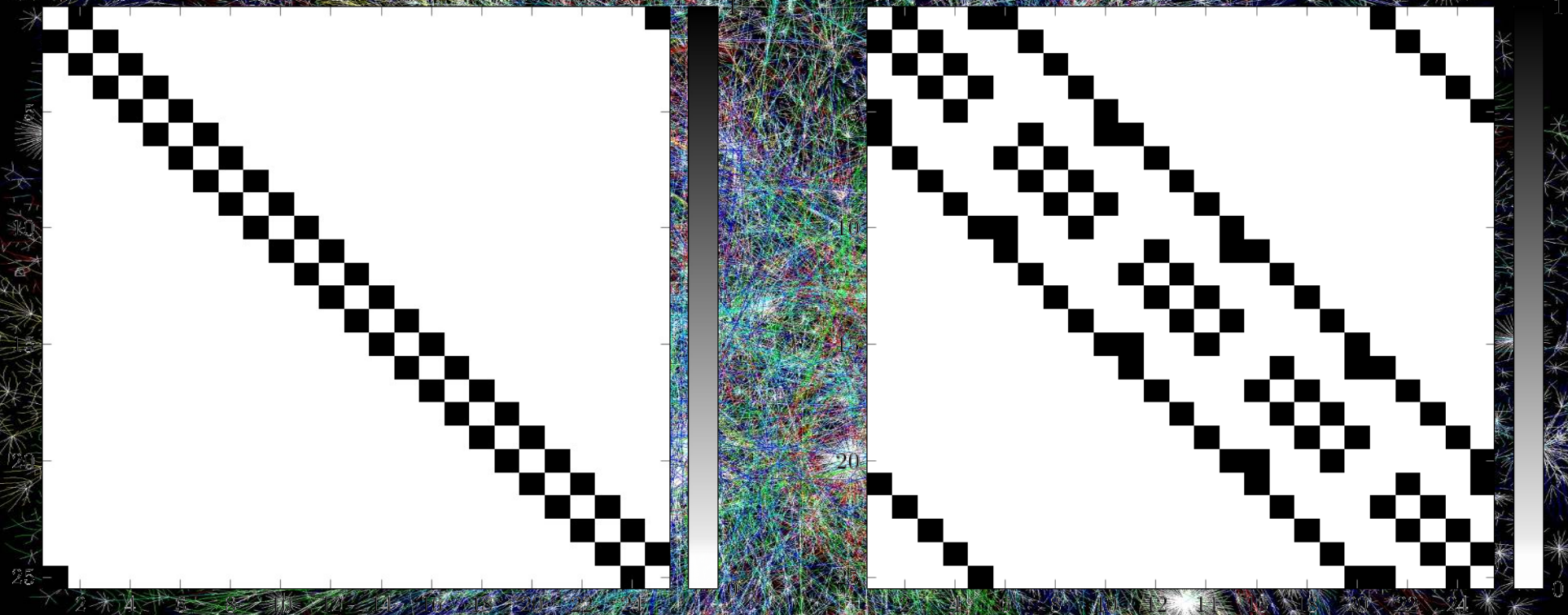
New datasets

Comparison Translation Operators for $2D$ and $3D$

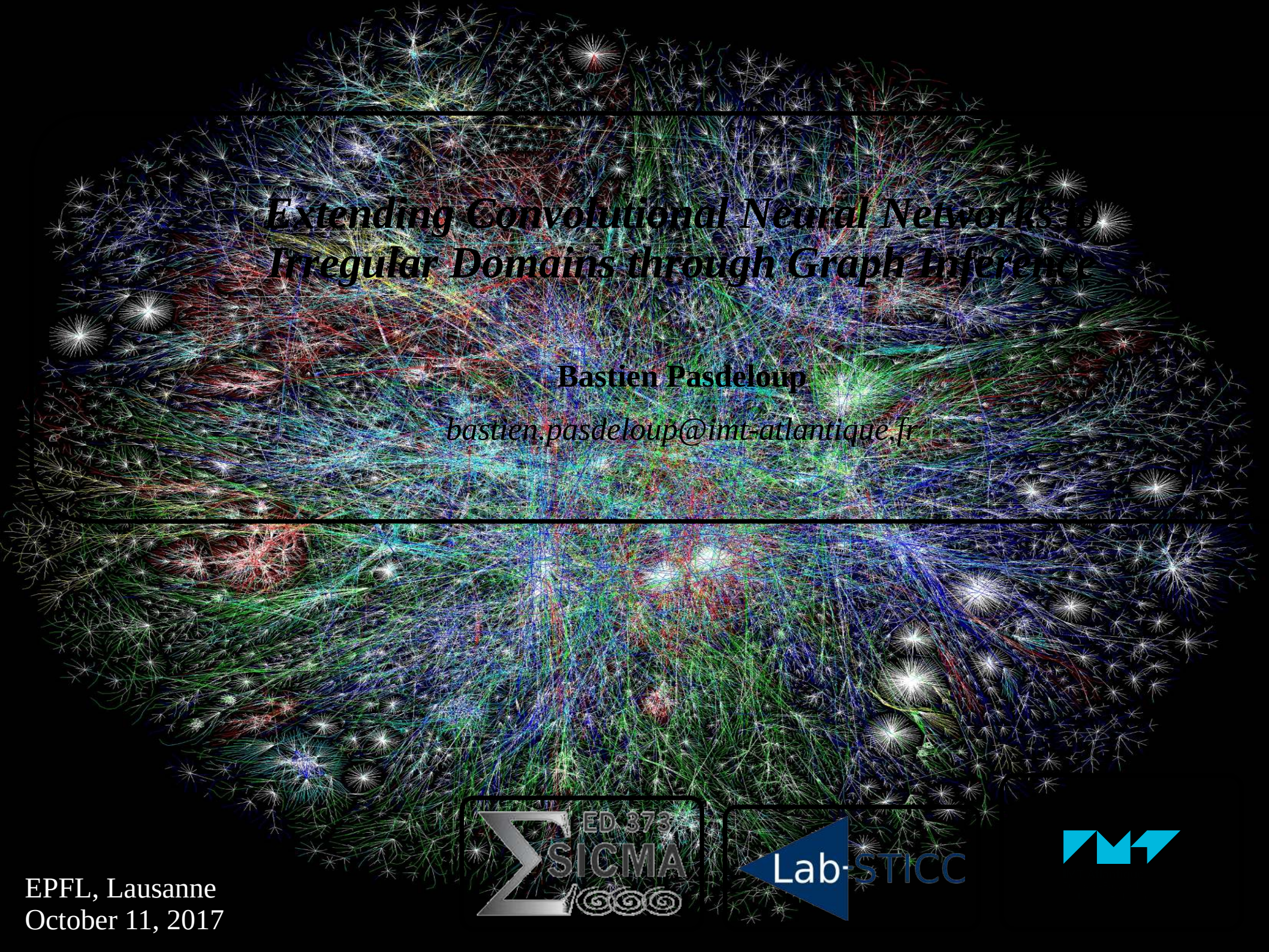


Complacency

Translation operators



If the adjacency matrix of a graph find a permutation P such that the corresponding matrix is the closest to I .



Extending Convolutional Neural Networks to Irregular Domains through Graph Inference

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